Uncertainty, Market Concentration, and the Entrepreneurial Funding Channel*

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Abstract

This paper studies why uncertainty shocks have persistent effects on economic activity. I first document that shocks to uncertainty about non-financial productivity depress firm entry and raise market concentration for longer than standard transmission mechanisms can account for. I then develop a macroeconomic model with a new entrepreneurial funding channel, in which risk-averse investors become less willing to finance prospective entrepreneurs when uncertainty rises. Firm entry falls in response, creating a missing cohort of firms. Because there is no compensating rise in entry when uncertainty subsides, the missing cohort persists and the number of firms recovers only slowly. As a result, output is persistently dampened and market concentration among incumbent firms rises. Transitory uncertainty shocks can therefore generate long-lived changes in the size of the firm population and in market structure.

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Productivity uncertainty shocks have been studied as a potential driver of various firm outcomes, such as investment, employment, and borrowing.[1] Most of these studies, however, focus on business cycle (i.e. short run) outcomes, and find that uncertainty-induced recessions are short lived, lasting a year to a year and a half.[2] Meanwhile, market concentration —of sales and output in particular— has long been a key industry outcome of concern to academics and policymakers alike, due to its potential impact on markups, policy efficacy, and long run productivity growth.[3] However, there is a lack of research directly examining the relationship between this important driver and outcome of firm behavior,[4] and, in turn, whether uncertainty shocks have more persistent effects on the economy than their short run impact on incumbent firms would suggest.

This paper studies why uncertainty shocks have persistent effects on economic activity. I show that uncertainty shocks (as an exogenous increase in the second moment of firm-level productivity shocks) cause a persistent decline in firm entry and a persistent increase in sales concentration. The standard transmission channels found in models of firm dynamics and uncertainty shocks, for example the Oi-Hartman-Abel (OHA), real option, and financial accelerator channels, cannot replicate the persistence of the observed relationship. Instead, the model must be extended to include an *entrepreneurial funding channel* of uncertainty shocks, by which uncertainty shocks decrease risk averse investors' willingness to fund startups, to replicate the persistence.

I first examine the empirical response of market concentration and firm dynamics to uncertainty shocks through the lens of a theoretically agnostic local projection framework and macroeconomic time series data. Two facts emerge from this analysis: 1) uncertainty shocks increase industry concentration, and 2) uncertainty increases concentration through its extensive margin (number of firms) but decreases concentration through its intensive margin

[1] See Bloom (2009) or Bloom et al. (2018) for well known studies focusing on the impact of uncertainty shocks on investment and employment, and Christiano et al. (2014) or Arellano et al. (2019) for examples with a greater focus on borrowing. The uncertainty literature has grown quite large and is discussed at greater length in Section 2.

[2] For example, the economy recovers from a one quarter uncertainty shock within one year in Bloom (2009) and Bloom et al. (2018), and within a year and a half in Alfaro et al. (2023). I discuss notable exceptions in Section 2.

[3] See recent examples, De Loecker et al. (2020) for a focus on the relationship between concentration and markups and Akcigit and Ates (2023) for a focus on the relationship between concentration and productivity growth. Similar to uncertainty, the concentration literature has grown quite large. I will highlight more works studying concentration in Section 2, but a true survey of the literature will be beyond the scope of this paper.

[4] This is not to say that the relationship has not been indirectly studied. From the uncertainty literature there are two clear examples. First, the Oi-Hartman-Abel (OHA) effect (named for Oi (1961), Hartman (1972), and Abel (1983)), mechanically increases output among firms with above-average productivity, while decreasing it among firms with below-average productivity, leading to an increase in output concentration. Second, there is an extensive literature discussing firm sensitivity to financial shocks by size, going back to at least the financial accelerator literature of Bernanke et al. (1994) and Bernanke et al. (1999). Recent works focusing on the macroeconomic impacts of uncertainty shocks have emphasized the phenomenon's role in creating credit disruptions, acting like a financial shock, as in Arellano et al. (2019). Therefore, uncertainty shocks increase the cost of credit, which hurts small firms relatively more than large firms, potentially leading to an increase in concentration. I detail both of these examples in more detail in Section 2. From the firm dynamics literature, Decker et al. (2020) studies the impact of increasing dispersion in firm-level productivity shocks (i.e. increasing firm-level uncertainty) on firm dynamism, but in this case the authors do not directly examine the relationship with sales concentration.

(the comparative size of firms). More precisely put, a one standard deviation shock to the Jurado et al. (2015) measure of macroeconomic uncertainty increases sales concentration (among US public firms) by as much as 7.65 percent relative to its pre-shock level within three years after impact. Concentration then remains more than 5.6 percent elevated five years after impact, and is projected to take approximately eight years to return to its pre-shock level. The same shock decreases the firm entry rate by nearly half a percentage point, with entry remaining depressed for at least five years, and temporarily increases the firm exit rate by approximately 20 basis points (increasing concentration along the extensive margin by reducing the number of firms) while decreasing the growth rate differential between large and small firms (decreasing concentration along the intensive margin of the firm sales distribution). I then focus more narrowly on industry-specific uncertainty shocks, and find that these empirical results hold at the within-industry level in a representative sample of US manufacturing firms. That is, the response of firm entry and concentration to an uncertainty shock far outlasts the uncertainty-induced recessions found in the extant literature. However, while these empirical exercises measure the impact of uncertainty shocks on entry and concentration, they cannot identify what transmission mechanisms explain its persistence.

Therefore, I next develop a quantitative model to better understand what transmission mechanisms relate uncertainty, firm entry, and concentration, and which are the most important in explaining the persistent effect of uncertainty shocks observed in the data. I begin with a baseline model that incorporates the firm dynamics and productivity uncertainty processes of Bloom et al. (2018) or Alfaro et al. (2023), adding the endogenous default risk and the financial accelerator mechanism of Arellano et al. (2019), and allowing firm entry to be determined by the investment decisions of a risk averse venture capitalist, capturing the forces motivating entrepreneurship in canonical works such as Evans and Jovanovic (1989). The venture capitalist is distinct from the risk neutral bank that lends to incumbent firms, so that uncertainty may separately impact the funding of new firms and the borrowing of existing firms. This framework allows for uncertainty —as time-varying volatility in firm-level productivity— to impact the economy through various standard channels, such as the OHA effect, the real options effect, and the financial accelerator. However, when I calibrate this model to US data, I show that these channels can only account for a temporary increase in concentration after an uncertainty shock, failing to generate the persistence observed in the data. In fact, when firm entry is not allowed to respond to uncertainty, concentration fully recovers within one year of the shock. I find that I can only replicate the persistent relationship once I include the entrepreneurial funding channel, by which uncertainty shocks decrease risk averse investors' willingness to fund startups, thus decreasing firm entry.

The importance of the entrepreneurial funding channel becomes clear when I examine other industry outcomes in response to an uncertainty shock. I find that an uncertainty shock decreases the venture capitalist's allocation of start up funding to new firms, which in turn leads to a temporary decline in entry. However, when uncertainty returns to a lower level, there is no compensating increase in new businesses. As a result, there is a "missing cohort" effect which leads to a smaller number of firms producing output, thus a rise in concentration. Because the missing firms are replaced by neither a subsequent increase in entry nor a decrease in exits, the number of firms only slowly recovers as the industry gradually turns over, and concentration is projected to remain elevated for approximately 23 years after a one period shock. The missing cohort and subsequent decline in the number of firms also leads to a persistent decline in overall output, assets, and borrowing in the economy. More precisely put, the number of firms initially declines by approximately one percent, which in turn translates to a nearly one percent decline in assets and borrowing, both of which remain half a percent below their pre-shock levels ten years after impact. Output likewise persistently declines, although less severely, and only after briefly expanding on impact due to the OHA effect.

However, before using the quantitative model to confidently draw conclusions on the source of this persistence, there are two potential concerns worth scrutinizing further.

First, the quantitative model relies on a non-standard firm entry process. While works such as Ates and Saffie (2021) likewise model firm entry in a macroeconomy as being determined by the venture funding decisions of a financial intermediary, the Free Entry condition is a far more common mechanism for determining entry in macroeconomic models (for example, Hopenhayn (1992), Restuccia and Rogerson (2008), and Clementi and Palazzo (2016)). Therefore, I compare the impact of an uncertainty shock on the firm entry rate in my calibrated model and an identical model with free entry. In turn, I find that the free entry model actually predicts an increase in firm entry in response to an uncertainty shock —a finding inconsistent with the previously discussed empirical relationship. The reason for the difference is that while an uncertainty shock decreases the Sharpe ratio of venture funding (i.e. it decreases the risk-adjusted expected return on investing in startups) the expected value of startups increases due to the OHA effect and limited liability. Thus, entry determined by a risk averse investor decreases in response to an uncertainty shock, while entry determined by its expected value alone (as under free entry) increases. The response of firm entry therefore acts as a separating moment between the two competing models of entry, and the data favor entry determined by a risk averse investor.

Second, the fact that declining entrepreneurial activity is necessary to generate persistence in the response to uncertainty shocks in the model raises the possibility that wealth or other financial shocks, rather than uncertainty shocks per se, may actually explain the empirical results.[5] To answer this question, I compare the impact of firm-level productivity uncertainty and investor wealth shocks (a financial shock that constrains entrepreneurs) on the cross-section of firm leverage in my model.[6] Only the firm-level uncertainty shock can simultaneously match both macro and micro empirical moments.

The quantitative model additionally predicts that the persistent effect of uncertainty shocks depends on the pre-existing level of concentration: the *concentration dampener effect*. I find that when the economy receives an uncertainty shock while relatively concentrated (i.e. has a high level of pre-existing concentration) then the subsequent recession is more shallow and considerably less persistent than in the model calibrated to US data. A back of the envelope calculation shows that an economy whose concentration is approximately 40 percent higher than the baseline model recovers from an uncertainty shock 52 percent faster and the initial declines in assets and borrowing are six percent shallower. The reason for the concentration dampener effect becomes apparent when considering how concentration interacts with the financial accelerator. Credit constraints are often more binding for young and small firms relative to old and large firms. This is true in observed data, and as I will show, in my model as well.[7] A more concentrated economy has a larger share of its output controlled by large firms who do not face binding credit constraints. As a result, when a more concentrated economy faces a credit supply disruption —like the kind caused by uncertainty shocks— then a smaller amount of its aggregate output is produced by firms affected by the disruption, dampening the overall effect of the shock.

I lastly turn to discussing four noteworthy implications of these findings. The first two focus on the impact and cost of cyclical shocks, while the second two concern secular trends in the US. First, the relationship between uncertainty shocks, entrepreneurship, and concentration provides a transmission mechanism by which cyclical uncertainty shocks may have a permanent effect on economic outcomes in models of long run growth. For example, uncertainty shocks that decrease entry and increase concentration may in turn slow the creative destruction process and permanently decrease endogenous productivity growth in product quality

[5] Evans and Jovanovic (1989) find that almost all entrepreneurs face binding financial constraints, and much attention has been paid since to the role of collateral and household wealth on entrepreneurial activity. Because some of my empirical exercises utilize asset pricing based proxies for uncertainty, there may be a concern that a financial shock is actually acting as a lurking variable, simultaneously increasing expected financial volatility and dampening firm entry by tightening financial constraints on entrepreneurs, thus increasing concentration.

[6] Empirical uncertainty shocks are correlated with a larger decline in leverage for small firms relative to large firms. I estimate this response using a binned linear regression and a panel of US public firms.

[7] More precisely, I will show that financial constraints are tighter for small firms than large firms in my model, since I will abstract from firm age in my quantitative exercises.

ladder models, such as Klette and Kortum (2004), Akcigit and Kerr (2018), and Akcigit and Ates (2023). I will note, however, that the business cycle model studied in this paper does not itself have a mechanism for cyclical shocks to have a truly permanent effect, and I leave this extension to future research. Second, the concentration dampener effect presents a counter-argument to the hypothesis that concentration may increase volatility in an economy through the increased importance of granular shocks, as in Gabaix (2011). Third, the entrepreneurial funding channel represents a channel through which the secular rise in firm-level uncertainty may be a cause of the secular decline in entrepreneurship, thus business dynamism, in the United States. While this is not explicitly contradictory to studies such as Decker et al. (2020), that argue the opposite effect of the increasing dispersion in firm-level productivity shocks, it does suggest that further work needs to be done in accounting for the various channels by which increasing firm-level productivity uncertainty may impact business dynamism to understand its net effect. Fourth, the concentration dampener effect suggests a possible connection between two contemporaneous secular trends in the US, increasing concentration and the Great Moderation, as increasing concentration may dampen the impact of financial shocks and in turn decrease the aggregate volatility experienced by the economy.

Roadmap. The remainder of the paper is organized as follows: Section 2 discusses the related literature and enumerates this study's contributions in detail, Section 3 presents empirical evidence of the persistent response of firm entry and concentration to uncertainty shocks, Section 4 presents the general equilibrium model, Section 5 turns to quantitative exercises, and Section 6 concludes.

2 Contributions and Related Literature

This study's first contribution is to identify the extremely persistent effects of cyclical uncertainty shocks. The uncertainty literature has grown quite large, a fact evident in recent surveys of the literature, such as Cascaldi-Garcia et al. (2023) which focuses on measurement, Fernández-Villaverde and Guerrón-Quintana (2020) which focuses on theory, and Dew-Becker and Giglio (2023) which focuses on financial markets. One may also consult Jefferson (2023) for a recent, broad, review of uncertainty and its importance to policymakers. Studies have considered a wide range of economic outcomes, such as investment decisions (see for example, Bernanke (1983), Bloom (2009), and Bloom et al. (2018)), employment decisions (Leduc and Liu (2016)), markups (Basu and Bundick (2017)), and policy efficacy (Vavra (2014)).

However, most studies of uncertainty shocks focus on business cycle (i.e. short run) outcomes, and find that uncertainty-induced recessions last for a year to 1.5 years (for example, Bloom et al. (2018) and Alfaro et al. (2023)). Notable exceptions include Bonciani and Oh (2019) and Carriero et al. (2023) which focus on the long run cost of uncertainty shocks via a reduction in R&D expenditures. I find that uncertainty shocks, by disrupting the firm entry process, induce extremely long lived recessions (with full recovery taking up to 23 years in the absence of subsequent expansionary shocks in my quantitative exercises). The finding that uncertainty shocks have persistent effects due to their impact on entrepreneurship relates this work to other macroeconomic studies that focus on crises prolonged by declining entry, such as Clementi and Palazzo (2016) and Ates and Saffie (2021), the literature on hysteresis (see Aikman et al. (2022) for a detailed discussion), and the literature on slow recoveries after financial crises more specifically (see Jordà et al. (2017) for a review).

Relatedly, this study focuses on examining the persistent effects of uncertainty shocks through the lens of a "standard" uncertainty-driven business cycle model, that does not have a mechanism for cyclical shocks to have a true permanent effect. However, by identifying the relationship between uncertainty shocks, entrepreneurship, and concentration, my work suggests that future research embedding this relationship into an endogenous growth model could exhibit long run effects of cyclical uncertainty shocks, by impacting either firm entry or concentration. With further extensions of such a model, uncertainty shocks that permanently increase concentration may also permanently impact markups (De Loecker et al. (2020) and Edmond et al. (2023)), the labor share of profits (Autor et al. (2020)), policy efficacy (Colciago and Silvestrini (2022)), and business dynamism more broadly.

My second contribution is to incorporate the entrepreneurial funding channel into a modern heterogeneous firm model with productivity uncertainty shocks.[8] Much work since Evans and Jovanovic (1989) has focused on financial constraints as a key determinant of entrepreneurship and the most important friction to firm entry.[9] However, there is also a history of work that emphasizes risk aversion and uncertainty as a barrier to entrepreneurship. Kihlstrom and Laffont (1979) present a general equilibrium model in which firm entry is determined by entrepreneurship, which itself is determined by the distribution of risk aversion among households.[10] That is, entrepreneurship is determined by agents' willingness to bear uncertainty. The authors highlight that their model is merely a formalization (and special case) of intuition first laid out in Knight (1921). As such, it is clear that I am not the first to

[8] There is of course a literature considering how firm entry impacts concentration and vice versa, see, for example, Bresnahan and Reiss (1991) as a discussion of how entry impacts competitiveness and a history of the topic.

[9] I will direct readers to Quadrini (2009) for a more extensive review of entrepreneurship in macroeconomics.

[10] Cressy (2000) shows that, in fact, the Evans and Jovanovic (1989) finding that financial constraints are typically binding on entrepreneurs can also be explained by decreasing (in wealth) relative risk aversion, as in Kihlstrom and Laffont (1979).

link uncertainty and entrepreneurship. Rather, my contribution is to bring this relationship into a modern heterogeneous firm model with uncertainty shocks, building on work such as Bloom et al. (2018), Arellano et al. (2019), and Alfaro et al. (2023), and to document in my model that entrepreneurship is quantitatively important to explaining the persistent impact of uncertainty on concentration.[11]

This study's third contribution is to directly measure the relationship between cyclical uncertainty shocks and the level of sales concentration. Despite the wide range of economic outcomes considered by the uncertainty literature, no existing work has directly examined uncertainty's impact on concentration.

That is not to say that the relationship has not been indirectly studied through various channels. First, the OHA effect of uncertainty shocks, named for Oi (1961), Hartman (1972), and Abel (1983), describes the fact that a second moment productivity shock will increase output for firms with above average firm-level productivity, and decrease output for firms with below average firm-level productivity. As a result, output will rise overall (the OHA effect) and will become more concentrated in firms with above average productivity. Second, an extensive literature shows that uncertainty shocks decrease the supply of credit, thus increase the cost of borrowing in the economy; see for example Christiano et al. (2014), Gilchrist et al. (2014), Arellano et al. (2019), Berger et al. (2020), Fernández-Villaverde and Guerrón-Quintana (2020), and Akinci et al. (2022). At the same time, there is also extensive literature documenting differences by firm size in the sensitivity to financial shocks (i.e. shocks that increase the cost of credit, like an uncertainty shock). Small firms are more severely impacted by these shocks, as shown in the early financial accelerator literature (Bernanke et al. (1994) and Bernanke et al. (1999)) and recent empirical work including Chodorow-Reich (2014), Siemer (2019), Aruoba et al. (2022), and Davis and Haltiwanger (2023). Together, these two literatures imply that an uncertainty shock will cause a credit tightening, increasing credit costs more for small firms than larger firms, and potentially increasing the concentration of economic activity in large, less impacted, firms.

This work's fourth contribution is to introduce the concentration dampener effect and to show how it influences the transmission of cyclical shocks. The financial accelerator literature — connecting the supply of credit and financial intermediation to real economic outcomes— has a long history, including seminal work based on moral hazard problems, as in Hart and Moore (1994) and Kiyotaki and Moore (1997), or costly state verification (CSV) problems, as in Bernanke and Gertler (1989), Bernanke et al. (1999), and Carlstrom and Fuerst (1997).

[11] Brand et al. (2019) also study the impact of uncertainty shocks on firm entry, but do so in a representative agent setting and do not extend the analysis to consider the subsequent impacts on concentration.

A common modeling concern in quantitative analyses of the financial accelerator is that the economy or individual agents may outgrow financial constraints, as in Moll (2014).[12] This paper's contribution to the financial frictions literature is to highlight that the aggregate size of the economy is not a sufficient statistic for understanding how consequential financial frictions may be to economic outcomes; the concentration of resources across firms matters as well. This finding is similar to the key result of the misallocation literature (Hsieh and Klenow (2009), Bartelsman et al. (2013), Gopinath et al. (2017) and the many works that followed), that the distribution of productivity distortions across firms influences the transmission of cyclical shocks. The Heterogeneous Agent New Keynesian (HANK) literature, as in Kaplan et al. (2018), has made similar arguments regarding the importance of the household wealth distribution in determining the transmission of monetary policy shocks, with a similar logic regarding the effective tightness of financial constraints in the economy.

3 Empirical motivation

I first examine the empirical response of firm entry and market concentration to uncertainty shocks through the lens of a theoretically agnostic and flexible local projection framework. Two facts emerge from this analysis: 1) uncertainty shocks increase industry concentration, and 2) uncertainty increases concentration through its extensive margins but decreases concentration through its intensive margin.

I additionally show that this relationship is robust to estimation in a structural Bayesian vector-autoregression (SVAR) with an identification strategy based on sign restrictions informed by economic theory. Moreover, the SVAR allows me to compare the effects of uncertainty shocks to the effects of other standard business cycle shocks (e.g. monetary policy, aggregate demand, and aggregate supply shocks), and I find that uncertainty shocks have the largest effect on market concentration of any tested shock and explain a plurality of the cyclical variation in concentration over the business cycle. However, I leave the SVAR analysis to Appendix B to economize on space in the main text.

3.1 Macro evidence

I will begin by examining the relationship between uncertainty and concentration at the aggregate level. To do so, I first estimate the response of sales concentration among US

[12]The intuition behind this challenge is straightforward: Agents become wealthy enough that they can pay off all their (optimally allocated) debt, and the constraint on their borrowing limit does not bind, ameliorating the effects of financial frictions in the model. Models with moral hazard circumvent this problem by either assuming or engineering a micro-foundation which yields an always- or occasionally-binding collateral-constraint. Models featuring a CSV problem have alternatively taken to assuming the stochastic "death" of agents to prevent them from becoming too wealthy. Agent deaths are alternatively expressed as an exogenous shock to transition into an alternative role, such as going from a bank to a typical household.

public firms to a one standard deviation macroeconomic uncertainty shock, then move to examining the impact of the same shock across the extensive and intensive margins of the firm sales distribution to better understand the transmission mechanisms at work.

These responses will be estimated via Jordà (2005) style local projections, and the baseline model can be written as:

$$log(HHI_{t+h}/HHI_{t-1}) = \alpha^h + \beta^h(uncertainty_t) + \theta^h X_{t-1} + \epsilon_t^h \tag{1}$$

where t is the time index and h the forecast horizon; concentration will be measured as the Herfindahl–Hirschman index (HHI) of sales among all US public firms, uncertainty will be proxied by the Jurado et al. (2015) measure of macroeconomic uncertainty, and X is the vector of lagged controls: real GDP growth rate, core PCE inflation, one year Treasury yield as a proxy for the policy rate (as in Gertler and Karadi (2015)), and lagged concentration.[13] Data is annual from 1990 through 2019.[14]

I attempt to use the most standard measures of uncertainty and concentration possible. I use the HHI as my baseline proxy for concentration because it is one of the two most widely used measures of concentration in the macroeconomics literature (as noted in Syverson (2019) and Ahern et al. (2024)) and the measure used by policymakers when considering anti-trust enforcement (see the DOJ and FTC (2023) merger guidelines, page 5).[15] My measure of concentration is presented in Figure 15 and discussed in further detail in Appendix A.

I likewise use the Jurado et al. (2015) (three-month ahead) macroeconomic uncertainty measure because it has become a standard measure for uncertainty based on real activity.[16] More common proxies for uncertainty exist, such as the CBOE VIX and the Ludvigson et al. (2021) financial uncertainty series. However, these measures are based on expected

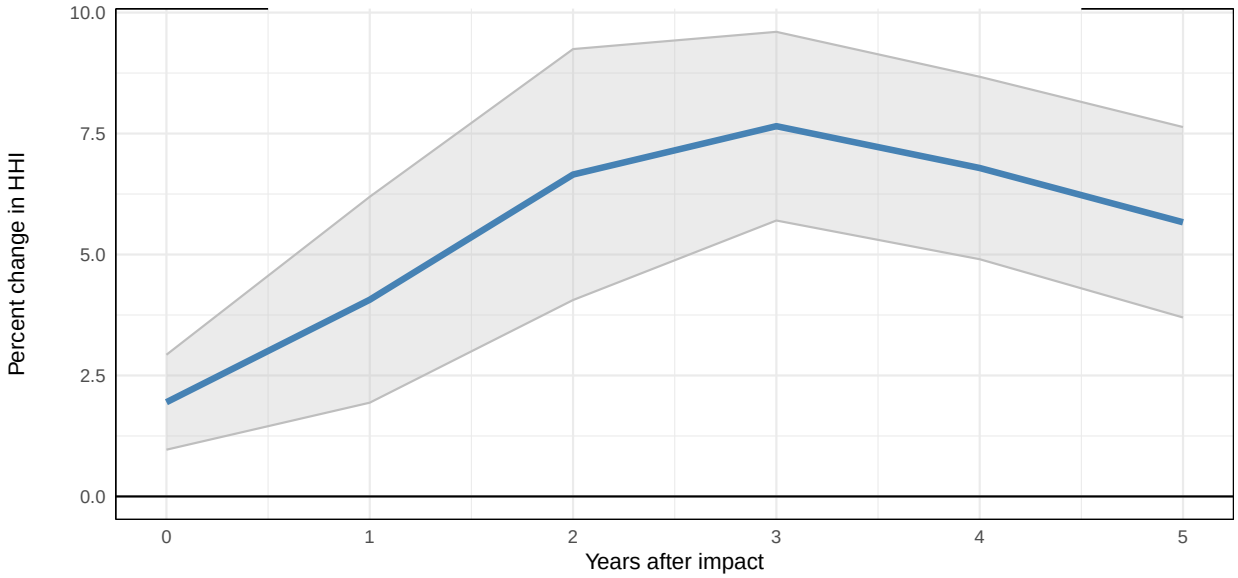
[13] I will note, and discuss in more detail in Appendix A, that basing my measure of concentration on public firms is currently a limitation of this paper that should be addressed by future research.

[14] I begin the sample in 1990 because this is when the aggregate HHI changes from a secular decline to stationary if not secular incline, see Figure 15 in Appendix A. Alternatively, Covarrubias et al. (2020) argues that the US transitions from "good" concentration to "bad" concentration in 2000, and I could use this regime shift to mark the beginning of my sample. However, I would then only have 20 years of annual data, which is too little to meaningfully estimate the IRFs of interest.

[15] The other most commonly used proxy for concentration in macroeconomics is the concentration ratio, in which one measures the percent of the total activity of interest accounted for by the n largest firms in a group. These measures are used in well known works such as De Loecker et al. (2020) and Autor et al. (2020), but are not commonly used by regulators, so I adopt the HHI as my baseline measure and will use concentration ratios in future robustness checks.

[16] A potential limitation to using the Jurado et al. (2015) macroeconomic uncertainty series is that Ludvigson et al. (2021) find it endogenously responds to fluctuations in business cycle conditions. This poses a threat to causal identification in my econometric model, as it suggests the possibility of lurking variables impacting both uncertainty and concentration. However, I ameliorate these concerns in Appendix C. I show that the response of concentration to uncertainty shocks qualitatively holds even when controlling for possibly contemporaneous business cycles shocks, namely productivity level and monetary policy shocks. Quantitative studies of endogenous uncertainty, such as Straub and Ulbricht (2023) or Bernstein et al. (2024), show that uncertainty is most likely driven by a mixture of endogenous responses to business cycle shocks and its own exogenous shocks. Therefore, by controlling for contemporaneous business cycle shocks, I isolate the variance in the uncertainty series to that driven by exogenous uncertainty shocks.

Figure 1: Response of sales concentration to an uncertainty shock

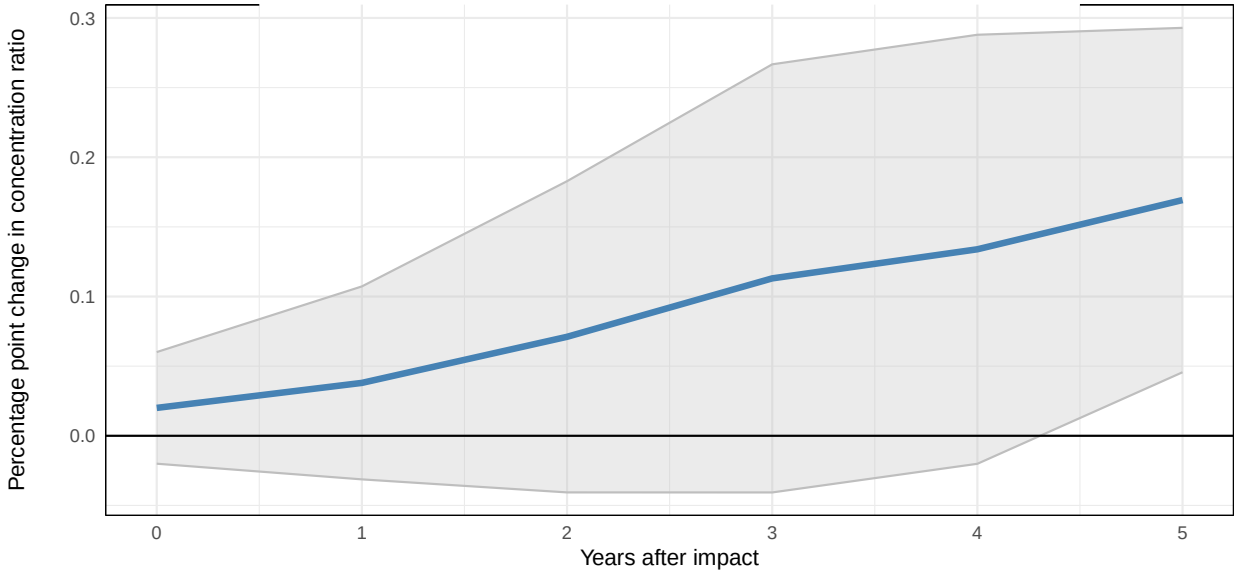


Notes: This figure shows the response of sales concentration to a one standard deviation uncertainty shock. Sales concentration is measured by the HHI of US public firms and uncertainty is proxied by the Jurado et al. (2015) (one quarter ahead) macroeconomic uncertainty series. The model includes lagged sales concentration, real GDP growth, core PCE inflation, and the one year Treasury yield as additional controls. The gray band is the Newey-West adjusted 90 percent confidence interval. Data is annual from 1990 through 2019.

volatility in financial markets which could stem from uncertainty regarding the economic fundamentals of the underlying assets (such as the productivity of firms) or uncertainty in investors future willingness and ability to participate in financial markets. While asset markets and the financial sector may be important sources of uncertainty in the economy, this project focuses on uncertainty regarding firm-level productivity shocks, thus requires a baseline measure of uncertainty more directly related to real activity. These three measures of uncertainty are presented in Figure 18 and are also discussed in further detail in Appendix A.

Fact 1: Uncertainty shocks increase industry concentration. Figure 1 shows that sales concentration among US public firms increases in both an economically and statistically significantly manner in response to a one standard deviation uncertainty shock. Concentration responds immediately on impact, rising by approximately 2 percent, and increases to being 7.65 percent greater three years after the shock. Concentration then begins to recover, but remains more than 5.6 percent elevated relative to its pre-shock level five years after impact. Extrapolating the (linear) rate of recovery between three and five years after impact, concentration will not return to its pre-shock level until approximately eight years after the shock.

Figure 2: Response of employment concentration to an uncertainty shock

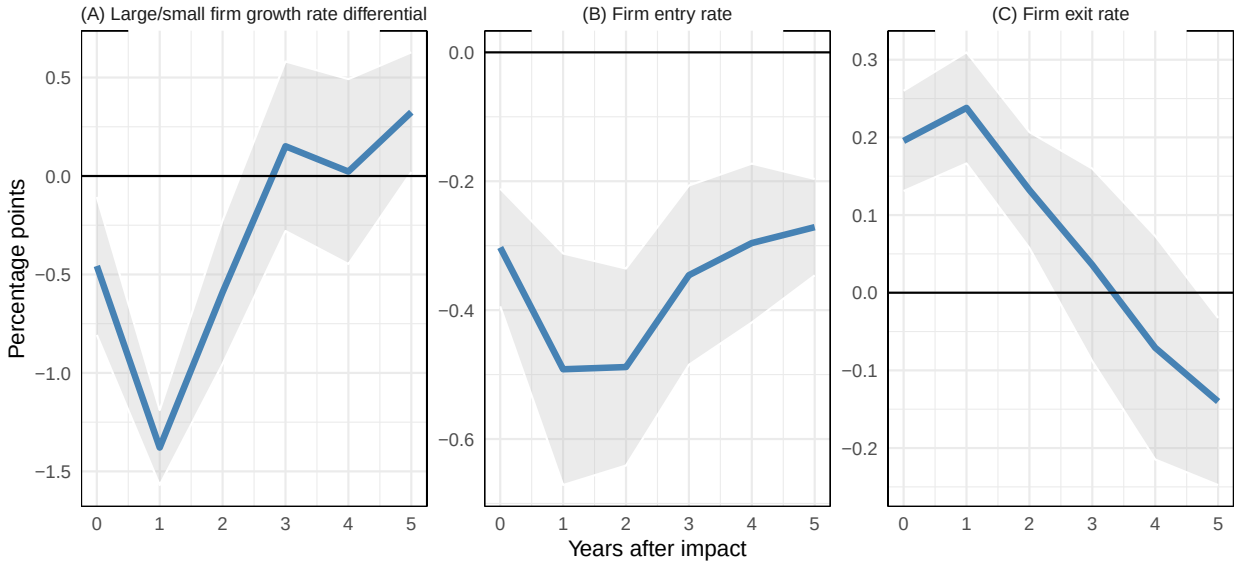


Notes: This figure shows the response of employment concentration to a one standard deviation uncertainty shock. Employment concentration is measured by the percent of employees hired at firms with 500 workers or more in the given year and uncertainty is proxied by the Jurado et al. (2015) (one quarter ahead) macroeconomic uncertainty series. The model includes lagged sales concentration, real GDP growth, core PCE inflation, and the one year Treasury yield as additional controls. The gray band is the Newey-West adjusted 90 percent confidence interval. Employment data is from the Census Business Dynamism Statistics and is representative of the universe of US firms. Data is annual from 1990 through 2019.

This finding is robust to concentration measure and firm sample. Figure 2 shows the response of employment concentration (the share of employees working in large firms, those with 500+ employees in a given year) among the universe of US firms (sourced from the Census Bureaus' Business Dynamism Statistics dataset). Employment concentration increases in response to an uncertainty shock, albeit more muted and at a slower pace than sales concentration. Further robustness checks across model specification, uncertainty measures, and data frequency are presented in Appendix C.

However, observing that concentration rises in response to an uncertainty shock does not reveal what transmission mechanisms relate the two phenomena, so I next turn to estimating the impact of uncertainty shocks on the intensive and extensive margins of concentration. That is, I estimate the impact of uncertainty shocks on firm entry and exit (the extensive margin of the firm sales distribution), as well as on the relative (employment) growth rate between large and small incumbent firms (a proxy for the intensive margin of the concentration). I will likewise estimate these responses with the model specified by Equation 1, but replacing the outcome variable of interest and its autoregressive control with the three margin measures just described. These data are also annual from 1990 through 2019, but

Figure 3: Response of the intensive and extensive margins of concentration to an uncertainty shock



Notes: This figure shows the response of the firm size distribution to a one standard deviation uncertainty shock. The difference in employment growth is measured by the percentage point difference in the average net employment growth rate between continuer establishments owned by large firms (those with 500+ employees in the given year) and small firms (those with less than 500 employees in the given year). Uncertainty is proxied by the Jurado et al. (2015) (one quarter ahead) macroeconomic uncertainty series. The model includes lagged sales concentration, real GDP growth, core PCE inflation, and the one year Treasury yield as additional controls. The gray band is the Newey-West adjusted 90 percent confidence interval. Employment and firm data is from the Census Business Dynamism Statistics and is representative of the universe of US firms. Data is annual from 1990 through 2019.

come from the Census Bureau's Business Dynamism Statistics dataset, thus reflect a representative sample of US firms.

Fact 2: Uncertainty increases concentration through its extensive margins but decreases concentration through its intensive margin. Figure 3 first shows that the difference between the average continuer establishment growth rate for large firms and small firms is at first negative, interpreted as an increase in employment growth among small firms. As smaller firms grow faster than large firms, the difference in size, thus market share, between the firms decreases. This also marks a decline in concentration. Figure 3 then shows that the firm entry rate declines by approximately 30 basis points one year after impact and then declines further to nearly a half a percentage point lower than its pre-shock level two years after impact. The firm entry rate remains depressed by at least a quarter of a percent for five years after impact. Simultaneously the firm exit rate increases by approximately 20 basis points on impact before recovering and becoming statistically indistinguishable from zero within four years. Therefore, when an uncertainty shock occurs, the number of firms in

the economy declines and is not subsequently recovered by a compensating increase in firm entry or slow down in exits. As output is controlled by fewer firms, concentration increases.

Given that uncertainty shocks increase concentration, it follows that the response of the extensive margin must dominate the intensive margin. It also follows that any theory attempting to explain the persistent effects of uncertainty shocks needs to account for the impact of uncertainty shocks on entrepreneurship and exits.

3.2 Industry-level evidence: a case study of US Manufacturing

While studying aggregate time series is informative and provides initial evidence that uncertainty shocks increase concentration, it cannot clarify whether the increase in concentration is due to a rise in inter-industry concentration, intra-industry concentration, or both. Therefore, I next turn to examining the relationship between industry-level uncertainty and concentration.

Methodology and data

This analysis proceeds in two steps. First, I estimate a set of industry-specific firm-level productivity uncertainty shocks.[17] Following Bloom et al. (2018), I postulate that productivity uncertainty follows an AR(1) process with time-varying volatility:

$$A_{f,t} = \mu_i + \rho_i A_{f,t-1} + \sigma_{i,t}\epsilon_{f,t} \tag{2}$$

$$\sigma_{i,t} = \mu_i^{\sigma} + \rho_i^{\sigma}\sigma_{i,t-1} + \nu_{i,t} \tag{3}$$

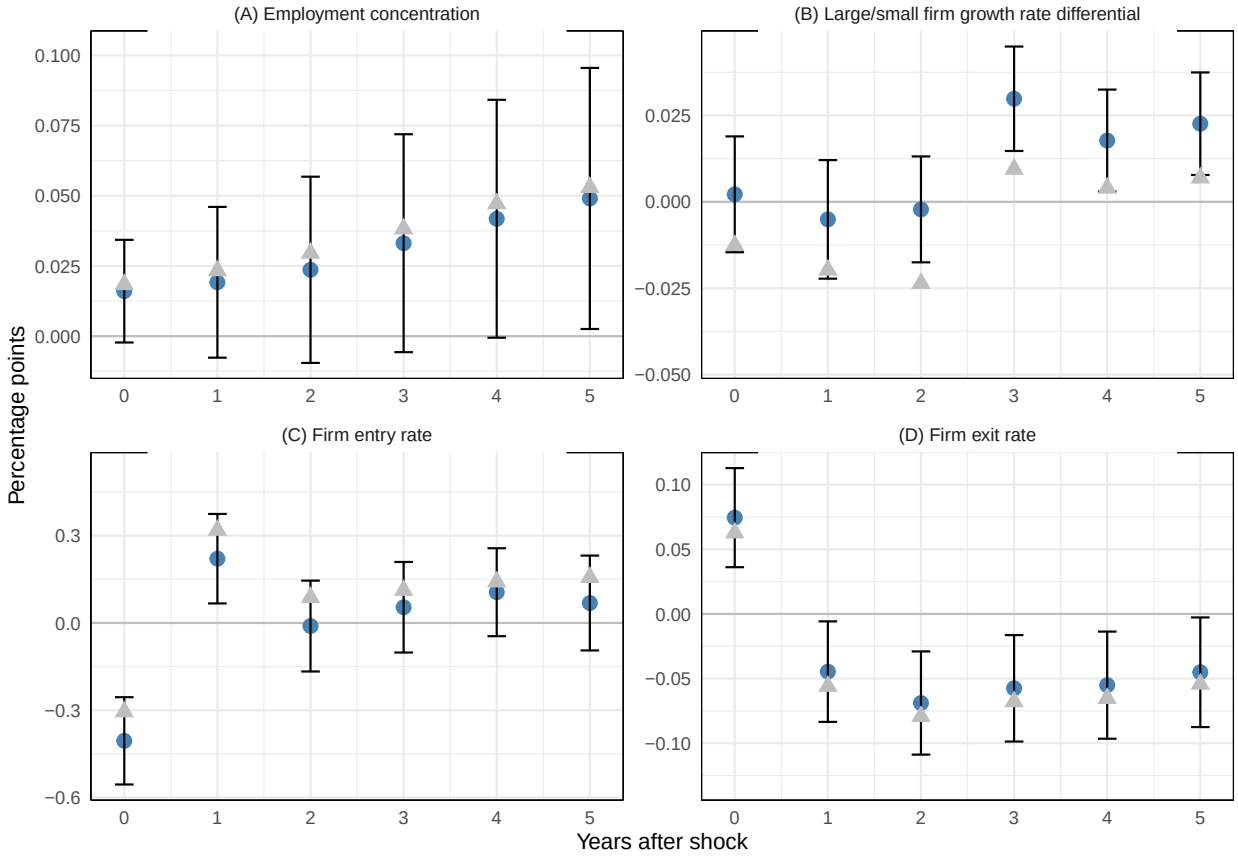
where A is firm f's total factor productivity in period t, and all firms in industry i follow the same productivity law of motion while drawing independent shocks $\epsilon_{f,t}$; $\sigma_{i,t}$ is in turn the variance of productivity shocks in industry i during period t and is naturally interpreted as the industry-level measure of productivity uncertainty, while $\nu_{i,t}$ is the industry-level uncertainty shock.

I estimate the set of uncertainty shocks, $\nu_{i,t}$, as residuals from industry-specific uncertainty AR(1) processes, where industry-level uncertainty is taken as the 90th minus 10th productivity dispersion measure reported by the Bureau of Labor Statistics' DiSP dataset, documented in further detail in Cunningham et al. (2023).[18] Notably, this dataset reports productivity

[17] This exercise has the additional benefit of focusing specifically on firm-level uncertainty shocks, the uncertainty of interest in this study. The Jurado et al. (2015) macroeconomic uncertainty measure is based on a panel of aggregate time series, which potentially obfuscates the distinction between firm-level and aggregate productivity uncertainty shocks.

[18] Note that the DiSP dataset reports the dispersion in the *level* of firm-level total factor productivity (TFP), not necessarily the dispersion in the *shocks* of firm-level TFP shocks. However, assume 1) that all firm-level productivity (within a a given industry) evolves according to the same AR(1) model, but experience iid random productivity shocks, and 2) that we can model the evolution of the 90th and 10th productivity level percentiles as representative firms that follow the same AR(1)

Figure 4: Industry-level concentration's response to an industry-level uncertainty shock



Notes: This figure shows the impact of an uncertainty shock on industry-level outcomes. Blue dots show the impact of a one standard deviation uncertainty shock on an industry outcome, controlling for lagged industry-characteristics. The gray triangles show the same pseudo-elasticity without controls. The black lines are 90 percent confidence intervals with standard errors clustered at the date-level. The sample contains 2924 observations, 86 industries at the 4-digit NAICS, spanning 1990 through 2019.

dispersion at the 4-digit NAICS level for all US manufacturing sub-industries, and is representative of the universe of US manufacturing firms. The estimated firm-level uncertainty shocks are characterized by Figure 19 in Appendix A.

Second, I use my collection of estimated industry-level uncertainty shocks to directly measure the response of concentration and its margins to uncertainty shocks in the following panel

productivity process as in Equation 2. It follows from the fact that an AR(1) can be written as an $MA(\infty)$ process that a firm representing the p-th percentile of the firm-level productivity distribution must have experienced, on average, productivity shocks equivalent to the p-th percentile of the productivity shock distribution. Writing the 90th and 10th percentile representative firms productivity as $A_{90,t}$ and $A_{10,t}$ respectively, the dispersion in TFP levels is $D_t^{90,10} = A_{90,t} - A_{10,t}$, which can be rewritten as $D_t^{90,10} = \rho[A_{90,t-1} - A_{10,t-1}] + [\epsilon_{90,t} - \epsilon_{10,t}]$. So then using $D_t^{90,10} = \sigma_{i,t}$, one can directly map $\nu_{i,t} = [\epsilon_{90,t} - \epsilon_{10,t}]$, and recalling that the p-th percentile representative firm must experience, on average, the p-th percentile productivity shock, $\nu_{i,t}$ is, on average, the difference between the 90th and 10th percentiles of the firm productivity shock distribution. Measuring uncertainty shocks as the difference between two percentiles of the empirical productivity shock distribution is a common practice in the productivity uncertainty literature, see for example Bloom et al. (2018).

local projection setting:

$$\Delta_t^h Y_{i,t} = \gamma_i^h + \gamma_t^h + \beta^h(\hat{\nu}_{i,t}) + \theta^h X_{i,t-1} + \epsilon_{i,t}^h \tag{4}$$

where $\Delta_t^h Y_{i,t}$ is the change in outcome Y between periods $t+h$ and $t-1$, γ_i^h is the industry-specific fixed effect, γ_t^h is the time fixed effect, $\hat{\nu}_{i,t}$ is the estimated industry-specific uncertainty shock, and X is the set of industry-specific lagged controls: total factor productivity, total employment, consolidated input costs, real output price, and the first lag of the outcome of interest. The regression is weighted by average industry employment share. Data is annual from 1990 through 2019 and industry-level controls are collected from the Bureau of Labor Statistics.

I will again study four outcomes of interest: concentration, the relative growth of large versus small incumbent firms, firm entry, and firm exit. However, I will now turn to using employment concentration, measured as the employment share in an narrow industry controlled by firms with at least 20 employees, rather than relying on the HHI of sales among public firms.[19] I change my measure of concentration for two reasons: first, to show this relationship of interest is robust to the way one measures concentration, and two, so I can use data reflecting a representative sample of US manufacturing firms rather than more narrowly focusing exclusively on public firms, which carries with it a host of measurement problems when studying concentration (see Appendix A for further discussion of documented measurement issues).

Results

Industry-level uncertainty shocks increase within-industry concentration and firm exits while decreasing firm entry. Figure 4 shows that the response to uncertainty shocks is consistent across both national and narrow industry levels. First, the employment concentration ratio increases by approximately 0.05 percentage points within five years after impact, is statistically significant at the 10 percent confidence level, and shows no signs of recovering to its pre-shock level. Moreover, since the regression is weighted by industry employment, this response can be interpreted as the aggregate increase in employment concentration in the US manufacturing sector in response to a narrow industry uncertainty shock. That is, a one standard deviation uncertainty shock in a narrowly defined manufacturing sub-industry leads to a 0.05 percent increase in employment concentration, on average, for the entire US manufacturing sector.

[19] It is more common to measure employment concentration as the percent of industry employment controlled by firms with at least 500 employees. However, I find that this definition yields nosier, but qualitatively similar, results compared to this definition.

Turning to the extensive margin of concentration, an industry-level uncertainty shock decreases the firm entry rate by approximately 0.48 percentage points (half a percent) on impact but then does not subsequently increase enough to make up for the initial decline. At the same time, the firm exit rate increase by 0.075 percentage points on impact and then recovers. The recovery in exits may be explained by an increase in the quality of incumbent firms. On the one hand, Ates and Saffie (2021) suggest that in response to a recessionary shock there is a decline in the number of entrants but an increase in the quality of entrants (i.e. higher productivity). On the other hand, Foster et al. (2001) empirically document that exiting firms have, on average, a significantly lower total factor productivity than incumbent or entering firms. Thus, as entrants decline and exits increase due to an uncertainty shock, there is a smaller number of firms in the affected industry, but the quality of these firms is relatively higher than prior to the shock, and in turn the exit rate declines.

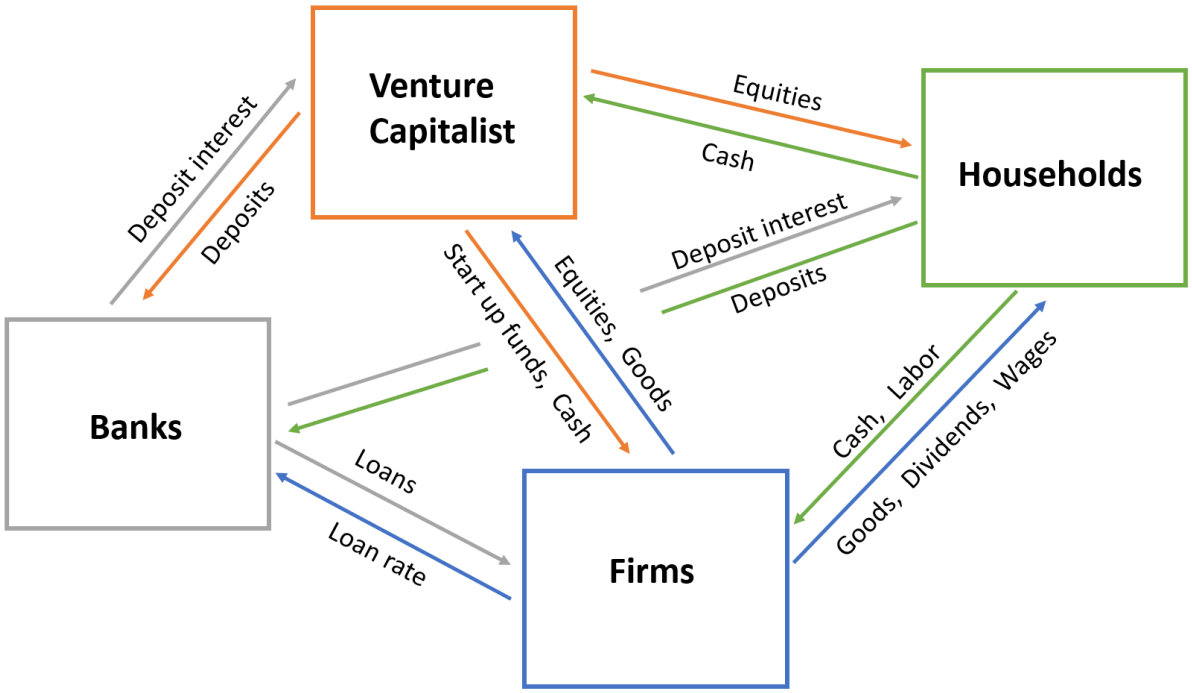
4 Model

I have empirically documented that uncertainty shocks cause a persistent rise in concentration. However, my empirical exercises can neither speak towards what transmission mechanisms explain this persistence. As a result, I next turn to writing a model which can be used in a series of quantitative experiments to analyze these two remaining questions.

The economy is made up of four types of agents: banks, firms, households, and venture capitalists. An industry of heterogeneous firms will combine capital and labor to create a homogeneous output, which in turn may be invested as capital for future production or bought by households for consumption. Firm-level productivity is heterogeneous and will evolve according an exogenous law of motion, whose innovations will vacillate between high and low volatility regimes. The state of the volatility regime will define the *level of uncertainty* in the economy, while the transition between volatility regimes will define the *uncertainty shocks*. Firms will be born small and fuel their growth via reinvesting profits or taking out loans from a bank. However, firms will be subject to exogenous exit shocks, as well as endogenous bankruptcy decisions. Simultaneously, new firms will enter the industry, funded by a venture capitalist. As a result, the level of firm entry will not necessarily equal the level of firm exits, and the overall size of the industry may vary over time. The flows in the economy are summarized as a diagram in Figure 5.

The model can alternatively be summarized as largely following the firm dynamics and uncertainty processes of Bloom et al. (2018) or Alfaro et al. (2023), with the endogenous default

Figure 5: Flows in the economy



Note: This diagram presents the flow of resources through the economy. There is no true money or "cash" in this model, but I use the term as a short hand for trade in goods denominated resources.

risk and financial accelerator mechanism of Arellano et al. (2019) or Straub and Ulbricht (2023), and an extension to allow firm entry to be determined by the investment decisions of a risk averse venture capitalist, capturing the forces motivating entrepreneurship in canonical works such as Evans and Jovanovic (1989).

I will next present each agent's problem before defining the competitive equilibrium and characterizing various market outcomes.

4.1 Firms

There is a distribution of risk neutral firms that combine durable capital and labor to produce a homogeneous consumption good that is invested as capital, sold to other firms as capital, or sold to households in a perfectly competitive goods market. Firms maximize the expected present discounted value (PDV) of their lifetime profits, which are distributed as dividends each period. Firms are subject to idiosyncratic and exogenous productivity paths, although firms can borrow resources from a financial intermediary to fund growth or smooth dividend payouts over their lifetime. Firms face exogenous exit shocks and an endogenous bankruptcy (i.e. exit) decision before producing. When firms exit their capital is destroyed.

A producing firm takes prices and the current distribution of firms, μ, as given and chooses investment, employment, borrowing, and dividend payouts according to the formal problem:

$$V^F(b_{i,t}, k_{i,t}; \epsilon_{i,t}, A_t, \sigma_t^\epsilon, \sigma_t^A, \mu_t) = \max_{b_{i,t+1},\ k_{i,t+1},\ l_{i,t},\ \pi_{i,t}} [V^A, V^N, V^E] \tag{5}$$

where V^A is the firm's value when it adjusts its capital holdings:

$$\begin{aligned} V^A(b_{i,t}, k_{i,t}; \epsilon_{i,t}, A_t, \sigma_t^\epsilon, \sigma_t^A, \mu_t) = & \\ \max_{b_{i,t+1},\ k_{i,t+1},\ l_{i,t}} & \left[\pi_{i,t} + \Xi_t(1-\lambda)\mathbb{E}V^F(b_{i,t+1}, k_{i,t+1}; \epsilon_{i,t+1}, A_t, \sigma_t^\epsilon, \sigma_t^A, \mu_{t+1})\right] \\ \text{s.t.} \quad & \pi_{i,t} = A_t \epsilon_{i,t}(l_{i,t}^\alpha k_{i,t}^{1-\alpha})^\chi - w_t l_{i,t} - \phi(i_{i,t}, k_{i,t}) - b_{i,t} + \frac{b_{i,t+1}}{R_{i,t}} \\ & k_{i,t} = (1-\delta)k_{i,t-1} + i_{i,t-1} \\ & \pi_{i,t} \geq 0 \end{aligned} \tag{6}$$

V^N is the firm's value when it does not adjust its capital holdings:

$$\begin{aligned} V^N(b_{i,t}, k_{i,t}; \epsilon_{i,t}, A_t, \sigma_t^\epsilon, \sigma_t^A, \mu_t) = & \\ \max_{b_{i,t+1},\ l_{i,t}} & \left[\pi_{i,t} + \Xi_t(1-\lambda)\mathbb{E}V^F(b_{i,t+1}, (1-\delta)k_{i,t}; \epsilon_{i,t+1}, A_t, \sigma_t^\epsilon, \sigma_t^A, \mu_{t+1})\right] \\ \text{s.t.} \quad & \pi_{i,t} = A_t \epsilon_{i,t}(l_{i,t}^\alpha k_{i,t}^{1-\alpha})^\chi - w_t l_{i,t} - b_{i,t} + \frac{b_{i,t+1}}{R_{i,t}} \\ & k_{i,t+1} = (1-\delta)k_{i,t} \\ & \pi_{i,t} \geq 0 \end{aligned} \tag{7}$$

and V^E is the firm's value when it decides to exit because it either cannot pay its debts, thus incurring a negative dividend payout (which is interpreted as requiring a capital injection to

survive), or the PDV of its lifetime dividend stream naturally drifts to be negative due to a series of poor productivity shocks:

$$V^E(b_{i,t}, k_{i,t}; \epsilon_{i,t}, A_t, \sigma_t^\epsilon, \sigma_t^A, \mu_t) = 0 \tag{8}$$

and parameters $\{\beta, \lambda, \alpha, \chi, \delta\}$ are the firm's discount factor, probability of receiving an exogenous exit shock, labor share of production, production curvature factor, and capital depreciation rate, respectively; endogenous states include loans $b_{i,t}$ and capital $k_{i,t}$, which the firm owns and accumulates with a standard law of motion via investment, $i_{i,t}$, subject to an adjustment cost $\phi(i,k)$; the common productivity-level, A_t, and firm-level productivity state, $\epsilon_{i,t}$ are exogenous and evolve through time; the idiosyncratic productivity component evolve with time varying volatility, σ_t^ϵ; the choice variable, labor, $l_{i,t}$, is hired at wage rate w_t. Prices include the wage rate, w_t, which clears the labor market, the firm-specific loan rate, $R_{i,t}$, household's stochastic discount factor, Ξ_t, and the price of the output good, which is the numeraire. The value functions V^N and V^A are zero and the firm must exit if no available set of choices satisfy the non-negative dividend constraint.

Distribution of firms. There is a joint distribution of firms across capital, debt, and productivity levels, μ, which is defined over the space $\Delta = \mathbb{R}^+ \times \mathbb{R}^+ \times \mathbb{R}^+$. The distribution evolves according to the law of motion:

$$\mu_{t+1} = \Gamma(A_t, \sigma_t^A, \sigma_t^\epsilon, \mu_t) \tag{9}$$

In an abuse of terminology, the joint distribution of firms does not have to equal to one. Instead, the "distribution" can be thought of as tracking the population mass of firms in a given state, and the sum of all these state-specific masses is the size of the industry in period t. I use this notation for two reasons, 1) it closely follows the notation of previous works in this literature and makes comparison with these papers easier, and 2) it emphasizes that firms are atomistic and cannot individually influence aggregate outcomes, but the total number of firms can vary.

Technology. Firms produce output by combining labor and capital with a Cobb-Douglas technology and overall decreasing returns to scale, set by curvature $0 < \chi < 1$. Firms face an exogenous productivity path, determined by idiosyncratic, $\epsilon_{i,t}$, and common, A_t, components that follow independent AR(1) processes:

$$\epsilon_{i,t} = (1 - \rho_\epsilon)\mu_\epsilon + \rho_\epsilon \epsilon_{i,t-1} + \sigma_t^\epsilon e_{i,t} \quad e_{i,t} \sim \mathcal{N}(0,1) \tag{10}$$

$$A_t = (1-\rho_A)\mu_A + \rho_A A_{t-1} + \sigma^A e_t \quad e_t \sim \mathcal{N}(0,1) \tag{11}$$

where μ_x is the long run average, ρ_x the persistence, and σ_t^x the volatility of the $x \in \{\epsilon, A\}$ productivity component. Following Bloom et al. (2018), Arellano et al. (2019), Alfaro et al. (2023), I assume that evolution of the firm-level conditional volatility, σ_t^ϵ, is time-varying and follows a two-state Markov process, governed by transition probabilities

$$\sigma_t^\epsilon \in \{\sigma^L, \sigma^H\}, \quad \text{where} \quad Pr(\sigma_t^H | \sigma_{t-1}^H) = \iota^H, \quad Pr(\sigma_t^L | \sigma_{t-1}^L) = \iota^L \tag{12}$$

where $0 < \sigma^L < \sigma^H$.[20]

Capital accumulation. The firm owns capital and accumulates it through investments subject to both convex and fixed costs. The functional form of the capital adjustment cost is motivated by Cooper and Haltiwanger (2006), and is a combination of three terms: the cost of new capital, an increasing and convex cost, and a fixed cost of adjustment. The presence of the fixed cost results in the following piece-wise specification:

$$\phi(i,k) = \begin{cases} i + \frac{\zeta}{2} i^2 + \gamma k & i \neq 0 \\ 0 & i = 0 \end{cases} \tag{13}$$

where ζ is the quadratic cost of capital and γ is the (scalable) fixed cost of capital. I include the fixed cost, despite the resulting increase in the computational complexity of the firms problem, because it introduces the "real option" channel of productivity uncertainty shocks — the channel operative in works such as Bernanke (1983), Bloom (2009), and Bloom et al. (2018).

The capital adjustment costs have a second a interpretation. In the presence of endogenous bankruptcies but only a numeraire capital good, then the convex adjustment cost reflects a fire sale discount on the price of the capital liquidated to avoid bankruptcy. An extreme example of such a discount is the Fisherian debt deflation dynamics characterizing the Sudden Stop events described in Mendoza (2010) or Bianchi (2011). Additionally, the non-convex adjustment cost may reflect the differential inside versus outside value of a firm's capital or a recovery cost of capital for lenders.

Firm exit. Incumbent firms can choose to exit, as well as face an exogenous exit shock. First, firms may choose to default on their loans if their possible earnings and liquidated capital cannot cover operating and borrowing costs. The possibility of insolvency arises from

[20] While I follow the more computationally tractable strategy of Arellano et al. (2019) and Bloom et al. (2018) by focusing on a single source of time-varying productivity uncertainty, one can trivially extend the process to include aggregate uncertainty. However, Bloom et al. (2018) present evidence that aggregate and firm-level uncertainty are highly correlated and can be treated as the same process in a quantitative setting, so the marginal benefits of including the additional series is limited.

the fact that firm take out non-state-contingent loans in period t, but do not learn their productivity, thus revenue and ability to repay debt, until period $t+1$. This default risk is priced into a firm's loan rate schedule offered by the bank, and is described in more detail later. Second, all firms are subject a probability, λ, of being forced to exit each period. If a firm exits then its capital is destroyed. The presence of exogenous exits is helpful in creating a non-degenerate distribution of firm sizes in the economy (see, for example, Hopenhayn (1992)), which is necessary for a meaningful examination of sales concentration. Meanwhile, the presence of endogenous firm exits are key in motivating the asymmetric response in loan rates to a financial uncertainty shock, which is potential channel through which concentration and uncertainty may be related.

Firm entry. A mass of new firms enter the industry at the end of each period. There is an unlimited number of ex ante identical potential entrants prepared to enter the production market. However, new firms must pay a fixed cost of entering, c^e, and this start up cost must be funded by a Venture Capitalist. As a result, the number of new entrants evolves according to:

$$E_t = \frac{F_t}{c^e} \tag{14}$$

where F_t is the Venture Capitalist's allocation of wealth to entrepreneurial activities in period t. Although this is a departure from the more standard Free Entry condition or matched entry/exit conditions that typically dictate firm entry in macroeconomic models, I will discuss in Section 4.2 that this model more closely follows the entrepreneurial process in canonical works such as Evans and Jovanovic (1989). One may also see Quadrini (2009) for review of entrepreneurship in macroeconomics.

All new firms are are endowed with the same level of initial capital, k_0, are born debt free, and draw an initial idiosyncratic productivity value from the distribution described in Equation 10. Since firms are born at the end of period t, they will transition to being an incumbent firm at the beginning of period $t+1$, and be given the opportunity to exit or produce like any other firm. I outline the full timeline of a period in Section 4.5.

4.2 Venture Capitalists

Venture Capitalists, who can be thought of as the investors financing prospective entrepreneurs, are born with wealth, are risk averse, and earn income each period by providing deposits to a bank for a risk-free deposit rate or investing in startups for a risky return.

Each period, the VC apportions a percentage of its financial portfolio to funding the start

up costs for new firms. In exchange for funding the firm, the VC owns the entrant. However, instead of managing the firm, the VC will instead sell its portfolio of new business to Households via the stock market. As such, the VC stands to earn a net return of the firm's equity value less start up costs. The realization of entrepreneurial returns will be described further in Section 4.7. In the meantime, note that the VC will take the distribution of risky returns on startups as given.

The VC is therefore solving a standard a dynamic portfolio allocation problem.[21] The VC's formal problem is written:

$$V^C(W_t) = \max_{\upsilon_t} U(W_t) + \beta \mathbb{E} V^C(W_{t+1}) \quad \text{s.t.} \tag{15}$$

$$W_{t+1} = R^*_{t+1} W_t, \quad \text{given } W_0$$

$$R^*_{t+1} = \upsilon_t R^e_{t+1} + (1 - \upsilon_t) R^d_{t+1}$$

where W_t is the VC's financial wealth at time t, υ_t is their portfolio allocation in the risky asset, R^* is the realized return on the VC's financial portfolio, R^e is the realized risky return from funding startups, R^d is the risk free return bank deposits. For concreteness, assume the VC has standard isoelastic utility, so that $U(W) = \frac{W^{1-\sigma}}{1-\sigma}$, and there is a restriction on short-selling and leverage so that $0 \leq \upsilon \leq 1$. The VC's problem is then standard and subject to several textbook treatments, so I leave a discussion of the solution to Appendix D.1.

By an accounting identity, the Venture Capitalists' gross financial investment flows can be written as:

$$D^C_t = (1 - \upsilon_t) W_t, \quad F_t = \upsilon_t W_t \tag{16}$$

where D^C_t is the deposits placed with the bank and F_t is the gross amount of start up funds made available.

I model the VC in this fashion because it is a tractable way to introduce the two factors that determine entrepreneurship in standard macro-models of the phenomenon, as well as allow a channel through which uncertainty may impact firm entry.[22] First, the VC's willingness to

[21] The VC is therefore a standard risk averse investor, maximizing their risk adjusted return each period, similar to the static problem examined by Markowitz (1959). One can augment the VC's problem with consumption, which would transform this into the case studied by Samuelson (1969) (or Merton (1969) in continuous time). However, these same works also show that the consumption-saving and portfolio allocation problems are completely separable. Additionally, I have found that including two agents in the model that consume firm output (in this example, the VC and the Household) makes computationally solving the general equilibrium much more difficult. Therefore, since I can abstract from the VCs consumption problem while preserving the nuances of its investment decisions, and at the same time make the computational exercises much more tractable, I do so.

[22] Similarly, Ates and Saffie (2021) write a model in which entry is mediated by a financial intermediary's willingness to invest in startups. However, unlike in their setting, the VC at the heart of this economy cannot sort between high- and low-quality (i.e. productivity) firms ex ante. Additionally, Bilbiie et al. (2012) present a model in which entry is determined by a

invest in new firms is directly related to its expected return from funding these new firms. Consequentially, the level of firm entry will be a function of the expected value of firms. Second, the VC's current wealth will determine the gross level of funds available to invest in startups. This second factor captures the role of financial constraints in determining the level of entrepreneurial activity in the economy, emphasized by classic works such as Evans and Jovanovic (1989) or Quadrini (2000), as well as recent empirical work such as Aruoba et al. (2022) and Davis and Haltiwanger (2023). Lastly, as the expected variance of the risky asset return increases, the VC, a risk averse investor, will decrease its allocation of funds to the risky asset, lowering firm entry. I will later discuss that if firm value is positively correlated with productivity, then productivity uncertainty shocks will increase the expected variance of entrepreneurial returns, opening a potential channel through which uncertainty may impact firm entry, thus industry size, and by extension, market concentration.

Moreover, the empirical exercises presented in Section 3 motivate the inclusion of the entrepreneurial funding channel of uncertainty shocks. Uncertainty is uncorrelated, if not positively correlated, with the intensive margin of concentration. That is, an increase in uncertainty is (un)correlated with small firm sales growth actually increasing relative to large firms for at least a year after impact, which would decrease sales concentration in the short run. However, Figure 1 shows that an uncertainty shock is correlated with an immediate and persistent increase in concentration. Therefore, to match the data, the model needs a channel through which uncertainty may increase concentration, on impact, other than through making small incumbent firms relatively smaller. Uncertainty decreasing the number of firms in the economy by decreasing the firm entry rate is one such channel. Moreover, Figure 3 shows that uncertainty is correlated with an immediate decline in firm entry, thus there is empirical evidence supporting the inclusion of this channel in the model.

4.3 Banks

A risk neutral Bank sources deposits from the Venture Capitalist and Households to extend risky loans to incumbent firms. Banks additionally have access to a storage technology that pays a risk free rate equal to the deposit rate. Formally, the Bank solves the problem:

$$V^B(N_t) = \max_{B_t, G_t} E_t \Pi_t^B(B_t, G_t) + \mathbb{E}_t \beta V^B(N_{t+1}) \quad \text{s.t.} \tag{17}$$

combination of Free Entry and a risk averse household's attitude towards investment, although these authors explicitly abstract from uncertainty in determining the entry rate.

$$\Pi_t^B = R_t^\diamond B_t + R^d G_t - R^d D_t$$
$$N_t = \Pi_{t-1}^B N_{t-1}$$
$$B_t + G_t \leq N_t + D_t$$

where B_t is the mass of risky loans, G_t is the mass of capital in storage, N_t is the Bank's equity, D_t is the mass of deposits sourced from the VC and Household, R^d is the gross deposit rate banks must pay Households, and $R^\diamond$ is the realized gross loan portfolio return. The Bank can only issue one-period non-state-contingent loans.[23] If a firm defaults on its loans then the Bank loses the entire principal of the loan.[24] I will assume that the Bank is sufficiently capitalized (i.e. has a large enough N_t) such that there is no risk of the Bank failing in its ability to pay back the deposits costs in the beginning of the next period and the funding constraint does not bind.

4.4 Households

The Household represents a large mass of identical households that maximize their expected discounted lifetime utility by supplying labor to firms for a wage, saving their wealth in bank deposits and firm equities, and consuming goods bought from firms in a perfectly competitive market.

The Household's recursive problem is:

$$V^H(S_{t-1}, D_{t-1}^H) = \max_{C_t^H, L_t, S_t, D_t^H} [U(C_t) + \beta \mathbb{E} V^H(S_t, D_t^H)], \quad \text{s.t.} \tag{18}$$
$$C_t + (S_t - S_{t-1}) + D_t^H \leq w_t + \int_{i \in \Delta} \pi_i s_{i,t} \mu_{i,t} \, di + R^d D_{t-1}^h$$
$$L_t \leq 1 \quad \forall t$$
$$S_t = \int_{i \in \Delta} q_{i,t} s_{i,t} \, di$$

where C is consumption, L is labor, D^H is the level savings held as deposits, S is the total value of the stock portfolio, $s_{i,t}$ is the share of firm i's equity owned by the Household, and $q_{i,t}$ is the price of firm i's equity. While any increasing and concave utility function is permissible, I will later assume log utility for concreteness. The budget constraint —the first optimizing constraint listed— states that the Household cannot consume and invest more

[23] The inability of agents to author state-contingent debt contracts is the fundamental source of financial frictions in this model.

[24] Firms will default if they cannot pay the loan's principal and interest. Firms do not explicitly post collateral, but they can liquidate capital to pay their loans if needed and this is taken into account when the Bank calculates a firm's probability of defaulting, thus it is reflected in their loan rate. See Section 4.7 for further discussion of firm-specific loan rates.

than the current period wage, deposit interest, and dividend income. Note that the integral over firm-level dividend payouts in the budget constraint is interpreted as the integral over the firm-level state space, defined over space Δ and indexed by i, where π_i is the dividend payout of a firm in state i, $s_{i,t}$ is the Household's percent ownership of the mass of firms in state i at time t, and $\mu_{i,t}$ is the mass of firms in state i at time t. This notation convention will be commonly used to describe aggregates of firm-level outcomes moving forward in the exposition.

4.5 Timeline of a period

The timeline of a single period is as follows:

- Productivity and uncertainty shocks are realized and known by all agents
- The Household makes its consumption decision and purchases equities from the Venture Capitalist
- The Bank pays interest on deposits
- The Venture Capitalist realizes the return on its previous periods investments and updates portfolio allocations for the current period
- Some firms exit
- The remaining firms produce and sell goods, invest in capital, pay off previous loans, borrow from the bank, and pay dividends to the Household
- New firms enter based on funding provided by Venture Capitalist

4.6 Competitive Equilibrium

The recursive competitive equilibrium is defined by aggregate quantity functions $\{D^C, F, C, L, S, D^H, B, G\}$, firm-level quantity functions $\{k_i, b_i, l_i, \pi_i\}$, value functions $\{V^C, V^H, V^B, V^F\}$, and price functions $\{w, q_i, R_i, R^d\}$, such that for a given joint distribution of firms, μ, and set of uncertainty and productivity states $\{\sigma^A, \sigma^\epsilon, A, \epsilon_i\}$,

- $\{F, D^C\}$ solve the Venture Capitalist's problem, Equation 15
- $\{C, S, D^H, L\}$ solve the Household's problem, Equation 18
- $\{B, G\}$ solve the Bank's problem, Equation 17
- $\{k_i, b_i, l_i, \pi_i\}$ solve firm i's problem, Equation 7

and market clearing conditions hold. The Labor market clears if

$$1 = L_t = \int_{i \in \Delta} l_{i,t} \mu_{i,t} \, di$$

the goods market clears if

$$C_t = \int_{i \in \Delta} [A_t \epsilon_{i,t} (k_{i,t}^{(1-\alpha)} l_{i,t}^{\alpha})^{\chi} - \phi(i_{i,t}, k_{i,t})] \mu_{i,t} \, di$$

the loan market clears if

$$B_t = \int_{i \in \Delta} b_{i,t} \mu_{i,t} \, di$$

and the equity market clears if

$$S_t = \int_{i \in \Delta} q_{i,t} \mu_{i,t} \, di$$

while the aggregate capital stock and dividends are simply the sum of the corresponding firm-level allocations. Aggregate bank deposits are the sum of VC and Household deposits. The joint distribution of firms evolves according to the time-consistent law of motion 9.

4.7 Characterizing prices and other market outcomes

Households determine the supply of labor and wages. The Household will inelastically supply one unit of labor each period, $L = 1$. Given the fixed supply of labor the market clearing condition and prevailing wage rate can be determined by the condition:

$$1 = \int_{i \in \Delta} l_{i,t} \mu_{i,t} \, di \tag{19}$$

where the firm-level labor demand is standard in this frictionless labor-market: $l_i = l(k_i, b_i; \epsilon_i) = A_t \frac{\epsilon_i \alpha \chi}{w_t} k_i^{(1-\alpha)\chi}$. Plugging the labor demand into the market clearing condition, we can find the market clearing wage:

$$w_t = \int_{i \in \Delta} A_t (\epsilon_i \alpha \chi K_i^{(1-\alpha)\chi}) \mu_{i,t} \, di \tag{20}$$

which is a function of the productivity-weighted aggregate firm capital stock.

Households also determine a firm's stock price. The Household represents a large mass of identical households and owns all firm equity in general equilibrium, thus a firm's stock will be priced equal to the expected present discounted value of its lifetime dividends, discounted by the Household's stochastic discount factor. There is no expected capital gains, as any price above the total discounted lifetime dividends would yield a negative payoff and violate

a household's implicit participation constraint, and any price below the expected lifetime dividend payout would be arbitraged away by the large mass of households.[25] Therefore, the (cum dividend) price of a firm's stock is given as:

$$q_{i,t} = \mathbb{E}_t \sum_{s=0}^{\infty} (1-\lambda)^s \ \Xi_{t+s} \ \pi_{i,t+s} = V_{i,t}^F \tag{21}$$

where Ξ_{t+s} is the Household's stochastic discount factor between period t and $t+s$. The Household's stochastic discount factor can be easily solved and shown to be standard, such that $\Xi_{t+1} = \beta \mathbb{E}_t \frac{U'(C_{t+1})}{U'(C_t)}$. Also note that the last equivalence of Equation 21 emerges as one recognizes that the middle term is simply the series representation of the solution to the firm's recursive problem described by Equation 7.

Loans and deposits are priced in perfectly competitive credit markets. Therefore, each firm is subject to its own idiosyncratic price schedule dependent on its probability of default:

$$R_{i,t} = \frac{1}{\beta(1-\lambda-\theta_{i,t+1})} \tag{22}$$

where $\theta_{i,t+1}$ is the firm-specific probability of endogenous bankruptcy in period $t+1$, λ is the exogenous probability of exit. If $(1-\lambda-\theta_{i,t+1}) \leq 0$ then the bank will not lend to the firm. The storage technology will pay the risk-free rate $R^d = \frac{1}{\beta}$. Therefore the Bank is indifferent between lending to firms or utilizing the storage technology. The Bank likewise offers a deposit rate as if priced in a perfectly competitive bank funding market, thus pay the risk-free rate $R^d = \frac{1}{\beta}$.

The Venture Capitalist's **gross return on entrepreneurial funding is the equity value of startups divided by start up costs**. Therefore, the Venture Capitalist's gross return on entrepreneurial funding is determined by the realization of aggregate and firm-level states in period $t+1$ as follows:

$$R_{t+1}^e = \frac{1}{F_t} \int_0^{\infty} q_{j,t+1} \ \mu_{t+1}(k_0, 0, \epsilon_j) \ dj \tag{23}$$

$$= \frac{1}{F_t} \int_0^{\infty} V_{t+1}^F(k_0, 0, \epsilon_j, A_{t+1}, \sigma_{t+1}^A, \sigma_{t+1}^{\epsilon}, \mu_{t+1}) \ \mu_{t+1}(k_0, 0, \epsilon_j) \ dj \tag{24}$$

where k_0 is the initial capital endowment of starting firms, ϵ_j is the initial firm-level productivity draw, and $\mu(k_0, 0, \epsilon_j)$ is the mass of entrants with the initial productivity draw

[25] The combination of infinitely lived households and arbitrage free equity markets also implies that there is no possibility for rational asset bubbles to emerge and influence stock market pricing in this economy, Tirole (1982), unlike an the analogous overlapping generations model, Tirole (1985).

ϵ_j.[26] The second formulation of the entrepreneurial return process highlights the connection between firm value and financial returns. Moreover, it makes clear that if firm values are positively correlated with productivity, then heightened productivity uncertainty will increase the variance of expected entrepreneurial returns, and there exists a potential channel through which productivity uncertainty may impact Venture Capitalist's investment decisions.[27]

4.8 Uncertainty transmission channels

Before moving on to study how the economy responds to an uncertainty shock, I will first highlight the four possible transmission channels through which uncertainty may influence economic outcomes in this model. First, the the OHA effect, which describes the fact that an increase in the second moment of productivity will make firms with above average firm-level productivity even more productive, and firm below average even less productive. This effect has been studied in its eponymous works, Oi (1961), Hartman (1972), and Abel (1983), and occurs in this model due to the presence of firm-level uncertainty shocks. Note that in the language of Fernández-Villaverde and Guerrón-Quintana (2020), this model actually only has a "partial OHA" effect because the fixed adjustment cost of capital dampens firms ability to endogenously expand or contract to their frictionlessly optimal size in response to uncertainty shocks.

Second, the real-option channel introduced by the fixed capital adjustment cost, which produces a wait-and-see effect that delays capital investment. This is the channel examined in well known works such as Bernanke (1983) and Bloom (2009). Although I will note that many models in this tradition, such as Bloom et al. (2018), and Alfaro et al. (2023), impose fixed adjustment costs on both capital and labor, while I only include a fixed cost of capital to allow for a partial OHA effect, as previously noted.

Third, the financial accelerator channel of uncertainty shocks, due to the presence of endogenous defaults. This effect has been well explored in works such as Christiano et al. (2014) and Arellano et al. (2019). As productivity uncertainty increases, then the probability of a firm receiving a worse productivity shock increases, causing the probability of default to increase for a given level of debt, thus a firm's borrowing cost commensurately increases. The increase in borrowing cost decreases borrowing, ceteris paribus, which slows growth among high productivity, small, young firms in particular, leading to a slow down in growth and the deteriorating economic conditions further increasing the probability of default, leading

[26] Note that it is easy to confirm that firms will not remain in the state $(k_0, 0, \epsilon_j)$ after being born, so the mass firms in this state is equal to the mass of entrants in a given period.

[27] Firm values are positively correlated with firm productivity in heterogeneous firm models following the Hopenhayn (1992) tradition, a conclusion that Hopenhayn proves in his original work (Proposition 3). However I do not formally prove this conclusion in the current setting.

to the classic financial accelerator amplification and persistence effects.

Fourth, the entrepreneurial funding channel of uncertainty introduced in this work. Uncertainty shocks will increase the variance of expected entrepreneurial returns (by increasing the expected variance of startup firm values), thus decrease the risk averse Venture Capitalist's willingness to fund starts ups, and in turn decrease the overall rate of firm entry in the economy. Appendix D presents a numerical example of how uncertainty shocks may impact the Venture Capitalist's portfolio allocation problem.

5 Quantitative analysis

I next turn to examining the persistent effects of uncertainty shocks and concentration through a series of quantitative exercises, then conclude by discussing its possible implications for the impact of cyclical shocks and secular trends.

5.1 Numerical methods and calibration

I first describe the model solution method, parameter calibration, and quantitative exercises utilized to study the relationship between uncertainty and concentration.

Uncertainty shocks and generalized IRFs. To quantitatively assess the impact of uncertainty shocks in this economy, I estimate a series of standard general impulse response functions, as in Bloom et al. (2018) or Alfaro et al. (2023).[28] In this approach, I simulate a large number of economies, each characterized by an independent path of productivity level and volatility regime switching shocks. I then treat a subset of the economies with an uncertainty shock —a transition to or remaining in the high level of firm-level productivity uncertainty— after a specified burn in period. The complement of the treated set of economies is then the control group, and the difference in average outcomes between the two groups of economies is the impulse response due to an uncertainty shock. It is important to note that in the periods after the uncertainty shock, economies continue to experience shocks drawn from the productivity processes, but these shocks are now identical across economies.

Calibration. However, I have to first fix or calibrate several key parameters and processes. To do so, I divide parameters into three groups: standard values set in line with the extant literature, parameters dictating the exogenous productivity processes, and parameters governing firm entry and exit rates. For the first group, I primarily assign parameters according to their corresponding values in the baseline model of Khan and Thomas (2013) (which is

[28] See Koop et al. (1996) for a more technical discussion on generalized impulse response functions.

Table 1: Fixed Parameters

Parameter	Value	Name	Source
σ	2.00	intertemporal elasticity of substitution	Arellano et al. (2019)
β	0.96	discount rate	Khan and Thomas (2013)
α	0.60	labor share of income	Khan and Thomas (2013)
χ	0.87	production size-scale parameter	Khan and Thomas (2013)
δ	0.065	capital depreciation rate	Khan and Thomas (2013)
ϕ	2.00	quadratic investment cost multiplier	Cooper and Haltiwanger (2006)
γ	0.01	fixed adjustment cost	Cooper and Haltiwanger (2006)

also largely followed by Bloom et al. (2018)). Although I refer to Cooper and Haltiwanger (2006) when setting capital adjustment costs. The full set of parameters are presented in Table 1. For the second group, I directly estimate the productivity law of motion Equations (10 - 12) with econometric methods and an empirical measure of firm-level TFP shocks. Lastly, I calibrate the remaining parameters, namely the firm entry cost and probability of an exogenous exit shock, with a Simulated Method of Moments (SMM) approach. The model is calibrated to an annual frequency. I next discuss the results of the calibration and leave a detailed presentation of the data and methods to Appendix F.

Table 2 reports the full set of estimated productivity processes parameters. Despite being estimated independent of other parameters, without regard for model-implied moments, and based on public firm data, the estimated exogenous productivity uncertainty process is strikingly similar to other estimates in the literature. For example, Arellano et al. (2019) calibrate a similar firm-level productivity uncertainty series using an SMM approach and find $\sigma_H^\epsilon/\sigma_L^\epsilon = 1.33$ with transition probabilities $\iota^H = 0.84$ and $\iota^L = 0.94$, where in comparison I find $\sigma_H^\epsilon/\sigma_L^\epsilon = 1.32$ with transition probabilities $\iota^H = 0.82$ and $\iota^L = 0.94$. In a less direct comparison, Bloom et al. (2018), similarly using SMM but bifurcating the uncertainty processes into an aggregate and firm-level component, find for the firm-level uncertainty process $\sigma_H^\epsilon/\sigma_L^\epsilon = 4.1$ and the aggregate process $\sigma_H^A/\sigma_L^A = 1.6$ with common transition probabilities $\iota^H = 0.94$ and $\iota^L = 0.974$.

However, the estimated productivity parameters deserve to be taken with caution. The panel of firm-level TFP used to estimate the productivity process are measured using only public firm data from Compustat and a revenue/cost based productivity measure (which I will refer to as TFPR). As noted in Bloom et al. (2018), which uses a similar approach in measuring uncertainty shocks, TFPR will reflect both productivity and demand shocks. Therefore my productivity process may alternatively be described as a more generic firm fundamentals process. In an alternative interpretation, Hsieh and Klenow (2009) use TFPR as a measure of productivity distortions, arguing that the measure does not necessarily reflect changes

Table 2: Estimated Productivity Processes Parameter Values

Parameter	Value	Name
ρ_ϵ	0.783	idiosyncratic productivity shock persistence
σ_L^ϵ	6.6	volatility of idiosyncratic productivity shocks (%)
$\sigma_H^\epsilon/\sigma_L^\epsilon$	1.32	ratio of idio. prod. volatility in high versus low uncertainty states
ι^H	0.8272	probability of staying in high uncertainty state
ι^L	0.9445	probability of staying in low uncertainty state
ρ_A	0.477	aggregate productivity shock persistence
σ_L^A	2.2	volatility of aggregate productivity shocks (%)

in technological efficiency at all. Future work should ameliorate concerns arising from this proxy for TFP by using control-function productivity measurement methods, which under certain technical assumptions can be shown to be proportional to a firm's fundamental productivity rather than distortions. The interested reader should consult Foster et al. (2016) for a more comprehensive discussion of methods for measuring firm-level productivity and their relative pros and cons.

Table 3 reports the estimated firm entry and exit parameters calibrated via SMM.[29] In this procedure I targeted five macro-moments based on US data: the average firm entry rate, average firm exit rate, average output growth, standard deviation of output growth, and the median firm-level leverage. Where the first two are meant to align the model with US entrepreneurial and bankruptcy dynamics, the second two proxy for the US business cycle, and the fifth targets the average level of financial vulnerabilities which in turn may inform both the level of bankruptcies and volatility of business cycle fluctuations. However, because I target more moments than parameters being calibrating, the system will be over-identified, and the targets will not necessarily be matched exactly.

Four of five empirical moments are well matched by the quantitative model. The poorly matched moment is the average output growth, which for the US economy is positive, but for the stationary quantitative model is zero. However, the standard deviation of output growth in the quantitative model —a useful summary statistic for the presence and severity of business cycle fluctuations— is close to its empirical analog, at approximately three and two percent respectively.

[29] **Method details.** The estimated parameter vector was the loss minimizing candidate parameter vector of 1000 random samplings. Quantitative moments are calculated based on four hundred years of simulated data after a 100 year burn in period. The loss function is the standard GMM moment: squared moment residuals weighted by the inverse moment matrix so that the method is interpreted as minimizing the average percent deviation from the targeted moments. **Data details.** Firm entry and exit data are collected from the Census Bureau's Business Dynamism Statistics dataset, which (almost) covers the universe of US firms. Output growth is measured as the annual percent change in real GDP. Firm leverage is measured as the median total liabilities to total (book) asset ratio among public US firms. Data is annual from 1980 through 2019. See Appendix F for more details on both method and data.

Table 3: Calibrated Parameters and Targeted Moments

(A) Parameter values

Parameter	Value	Name
λ	0.089	exogenous exit rate
c^e	0.010	fixed entry cost

(B) Targeted moments

Target	Data	Model
Average entry rate	0.098	0.091
Average exit rate	0.089	0.089
Average output growth	0.027	0.000
Std. dev. of output growth	0.020	0.031
Median firm leverage ratio	1.123	1.089

Solution method. The model is solved with non-linear global methods and a bounded rationality algorithm motivated by Krusell and Smith (1998), Khan and Thomas (2008), and Bloom et al. (2018). Implementation details are discussed in Appendix G.

5.2 Industry's persistent response to uncertainty shocks

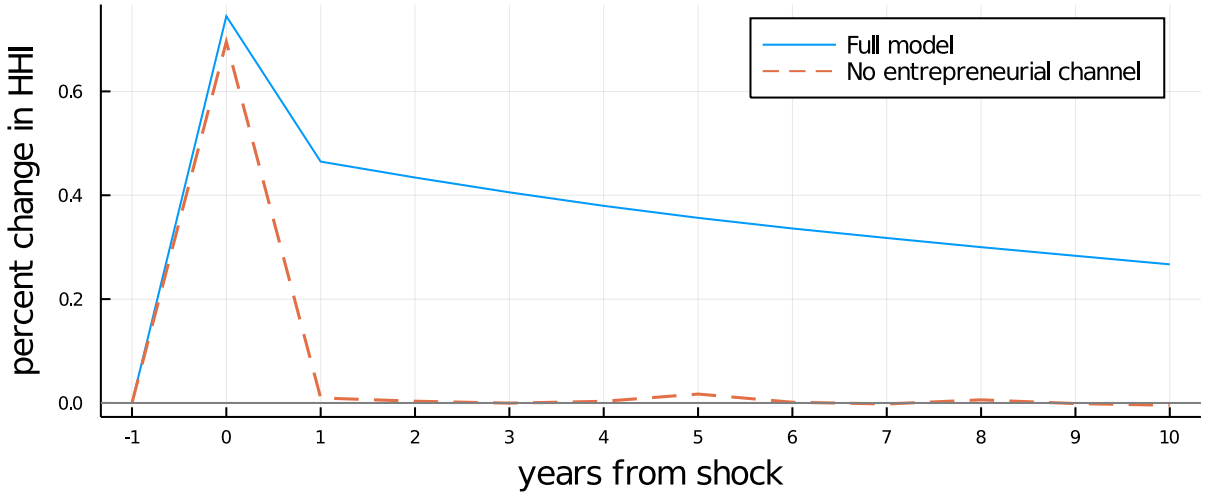
Uncertainty shocks persistently increase output concentration.[30] Figure 6 shows that a one period uncertainty shock raises concentration by approximately 0.75 percent on impact and remains persistently elevated over the next ten years. In fact, concentration remains approximately 0.27 percent greater than its pre-shock level ten years after impact, and assuming the response follows a linear recovery process, concentration will not recover to its pre-shock level before 23.8 years after the shock.[31] That is, the response of concentration to an uncertainty shock in the baseline calibrated model is qualitatively consistent with the positive and highly persistent empirical relationship documented in Section 3. Therefore, I next turn to asking which transmission channels are key to explaining this result.

The persistent rise in concentration is mostly driven by uncertainty's impact on entrepreneurial activity. Figure 6 compares the response of concentration to an uncertainty shock in the baseline model and an identical model in all but replacing the Venture Capitalist with a simple matched firm entry-exit condition, as in Bloom et al. (2018) or Arellano et al. (2019). The matched firm entry-exit condition imposes that the industry size (number of firms) is kept constant and shuts down the entrepreneurial channel of uncertainty shocks. I find that when uncertainty cannot impact firm entry, an uncertainty shock still increases concentration on impact, but there is no persistence in the effect, as concentration fully recovers within one year after the shock. This finding again qualitatively matches the empirical predictions discussed in Section 3: Uncertainty dampens the extensive margin of concentration, so when uncertainty cannot impact entrepreneurship, then uncertainty has a limited or negative im-

[30] There is no distinction between sales and output in this model, so I will use the two interchangeably, although I specifically used sales concentration in the previous empirical motivation.

[31] The 23 year response life is estimated by estimating a linear response based on the line drawn from one year after the shock ($x \approx 0.46$) and ten years after the shock ($x \approx 0.27$).

Figure 6: Response of concentration to an uncertainty shock



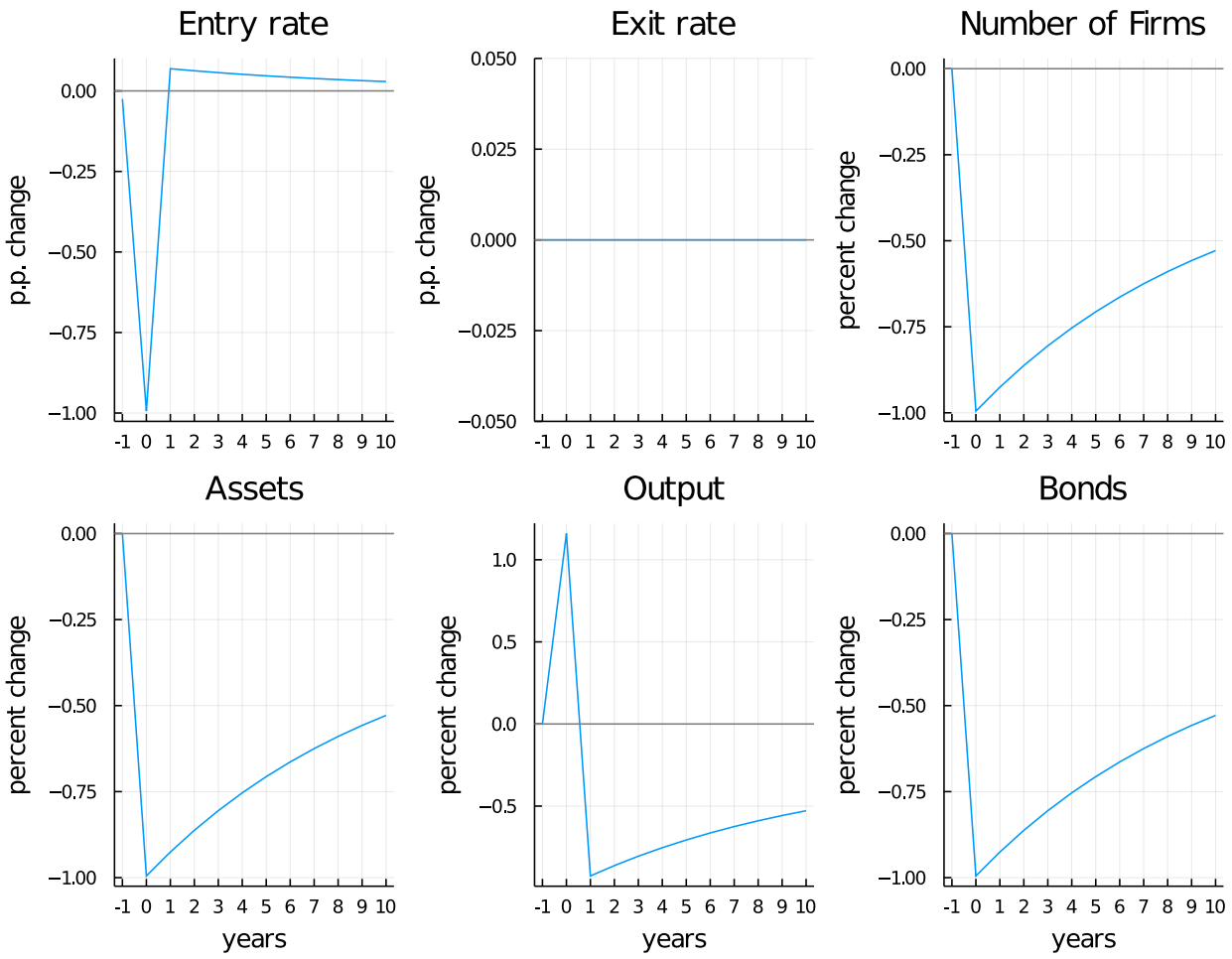
Note: This figure presents the impulse response function of output concentration to an uncertainty shock that moves the economy to a high uncertainty state for one period, keeping the aggregate and average firm-level productivity constant. The solid blue line presents the average response of 100 treated economies less the average response of 100 untreated economies. The dashed orange line presents a response calculated in the same manner, but in economies with matched firm entry and exit, shutting down the entrepreneurial funding channel of uncertainty shocks. Concentration is measured as the average of 5000 bootstrapped HHI values from an industry with $2000 \cdot |\mu_t|$ firms and sales drawn from the population weighted vector of possible firm sales.

pact on concentration.

Alternatively put, a "standard" model with only the OHA effect, real option effect, and financial amplification channel of uncertainty shocks cannot replicate the persistent rise in concentration observed in the data. The reason for the importance of the entrepreneurial funding channel becomes clear when one looks at how a broader set of industry outcomes respond to an uncertainty shock.

Uncertainty shocks lead to a broad based deterioration in industry outcomes. Figure 7 highlights the response of industry size, assets, output, and borrowing to an uncertainty shock. The shock first causes an immediate and persistent decline in the number of incumbent firms in the industry. This is because, on impact, the uncertainty shock leads to a decrease in firm entry that is offset by neither a future increase in the firm entry rate nor a compensating decrease in firm exits. This "missing cohort" of firms lead to a persistently smaller industry, thus fewer firms controlling output, and greater output concentration. The smaller industry in turn persistently produces less output, accumulates less capital, and takes out fewer loans. In fact, an uncertainty shock leads to an initial (approximately) one percent decline in the

Figure 7: Response of aggregate industry outcomes to an uncertainty shock

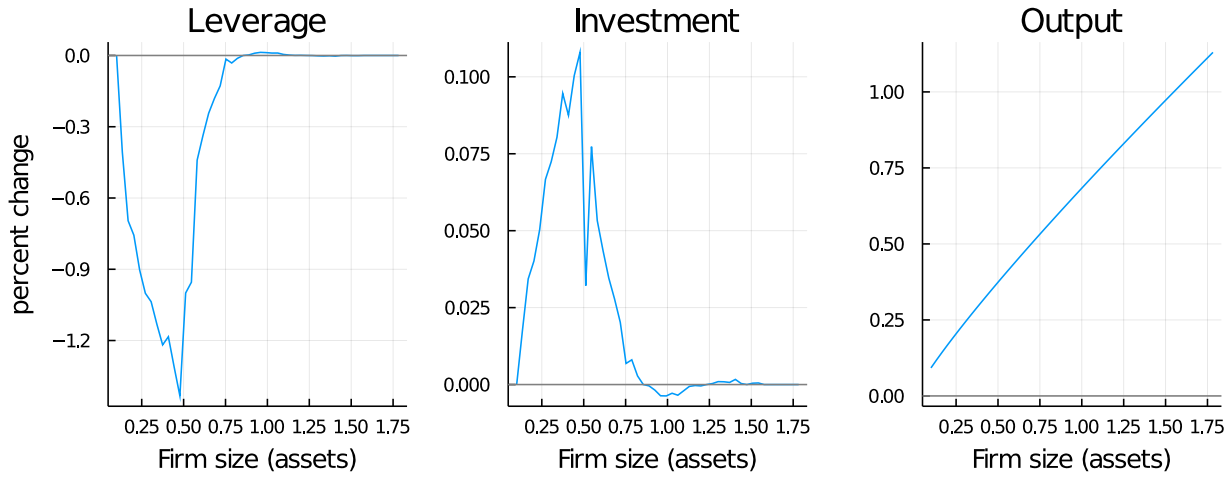


Note: This figure presents the impulse response functions of aggregate industry outcomes to an uncertainty shock that moves the economy to a high uncertainty state for one period, keeping the aggregate and average firm-level productivity constant. The solid blue line presents the average response of 100 treated economies less the average response of 100 untreated economies.

number of firms in the economy, which in turn translates to a nearly one percent decline in assets and borrowing. Output likewise declines, however immediately and not as severely as assets and borrowing, as it first expands on impact of the shock due to the OHA effect dominating the real-option effect. Industry outcomes then only slowly recover as the number of firms recovers, with assets and borrowing remaining half a percent less than its pre-shock level even ten years after impact.

The cross-sectional response of firms and evidence of the standard uncertainty transmission channels. Even though it is uncertainty's effect on entrepreneurship that appears to drive aggregate outcomes in this calibrated economy, the three standard trans-

Figure 8: Cross-sectional response of firm behavior to an uncertainty shock by firm size



Notes: This figure shows the cross-sectional response in firm behavior to an uncertainty shock. The left panel shows the change in the firms' (marginal) leverage schedule, that is, a firm's leverage in period $t+1$ given its size in period t, averaging over the productivity and aggregate productivity-weighted capital margins of the firm's state space. The middle panel shows the change in the firms' (marginal) investment schedule and the right panel shows the change in the firms' (marginal) output schedule, both calculated in a similar manner to the change in the leverage schedule.

mission channels can still be observed at work in the economy by examining the cross-section of firms' responses to the uncertainty shock. Figure 8 shows the marginalized firm policy functions by firm size (measured in capital assets). First, The left panel presents the change in firms' leverage policy (i.e. the firms' borrowing policy divided by its capital holdings) in response to an uncertainty shock. Small firms (those borrowing to invest in growth) decrease their leverage in response to the uncertainty shock. All else equal, an increase in firm-level uncertainty indicates a greater probability of receiving a productivity shock from the left tail of the shock distribution (i.e. receiving a bad shock), increasing the probability of default. Therefore, as uncertainty increases, the price of loans increase, and firms' quantity of credit demanded decreases. This is evidence of the financial amplification channel of uncertainty shocks explored in Arellano et al. (2019). However, note that the larger firms, those that have collateral in the form of capital assets, do not change their lending in response to an uncertainty shock because they have outgrown their borrowing constraints.[32] The fact that firms can effectively outgrow their borrowing constraints is consistent with observed borrowing data (see for example Caglio et al. (2021)), but it does ameliorate the potential effect of the financial accelerator channel.[33]

[32] Firms cannot explicitly post collateral in this economy, but those with capital can sell it to pay off debt in the event of a poor productivity draw, effectively allowing them to outgrow their borrowing constraints. This behavior is also exhibited in standard heterogeneous firm models with firm-owned capital, such as Khan and Thomas (2013).

[33] Arellano et al. (2019) and Straub and Ulbricht (2023) for example find strong effects of uncertainty shocks working through financial accelerator mechanisms, but they also need to impose working capital constraints on all firms.

The middle panel shows that investment increases among small firms, decreases for medium size firms, and is constant for large firms. This pattern indicates the the OHA effect dominates the real option induced "wait-and-see" effect for small, high productivity, growing firms, and the reverse is true for medium-sized firms. Moreover, this pattern is consistent with well documented empirical facts about firm life-cycle growth patterns (see Decker et al. (2016) for example). First, the fastest growing firms in the US economy are young-small-highly productive firms. These firms know their optimal size is going to be very large, and expand to grow to that optimal size. This phenomenon is replicated in the model, as small highly productive firms become even more productive due to the uncertainty shock and in turn increase their investment.[34] In comparison, empirically, firms with relatively lower productivity will either shrink and exit or endogenously choose to only become a small or medium-size firm. For these firms, in the quantitative exercise, the real option effect dominates and investment declines. This can be seen by the decline in investment among the medium-size firms, however it is masked for the small firms as the increase in investment among highly productive firms increases the overall average investment for the small-firm cohort. Meanwhile, large-mature firms, as previously discussed, outgrow their financial constraints and in turn experience no change in their ability to invest and continue to grow to their optimal size conditional on the ergodic distribution of the economy and the previous productivity path.

Third, the right panel shows that the effective firm-level output increases by firm size in response to the uncertainty shock. This is because the most productive firms (of any size), who produce the majority of the output in the economy, become even more productive and increase output. This outcome is evidence of the OHA effect at work and is why aggregate output initially increases on impact. However, if labor was not a flexible input then one might observe a decline in output on impact, as seen in Bloom (2009) or Bloom et al. (2018), as the "wait-and-see" effect initially dominates the volatility effect (in the language of Bloom (2009) or Vavra (2014)). Also note that while output increase for large firms, their investment does not simultaneously increase because these firms are, on average, increasing their dividend payouts rather than trying to grow faster.

In summary, the OHA effect dominates small firms and they, on average, increase investment, but at the same time become more financial constrained and are forced to decrease their leverage. In comparison, large firms are relatively unaffected by uncertainty shocks, having outgrown their financial constraints.

[34] For these firms, their optimal investment remains outside of the expanded inaction bands so the wait-and-see effect does not dominate.

Persistence of the recession and comparison to the literature. Recessions induced by uncertainty shocks that disrupt entrepreneurial financing are far more long lived than the typical uncertainty driven downturn found in the extant literature. For example, in Bloom (2009) output recovers from a one quarter uncertainty shock within one year in the study's baseline quantitative partial equilibrium model. Then Bloom et al. (2018), in a general equilibrium extension of the previous work, similarly find that the economy recovers from a one quarter uncertainty shock within one year. Lastly Alfaro et al. (2023), adding financial frictions to this class of models, find the economy recovers to a one quarter uncertainty shock within 1.5 years after impact. One may object that these works study the economy through a quarterly frequency while I focus on an annual frequency so it is not a like-to-like comparison. However, if one normalizes the ratio between the recovery length and duration of the shock, then the papers cited find a recession to shock duration ratio of 4 or 6, while the economy with the entrepreneurial financing channel has a recession to shock duration ratio of 23.

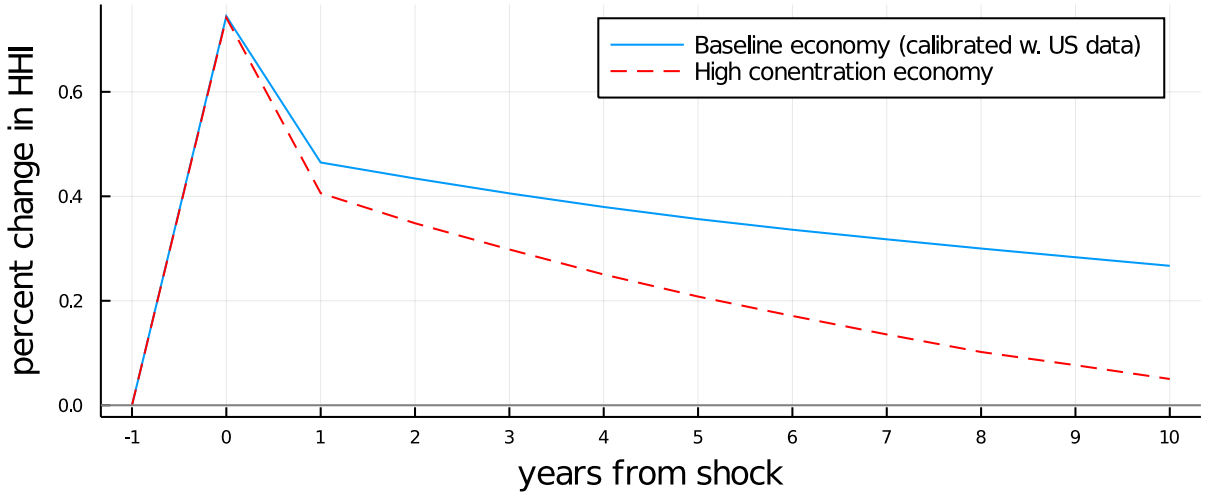
Related to the question of persistence, I also find that uncertain economies are concentrated economies. That is, I additionally find both empirical and quantitative evidence that economies (or industries) that experience a higher average level of uncertainty are also more concentrated in their steady state. However, as the focus of this paper is on impact of cyclical uncertainty shocks, I leave this discussion to Appendix E.

Robustness and limitations of the baseline model. The quantitative results are robust to various financial features of the models, for example excluding the debt subsidy or removing VC's limited liability over losses. However, the results are not robust to change in the entry condition, in particular one cannot replace VC mediated firm entry with the Free Entry condition —I elaborate on this point in Section 5.4.

Additionally, the limited impact of uncertainty through lending to incumbent firms may be due to the fact that this shock primarily works through the investment wedge, in the language of the business cycle accounting literature, such as Chari et al. (2007). Moving financial frictions to impact the labor wedge, for example with working capital constraints on labor, may alternatively yield larger responses in industry outcomes to uncertainty shocks, but I leave this extension to future research.

Lastly, the firm exit rate does not increase in response to uncertainty shocks, which is inconsistent with the empirical relationship presented in Section 3. This is because 1) firms

Figure 9: Response of concentration to an uncertainty shock, given the pre-existing level of concentration



Note: This figure presents the impulse response function of output concentration to an uncertainty shock that moves the economy to a state of high uncertainty state for one period, keeping the aggregate and average firm-level productivity constant. The solid blue line presents the response of the baseline economy calibrated to US data. The dashed red line presents the response of an economy with a steady state level of concentration 22.5 percent higher than the baseline model. Both responses are the average response of 100 treated simulated economies less the average response of 100 untreated simulated economies. Concentration is measured as the average of 5000 bootstrapped HHI values from an industry with $2000 \cdot |\mu_t|$ firms and sales drawn from the population weighted vector of possible firm sales.

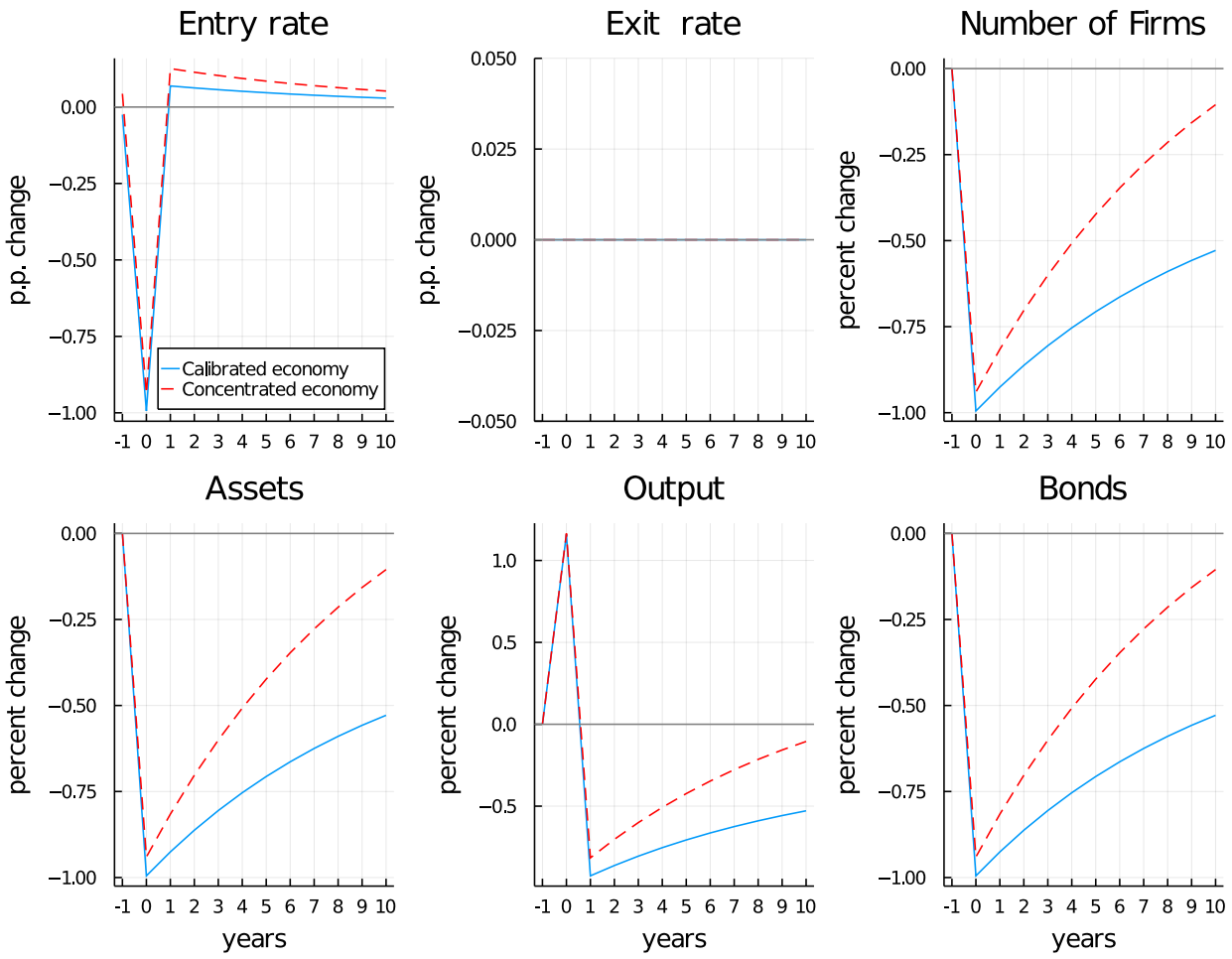
endogenously choose to limit their borrowing so that they are not in the position to go bankrupt should a shock occur, and 2) the PDV of lifetime dividend payouts will not drift below zero given the absence of a fixed cost of operating. I removed the fixed operating feature of the model make the calibration exercise more tractable, but future work should be done to reincorporate the cost to better match the response of exits to an uncertainty shock.

5.3 The pre-existing level of concentration and its dampening effect

I next turn to asking if the persistence of uncertainty shocks is state-dependent. The answer is yes. Concentration's response to an uncertainty shock, and the economy's response to the shock more broadly, strongly depends on the pre-existing level of concentration in the economy at the time of the uncertainty shock.

Higher levels of pre-existing concentration dampen the effects of uncertainty shocks. Figure 9 shows how concentration responds to an uncertainty shock when experienced in the baseline economy calibrated to US data versus in an economy with a 22.5 percent greater

Figure 10: Response of aggregate industry outcomes to an uncertainty shock, given the pre-existing level of concentration



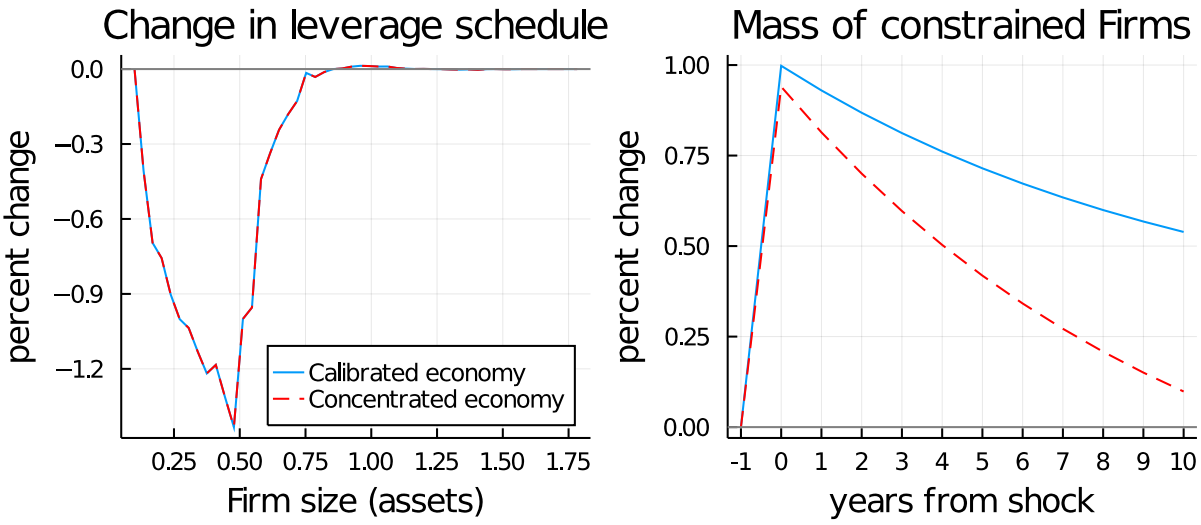
Note: This figure presents the impulse response functions of aggregate industry outcomes to an uncertainty shock that moves the economy to a state of high uncertainty state for one period, keeping the aggregate and average firm-level productivity constant. The solid blue line presents the response of the baseline economy calibrated to US data. The dashed red line presents the response of an economy with a steady state level of concentration 22.5 percent higher than the baseline model. Both responses are the average response of 100 treated simulated economies less the average response of 100 untreated simulated economies.

average level of concentration in its ergodic distribution.[35] Economies with a relatively high pre-existing level of concentration experience a significantly faster recovery to an adverse uncertainty shock —fully recovering within a projected 11.4 years after impact, compared to the baseline economy's projected 23 year recovery time.[36] Figure 10 shows that the hastened

[35] I create the more concentrated economy by decreasing the amount of wealth the VC is instantiated with, leading to lower entry rates. In the economy used for this exercise I decrease VC wealth by 50 percent, and the resulting economy is 41.4 percent more concentrated, as measured by HHI. I created a more concentrated economy in this manner because it leaves the agent's policy functions unchanged, so one can attribute the difference in responses to uncertainty shocks across economies as being solely driven by the difference in steady state concentration.

[36] The 11 year response life is estimated by estimating a linear response based on the line drawn from one year after the shock ($x \approx 0.406$) and ten years after the shock ($x \approx 0.05$).

Figure 11: Response of the firm leverage schedule and share of borrowing constrained firms to an uncertainty shock, given the pre-existing level of concentration



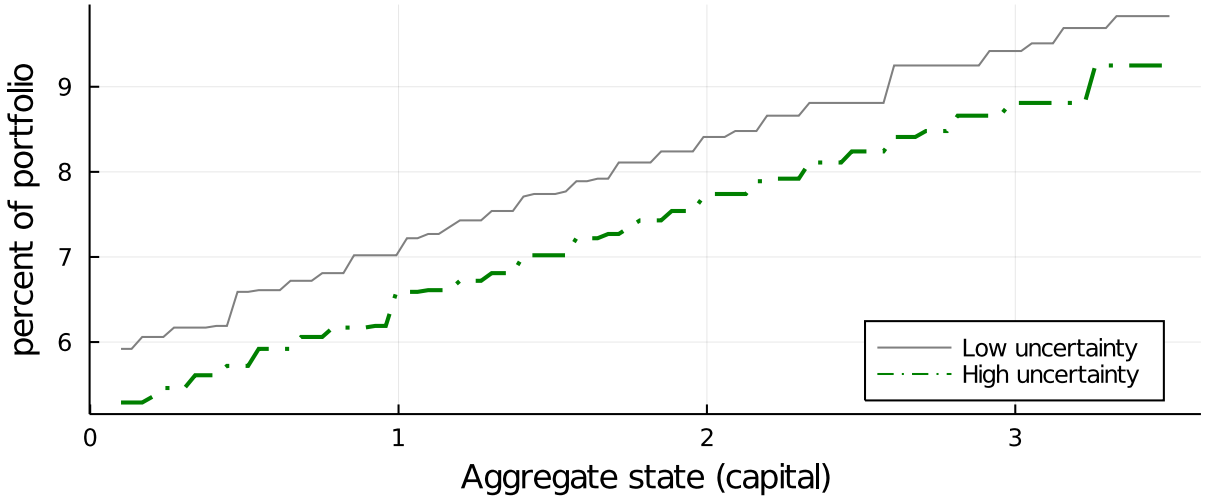
Note: This figures shows the cross-sectional response of firms' leverage schedule to an uncertainty shock, and the dynamic change in the percent of borrowing constrained firms in the economy after an uncertainty shock. The left panel shows the firms' (marginal) leverage schedule, that is, a firm's leverage in period $t+1$ given its size in period t, averaging over the productivity and aggregate productivity-weighted capital margins of the firm's state space. The right panel shows the percent of firms in the economy that have to change their leverage ratio when the economy moves from low to high uncertainty (the share of firms with assets less than 0.75). The solid blue line presents the response of the baseline economy calibrated to US data. The dashed red line presents the response of an economy with a steady state level of concentration 22.5 percent higher than the baseline model. Both IRFs are the average response of 100 treated simulated economies less the average response of 100 untreated simulated economies.

recovery in the concentrated economy is in fact an economy-wide recovery, with output and borrowing recovering much faster than in the baseline economy. A back of the envelope calculation then shows that a 41.4 percent increase in the average level of concentration leads to an approximately 52 percent faster recovery time.

The pre-existing level of concentration not only dictates the speed of the economic recovery, but also the depth of the recession. The concentrated economy experiences a relatively dampened response in industry outcomes to an uncertainty shock (except for, notably, in concentration). For example, Figure 10 shows that assets decline by approximately one percent in the baseline economy when the shock hits, while assets in the concentrated economy only fall by 0.94 percent. Alternatively put, the concentrated economy has a six percent weaker response to the recessionary shock than the baseline economy.

The concentration dampener —the effect by which the pre-existing level of concentration in an economy mediates the short run transmission and recovery from uncertainty shocks—

Figure 12: Venture funding allocation schedule



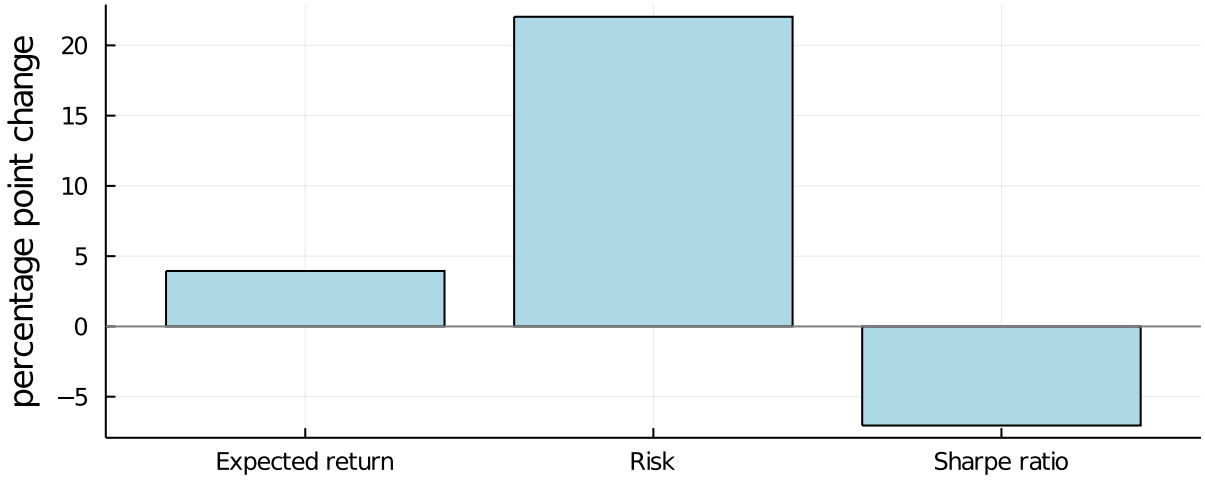
Notes: This figure shows the VC's venture funding allocation schedule when the economy is in a high or low uncertainty state, given the level of productivity-weighted aggregate capital.

exists because of concentration's effects on borrowing and the overall strength of the financial accelerator channel. The intuition is straightforward: Since firms borrow from banks to fund capital investments, it follows that as small firms are forced to cut their borrowing in response to an uncertainty shock (see Figure 11 left panel), they are also forced to cut their output growth. In a concentrated economy, a higher percentage of output is produced by large firms that do not have to decrease their leverage in response to a rise in uncertainty, thus the shock has an attenuated impact on output growth (see Figure 11 right panel). This in turn means that the expected aggregate state of the economy (measured in productivity-weighted capital) is less effected by an uncertainty shock in a relatively more concentrated economy. The VC's allocation of venture funding is an increasing function in the expected aggregate state of the economy (see Figure 12). Therefore, entry falls (slightly) less and recovers more quickly in the concentrated economy, leading to a faster recovery in the number of firms in the industry. As previously shown, the aggregate response to uncertainty shock is driven by the extensive margin, and since the more concentrated economy recovers to its pre-shock number of firms more quickly than the less concentrated economy, it also experiences a shallower and shorter-lived recession in asset accumulation, output, and borrowing.

5.4 Testing alternative shocks and entry models

I next turn to scrutinizing two key aspects of the quantitative exercise, and ask 1) can I replace risk averse VCs with the more standard Free Entry condition, 2) can I discriminate the effects of uncertainty shocks from financial shocks.

Figure 13: Response of venture capital risk to an uncertainty shock



Notes: This figure shows the response of expected return, risk, and Sharpe ratio of funding entrepreneurial ventures to an uncertainty shock. The Sharpe ratio is $\frac{r-r_s}{\sigma_r}$, where r is an expected asset return, r_s is the safe asset return, and σ_r is the variance of the risky asset return. The safe return used in the Sharpe ratio calculation is the safe asset return in the calibrated model: $r_s = \beta^{-1}$. The values are averaged across the calibrated aggregate state space.

5.4.1 Uncertainty shocks as a discriminating test of entry models

Firm entry mediated by the presence of a risk averse VC investor is a non-standard feature of this quantitative model. Moreover, since firm entry is the key economic activity explaining the persistent effects of uncertainty shocks, it is worth comparing how the economy would behave under a more standard model of entry.

The most common way for macro-models with dynamic firm entry to determine the level of entry is via the Free Entry condition (see for example, Hopenhayn (1992), Restuccia and Rogerson (2008), and Clementi and Palazzo (2016)).[37] In an economy operating with the Free Entry condition, new firms enter the market place each period until the expected value of creating a new firm is zero. In comparison, when a risk averse investor determines entry, then the entry rate does not necessarily increase with the expected value of firms entry, rather with the Sharpe ratio of the expected returns on venture funding.[38] The Sharpe ratio reflects the risk-adjusted expected return of an investment, so that even if the expected return of an asset increases, if the underlying risk increases more, than the Sharpe ratio falls and the investment is seen as less desirable to a risk averse investor. The fact that Free

[37] Although, notably, Gutiérrez and Philippon (2019) that argues that Free Entry is a poor assumption and that overall entry is set by regulation, in the spirit of the political economy literature.

[38] The Sharpe ratio is $\frac{r-r_s}{\sigma_r}$, where r is an expected asset return, r_s is the safe asset return, and σ_r is the variance of the risky asset return.

Entry and VC mediated entry ultimately rely on different investment metrics (the expected firm value versus Sharpe ratio, respectively) means that the two models do not necessarily need to yield the same qualitative response to economic shocks.

Free Entry would predict an increase in entry in response to a uncertainty shock in the baseline model calibrated to US data. Figure 13 shows the change in the expected value, risk (the standard deviation of the distribution of potential returns), and Sharpe ratio of firm entry in response to an uncertainty shock. That is, uncertainty actually increases the expected value of an entering firm, which in turn would lead to an increase in entry in a model with the Free Entry condition —a result that is not consistent with the empirical relationships presented in Section 3.[39] In comparison, the risk of return on funding entrepreneurs increased approximately four times as much as the increase in the expected return, causing the Sharpe ratio to decline. In an economy with entry determined by a risk averse investor, a decline in the Sharpe ratio for venture funding leads to a decrease in entry —a results that is consistent with the empirical relationships presented in Section 3.

Therefore, the response of firm entry to an uncertainty shock can act as a separating moment between two competing models of firm entry, and in this model calibrated to US data, the Free Entry condition is rejected in favor of entry determined by a risk averse VC investor.

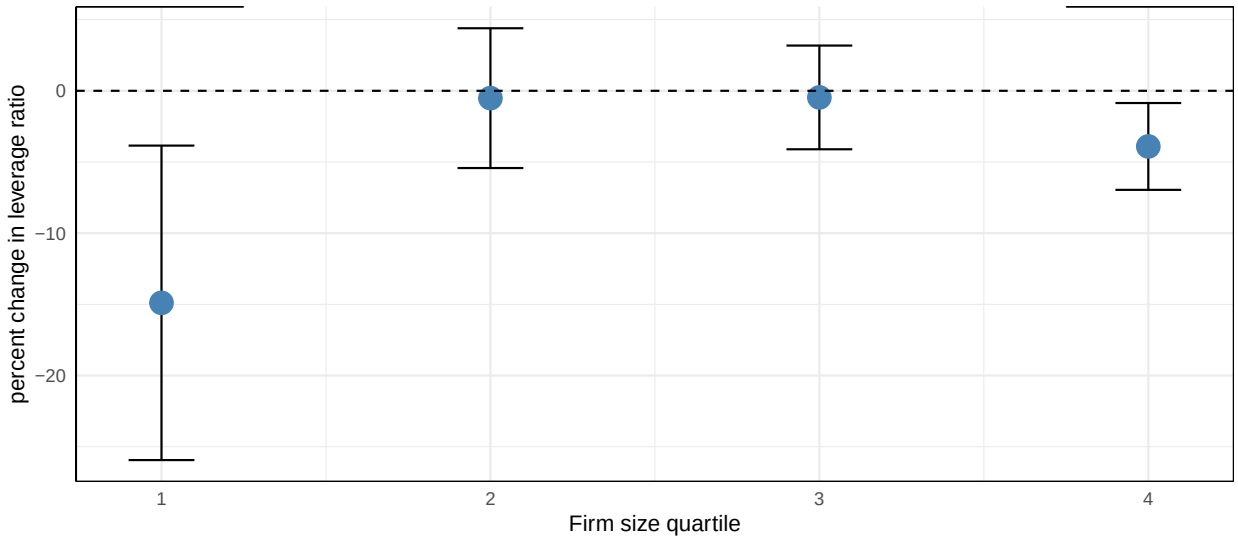
5.4.2 Firm-level leverage and output response as discriminating moments among shocks

The fact that entrepreneurial activity rationalizes the persistent relationship between uncertainty shocks and concentration in the model, raises the question whether alternative shocks, such as financial shocks, may actually explain the empirical results. For example, it is well known that fluctuations in house prices, which are used as collateral for small business loans, are linked to financial constraints entrepreneurial activity in the United States. See Davis and Haltiwanger (2023) or Aruoba et al. (2022) for recent empirical examinations of this relationship. If house prices, or fluctuations in VC wealth more generally, coincide with rises in the VIX (as a well known proxy for uncertainty), then the investor wealth shock could actually explain the empirical relationship between between uncertainty and concentration.

To discriminate among idiosyncratic uncertainty shocks and a VC wealth shock —which both decrease firm entry and increase concentration— I turn to comparing their impacts on firm-level borrowing decisions. I find that among the competing quantitative shocks,

[39]The expected value of entrepreneurship rising during an uncertainty shock reflects that relatively productive firms may be born with greater initial productivity, while losses from less productive firms exiting are mitigated by limited liability. That is, there is an unbounded potential return to entrepreneurship and bounded potential loss. This dynamic is discussed further in Quadrini (2009). Note that agents in this economy understand limited liability in so far as a VC cannot lose more than the seed money it invests in a start up.

Figure 14: Response of firm leverage to an empirical uncertainty shock by firm size



Notes: This figure shows the empirical response of firm leverage to an uncertainty shock. Blue dots are the firm leverage pseudo-elasticity with respect to uncertainty shocks, estimated by binned ordinary least squares. The black lines represent the 90 percent confidence intervals, clustered at the date level. Firm size is measured by the firm's asset ranking among all public firms in a given period. Leverage is a firm's debt to book asset value; changes are measured as period-to-period log differences. Uncertainty is proxied by the Jurado et al. (2015) measure of macroeconomic uncertainty. The regression additionally controls for one lag of the firm's change in leverage ratio, the firm's debt maturity composition, annualized real GDP growth, core PCE inflation, and the one year Treasury rate. Firm-level data is winsorized at the 2.5 and 97.5th percentiles. The sample contains approximately 61.6 thousand firm-date observations with 12.6, 17.3, 17, and 14.6 thousand observations in each ascending quartile respectively. Data is quarterly and spans 1990:Q3 through 2023:Q4. Firm level data is from Compustat.

only firm-level uncertainty shocks can simultaneously match both the aggregate response of concentration and the cross-sectional response of firm-borrowing. I will next describe how I measure the micro moment, characterize this object, and then compare the quantitative shocks' ability to replicate the data.

I measure the impact of an uncertainty shock on the cross-sections of firm leverage using a panel of US public firms, the Jurado et al. (2015) measure of macroeconomic uncertainty, and a binned linear regression. The model can be written as four independent linear regressions or as a single regression of the form:

$$\Delta leverage_{f,t} = \sum_{q=1}^{4} \mathbf{1}(i \in F_q)[\gamma_q + \gamma_f + \delta_q(uncertainty_t) + \theta_q^f Y_{f,t-1} + \theta_q^a X_{t-1}] + \epsilon_{f,t} \quad (25)$$

where γ_q is the fixed effect of a firm being in the q-th quartile of the firm assets distribution in a given date, $\mathbf{1}(i \in F_q)$ is the indicator function that firm i is in the q-th quartile in period

t, and γ_f is the firm fixed effect; $X_{f,t-1}$ is the vector of lagged firm-level controls, which includes the first lag of the change in leverage and the maturity structure of the firm's debt (defined as debt due within one quarter as a share of the firm's entire debt portfolio); X_{t-1} is the vector of lagged aggregate controls, including real GDP growth, core PCE inflation, and the one year Treasury yield as controls for the prevailing business cycle conditions.[40] I use the period-over-period percent change in book leverage and the Jurado et al. (2015) measure of macroeconomic uncertainty as measures for the two phenomena. The coefficient δ_q is the statistic of interest, and is interpreted as the psuedo-elasticity between firm-level outcomes and uncertainty, interacting with the firm's size category. Firm data are from Compustat, are quarterly from 1990:Q1 through 2023:Q4, and contain approximately 61.6 thousand observations unique firm-date.

Empirical uncertainty shocks cause a larger decline in leverage among small firms relative to large firms. Figure 14 shows that a one standard deviation uncertainty shock induces an approximately 15 percent decrease in the leverage ratio of the bottom quarter of firms (by book asset), no change in the middle 50 percent of firms, and a slight 4.6 percent decrease for the largest firms.

The empirical pattern is well matched by the cross-sectional response of firm borrowing to an uncertainty shock in the quantitative model. Figure 8 shows that leverage only decreases for small firms and then remains unchanged for medium and large firms in response to a quantitative uncertainty shock. In comparison, a VC wealth shock —a 10 percent decline in the VC's wealth in period t— does not change the firm leverage schedule. In fact, I do not present a new figure of the cross-sectional response of leverage to uncertainty and VC wealth shocks because it would simply reproduce the leverage panel of Figure 8, with the zero line labeled as the response to the VC wealth shock. The intuition behind the financial shocks null effect is that the VC wealth shock neither immediately impacts firm-level productivity nor the aggregate state of the economy (as firms enter and begin to produce in the subsequent period). Therefore, firms do not change their borrowing decisions.

I conclude that the quantitative model puts forth evidence that firm-level productivity uncertainty shocks (i.e. an increase in the dispersion of firm-level productivity shocks) are responsible for the empirical relationship between uncertainty and concentration found in the data. In comparison, a VC wealth shock (as a stand in for a generic financial shock) may lead to an increase in concentration in the quantitative model, but it cannot match other

[40] I am motivated to include a measure the firm's debt maturity structure by work such as Jungherr and Schott (2022), which argues that firms change their debt holdings based on the amount of debt they are rolling off their balance sheet in a given period and this may slow their response to aggregate conditions. It follow that a firm's change in borrowing due to an increase in uncertainty may then be moderated by its contemporaneous level of expiring debt, and I attempt to control for this effect.

moments associated with the empirical shock.

5.5 Discussion

Cyclical uncertainty shocks have very persistent effects by disincentivizing entrepreneurship, while the pre-existing level of concentration mediates their transmission by dampening the financial accelerator. While this relationship is noteworthy in its own right, it also extends our understanding of the consequences of cyclical shocks and the interaction of secular trends. The following will discuss four potential implications of this relationship.

Implications for cyclical shocks. Uncertainty's relationship with entry and concentration opens a channel for the cyclical shock to have very persistent, medium run, impacts on the economy. The literature studying cyclical uncertainty shocks, similar to most literature studying cyclical shocks, has focused on the short run transmission and consequence of the shock. However, an uncertainty shock leads to a decrease in firm entry for one period, and these "missing" firms are very slowly replaced, if ever, which leads to a smaller, more concentrated, industry. These results resemble the hysteresis observed after financial crises, see for example Aikman et al. (2022) and Jordà et al. (2017), and highlight uncertainty shocks' role in disrupting entrepreneurial activity and funding as a potential source of this hysteresis. This work is not the first to connect firm entry and hysteresis, see for example Clementi and Palazzo (2016), but it is the first to directly trace this behavior to a primitive uncertainty shock. Moreover, these findings create a clear link between transitory uncertainty shocks and productivity growth through their disruption of the firm entry process. It has been well documented, for example Foster et al. (2001), and theorized, as in Akcigit and Kerr (2018), that firm entry is a key driver of productivity growth. Therefore, uncertainty shocks disrupting firm entry means that they disrupt the creative destruction process, thus dampen productivity growth, which can have long run consequences. Additionally, as uncertainty increases concentration, it also impacts economic outcomes such as markups and innovation through this channel as well. However, this work utilizes a business cycle model, thus cannot speak directly towards the long run impacts of a cyclical shock. I leave incorporating the uncertainty, entrepreneurship, concentration relationship into models of long run growth for future research.

Second, the concentration dampener acts as a counterpoint to a hypothesis motivated by the granular shocks literature (e.g. Gabaix (2011) and Gabaix and Koijen (2020)) that rising concentration may increase volatility in an economy. The intuition is that as large firms become more important to the aggregate outcomes of an economy, then firm-specific shocks (i.e. granular shocks) will in turn have a larger impact on aggregate outcomes, thus poten-

tially increasing the volatility of these outcomes (if the firm-level shocks are more volatile than aggregate shocks). In comparison, the concentration dampener is an example of rising concentration decreasing the volatility of aggregate outcomes, by insulating the economy from either firm-specific or aggregate shocks that would otherwise be amplified through a financial accelerator type channel.

Implications for secular trends. Third, the rise in firm-level uncertainty may be a reason for the decline of business dynamism in the United States. Decker et al. (2020) argue that the secular decline in business dynamism, including entrepreneurship, in the US since the early 2000s is in part due to a decline in firm-level responsiveness to shocks rather than a change in the dispersion of firm-level shocks. This is because of the authors' finding that the dispersion of firm-level productivity shocks, i.e. uncertainty, has been increasing over the same time period.[41] The argument is that an increase in the dispersion of firm-level productivity shocks will in turn yield an increase in the rate at which firms need to create or destroy jobs, actually increasing dynamism.[42] However, I find that rising firm-level uncertainty is a factor contributing to the decline in entrepreneurship, thus business dynamism.[43] Since these channels appear to have conflicting impacts, it is therefore a matter of quantitative importance in determining whether firm-level uncertainty increases or decreases business dynamism on net.

Fourth, these two findings may potentially explain contemporaneous secular trends in the US, increasing concentration and the Great Moderation (the secular decline in the variance of aggregate outcomes in the US since the 1980s). Uncertainty in the 1970's, particularly during the Volker recessions, may have helped kick-start the persistent rise in (employment) concentration documented in the US since the 1980s. There is a large literature discussing alternative reasons for the secular rise in concentration (see Akcigit and Ates (2023) for a good overview) and I would not assert that uncertainty shocks are the only explanation, but this paper provides evidence to support that they are perhaps one of many reasons. Then, as concentration rose through the 80's and 90's, the impact of uncertainty shocks (or any shocks that primarily work through credit markets) were steadily dampened, leading to a decline in the volatility of aggregate outcomes, characterizing the Great Moderation. This paper has not attempted to rigorously study the interaction of these two secular trends, but does hint at a possible connection, and leaves further exploration of this to future work.

[41] Bloom et al. (2018) document a similar rise in firm-level uncertainty using a similar empirical procedure.

[42] The authors argue this relationship between the dispersion of firm-level shocks and employment dynamism in the context of a Hopenhayn (1992) style model of firm dynamics.

[43] See Decker et al. (2014) for a review of the relationship between entrepreneurship and business dynamism in the United States.

6 Conclusion

This paper studies why uncertainty shocks have persistent effects on economic activity. I first empirically show that uncertainty shocks induce a persistent decline in firm entry and a persistent rise in sales concentration among US public firms (and employment concentration among all US firms). I then turn to a quantitative heterogeneous firm model calibrated to US data to better understand what transmission mechanisms explain this persistence. However, standard channels of uncertainty shocks, such as the OHA effect, real option, and financial accelerator channels cannot alone replicate the persistence observed in the data. Instead, the model must be augmented with an entrepreneurial funding channel, by which uncertainty shocks decrease risk averse investors' willingness to fund startups, to fully capture the enduring effects of uncertainty shocks documented in the empirical exercises. The resulting "missing cohort" of firms is not replaced when uncertainty subsides, leading to a smaller and more concentrated industry that only slowly recovers. These effects are very persistent, but not permanent, as the long run equilibrium of the model is unchanged. Additionally, the model predicts that the pre-existing level of concentration mediates the transmission of uncertainty shocks to the broader economy by dampening the financial accelerator's amplification of the shock.

These findings highlight novel costs and benefits of uncertainty shocks and market concentration, respectively. However, this work has been exclusively positive in nature, and does not consider what impact these findings may have for normative policy recommendations in responding to either uncertainty or rising concentration. Moreover, because I study a business cycle model, this work cannot speak to whether uncertainty shocks have true long run effects, and I leave embedding the entrepreneurial funding channel in a model of endogenous growth to future research. The normative implications of the relationship between uncertainty and concentration may likewise be of interest for future research, particularly given the renewed interest in anti-trust policy and enforcement, see Babina et al. (2023) for further discussion.

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A Data

A.1 Measuring concentration

My discussion of concentration will proceed in two parts. First, I will define and characterize the measure of concentration used in this study. Second, I will discuss the challenges with measuring concentration more broadly and show common trends in measures of US concentration.

A.1.1 How concentration is measured in this study

I measure concentration as the aggregate Herfindahl–Hirschman index (HHI) of public firms in the United States. The HHI is calculated based on revenue data from Compustat, which has been winsorized at the 1 and 99th percentiles within the two-digit NAICs industry-date cross-section. The HHI is studied both at the annual and quarterly frequency. When the HHI is quarterly, then it studied as the four quarter moving average to smooth seasonality.

Figure 15 (top panel) shows the level of sales concentration (among public firms) in the US from 1980 through 2022. Concentration started in an elevated position in the 1980's, fell through the mid-1990's before plateauing, then began a secular trend upwards in the late 1990's and early 2000's. Second, Figure 15 (bottom panel) shows the year-over-year growth rate of concentration, and reveals large business cycle frequency changes in concentration which are not as readily apparent when focusing on the long term secular trends in the phenomenon. Moreover, two patterns emerge when focusing on concentration growth: 1) the series is heteroskedastic, with a greater variance in growth rates in the first half of the sample, and 2) there appears to be a positive trend in growth rates if fitting a line from 1980 through 2022. In summary, concentration in sales among US listed public firms has been increasing since the 1990s and the trend is accelerating.

I use the HHI as my baseline proxy for concentration because it is one of the two most widely used measures of concentration in the macroeconomics literature (as noted in Syverson (2019) and Ahern et al. (2024)) and the measure used by policymakers when considering anti-trust enforcement (see the see the DOJ and FTC (2023) merger guidelines, page 5 merger guidelines). However, there is a clear caveat to using this measure of concentration. The HHI is a function of both the intensive and extensive margins of sales concentration. Therefore, as I am currently restricted to using data from public firms, the HHI is acting as a function of initial public offerings and delisiting events, rather than the true entry and exit dynamics of the economy. Moreover, there is a well known disconnect between measures of concentration based exclusively on public firms compared to those based on all firms in the

US economy (see, for example, Decker and Williams (2023)).

Although it should be noted that I also use employment concentration ratios based on samples reflective of the universe of US firms, sourced from the US Census Bureau's Business Dynamism Statistics dataset. The aggregate employment concentration ratio is presented in the bottom panel of Figure 16, while Figure 17 presents the IQR of employment concentration among 4-digit NAICS US manufacturing sub-industries. Rossi-Hansberg et al. (2021) document the stark difference in trend between concentration at the national and sub-national level, arguing that the divergence stems from a small number of national firms increasing overall concentration.

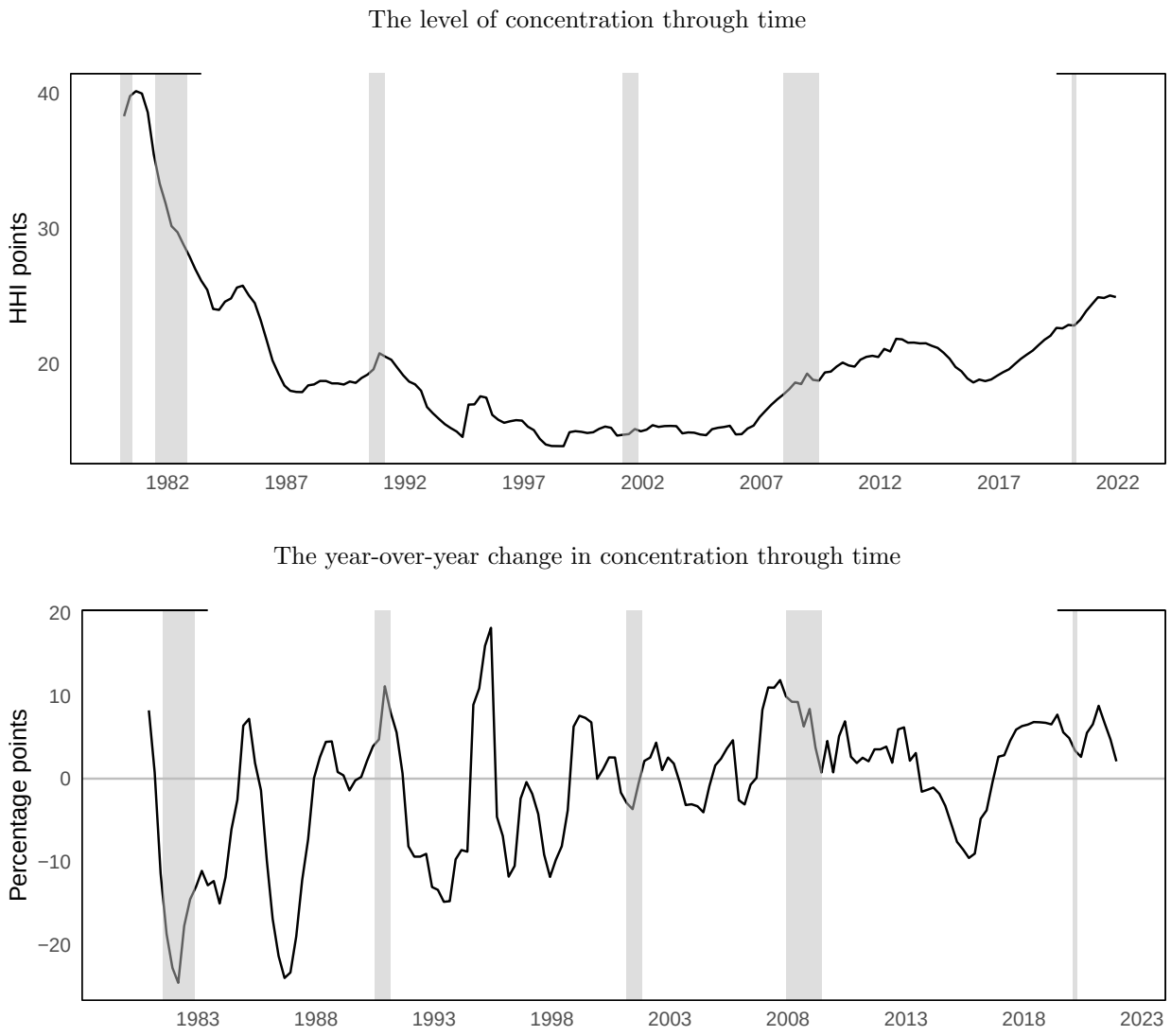
A.1.2 Difficulties of measuring concentration and common trends across methods

Measuring industry concentration is not straightforward, as one must make several choices, such as which data source, frequency, and summary statistic to use. Moreover, these choices are not innocuous. For example, Kalemli-Özcan et al. (2023) present in detail how different measurement choices and sample treatments can yield divergent conclusions about the trend of concentration in Europe. While Ahern et al. (2024) discuss how when one changes the weighting between intensive and extensive margins of concentration in measures such as the HHI, then conclusions regarding the secular rise in concentration in the US are reversed. I will next briefly compare and contrast the two most popular ways to measure concentration and two publicly available data sources, although a full accounting of how measures of concentration change based on modeling choices is beyond the scope of this study, and I would direct the interested reader to either of the two previously cited papers for lengthy discussions of the topic.

The macroeconomic literature typically relies on one of two different measures of concentration, the Herfindahl–Hirschman Index (HHI) and concentration rates (as noted in Syverson (2019)). The former is the sum of squared firm-level market shares within an industry, while the latter reports the percent of a given economic outcome of interest (e.g. employment, sales, or assets) accounted for by a group of firms. I will next examine examples of either measure.

Figure 16 (top panel) shows the aggregate HHI based on US public firm sales data. The solid red line is the unadjusted aggregate HHI, the dotted green line holds the 2-digit NAICS industry composition of the measure constant through time, and the blue dashed line holds the 3-digit NAICS industry composition of the measure constant through time. The latter two series are constructed by taking the quarterly weighted average across 2- and 3-digit

Figure 15: Sales concentration among US public firms



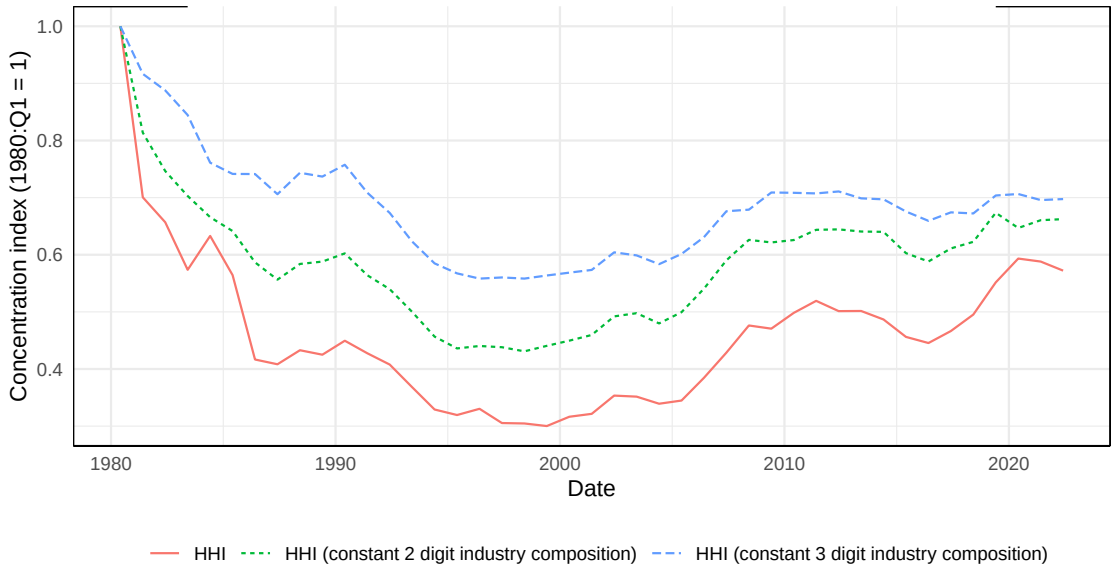
Notes: This plot presents the historical evolution of sales concentration in the United States, both in level and changes. Concentration is measured by the HHI across all public firms listed in the United States. Firm-level revenue has been winsorized at the date-industry (2-digit NAICS) level at the 1st and 99th percentiles, and the aggregate HHI presented is the four-quarter moving average. Firm-level data is from Compustat. Data is quarterly from 1980 through 2022.

NAICS industry-level HHI, where weights reflect the industry-level average share of sales in the historical sample.

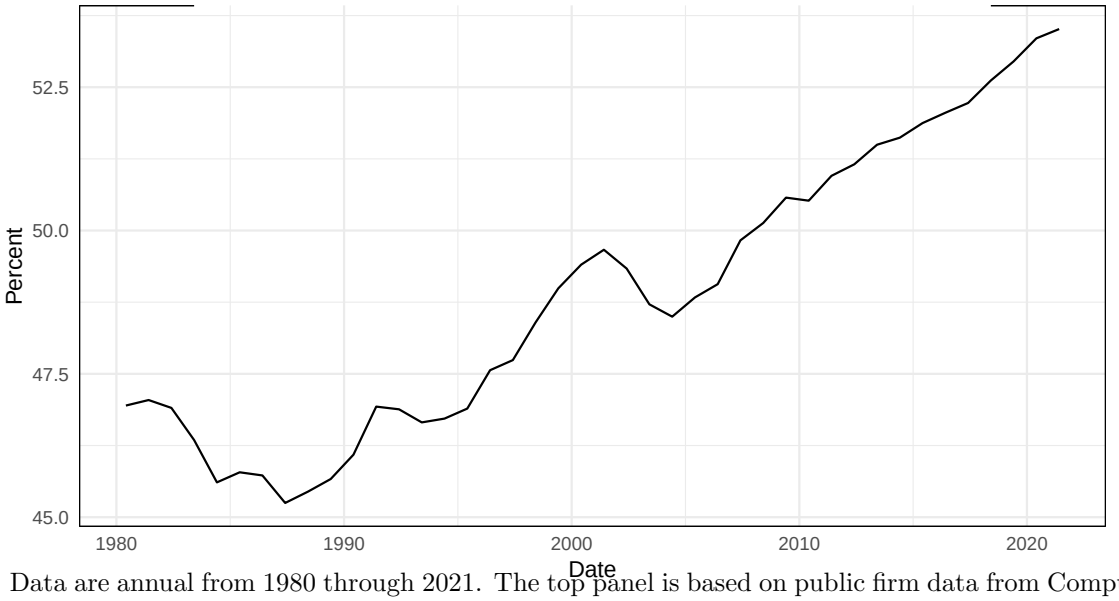
Public firm concentration has been rising since the late 1990s, whether or not one adjusts for industry composition, but has fallen overall since 1980. However, the decline in concentration relative to 1980 is dampened when only considering within-industry changes in

Figure 16: Measuring concentration

(Public firm) HHI



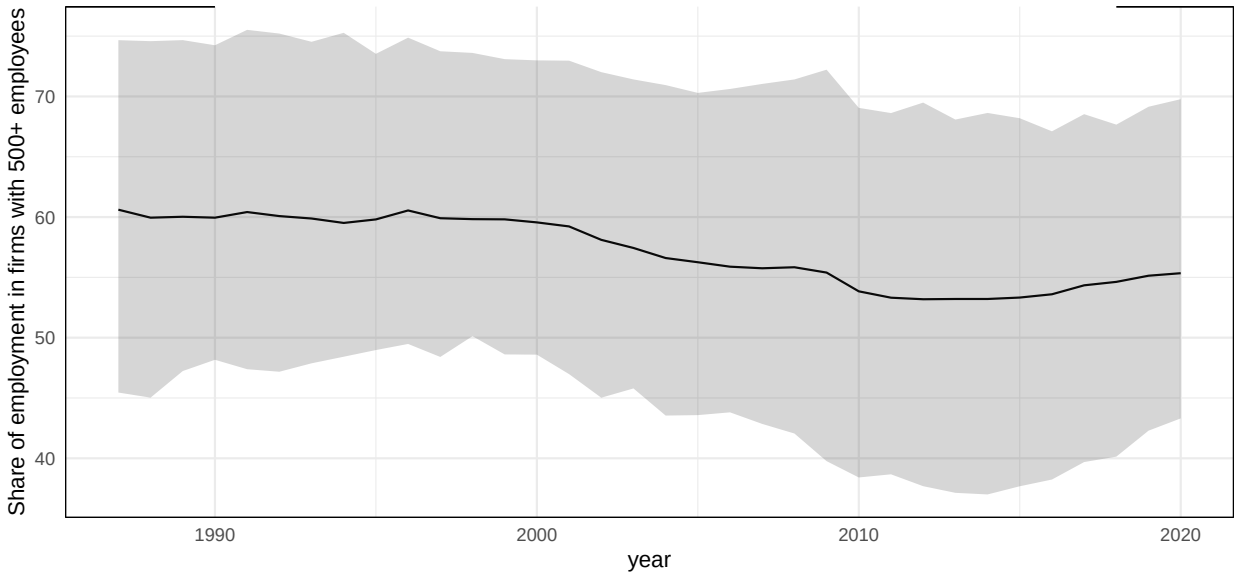
Share of employment in large firms



Note: Data are annual from 1980 through 2021. The top panel is based on public firm data from Compustat. The bottom panel is based on employment data from all firms, reported in the Business Dynamism Survey produced by the US Census Bureau. Starting HHI levels are 41.3, 237, and 571, for the HHI, 2-digit industry composition adjusted HHI, and the 3-digit industry composition adjusted HHI respectively.

concentration, i.e. holding industry composition constant. This indicates that the measured decline in concentration since the 1980s and rise since the 1990s is based on both increasing concentration within an industry, as well as sales moving from less to more concentrated

Figure 17: Dispersion of industry-level employment concentration



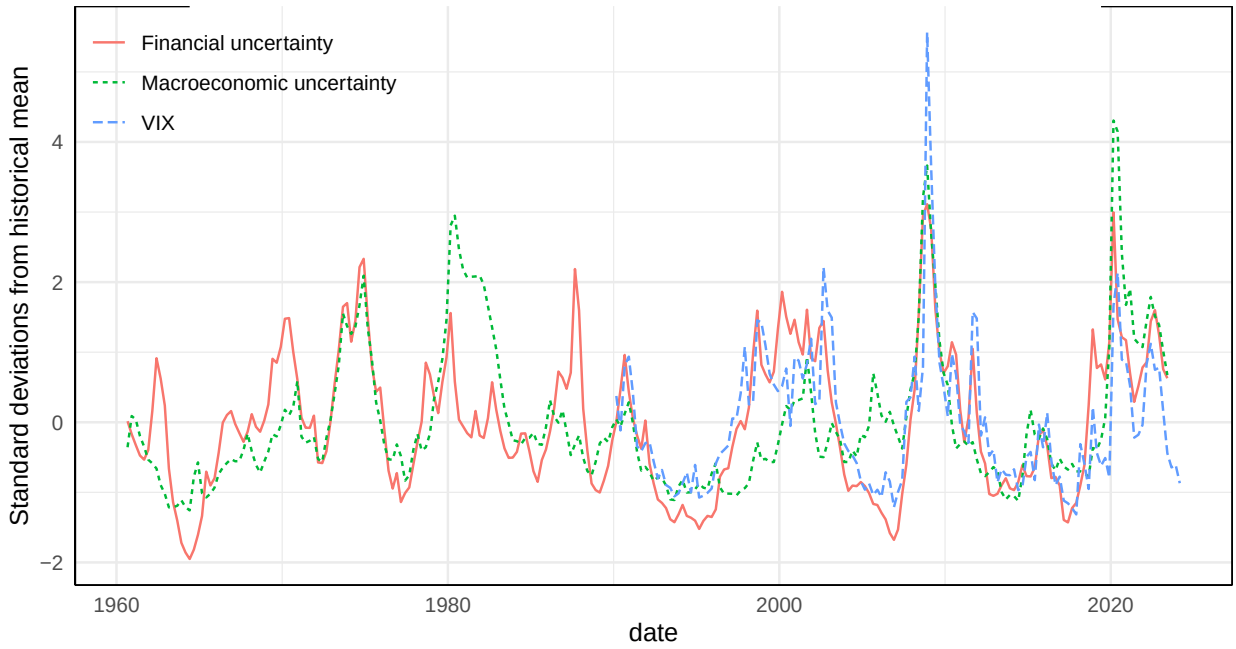
Notes: This plot shows the IQR of employment concentration for US manufacturing industries. There are 86 industries measured at the 4-digit NAICS level. Employment concentration is measured as the share of employment in firms with at least 500 employees in the given year. Employment data is from the Census Bureau's Business Dynamism Statistics dataset.

industries.

In comparison, Figure 16 (bottom panel) shows employment concentration, as measured by the percent of employment accounted for by large firms (500+ employees), has been rising since the late 1980s and indicates a greater level of concentration than in 1980. However, employment concentration also reflects a decline in concentration in the early 1980s. Note here that the differences between the top and bottom panel are not only the type of concentration statistic measured, but also the frequency (quarterly vs annual), scope of firms considered (only public firms vs all firms), and the economic activity of interest (sales vs employment). I raise these difference not to try and isolate the reason the two series are different, but rather to highlight the robustness of the trends that the two series have in common. Either measure of concentration suggests that concentration in the United States was falling at the beginning of the 1980s and has been rising since the 2000s.

Lastly, and given that I will use the public firm HHI as my baseline measure of concentration, I will note that Decker and Williams (2023) argue that studies of concentration are sensitive to the choice of using data based on all firms or only public firms than the choice between HHI or concentration ratio. In particular, these authors note that the correlation between concentration measures based on public firms are only weakly correlated with measures based

Figure 18: Comparing measures of uncertainty



Notes: This plot depicts the historical evolution of macroeconomic (ie real activity) uncertainty measured as in Jurado et al. (2015), financial uncertainty shocks measured as in Ludvigson et al. (2021), and the CBOE VIX. Data is quarterly from 1961 through 2022. Series have been standardized with mean zero, variance one, to facilitate the comparison.

on all firms.

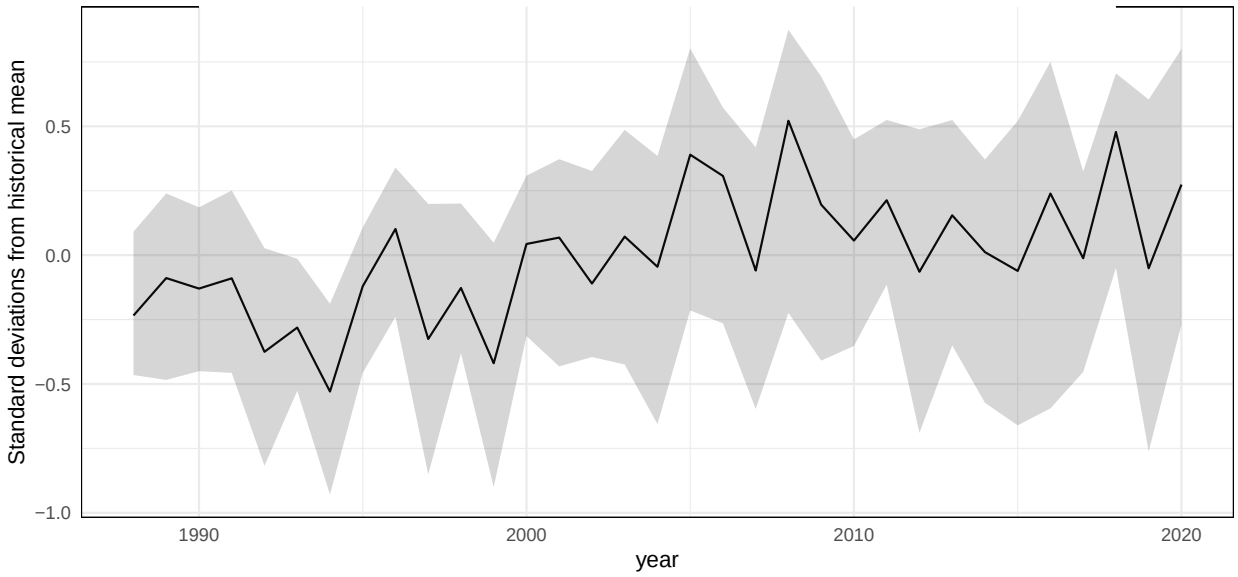
A.2 Measuring uncertainty

Like concentration, measuring uncertainty is not an easy task. While I will limit the discussion here to the measures of uncertainty used in this study, Cascaldi-Garcia et al. (2023) recently survey the broad literature on measuring uncertainty, and I direct interested readers to this paper for a lengthy discussion of the topic.

A.2.1 Macro uncertainty measures

Throughout this paper and its appendices I use three standard macro-level measures of uncertainty: the CBOE VIX, the three-month ahead financial uncertainty series from Ludvigson et al. (2021), and the three-month ahead macroeconomic uncertainty series from Jurado et al. (2015). The VIX is the expected volatility of the S&P 500 over the next 30 days. However, there is a certain opacity to source of uncertainty the VIX is measuring, which may be driven by either uncertainty in the underlying economic fundamentals of public firms (e.g. productivity uncertainty) or uncertainty regarding financial intermediaries and investors'

Figure 19: Dispersion of industry-level uncertainty shocks



Notes: This plot shows the IQR of uncertainty uncertainty shocks US manufacturing industries. There are 86 industries measured at the 4-digit NAICS level. Uncertainty is measured as the AR(1) innovations to the dispersion in the within-industry 90th minus 10th firm-level productivity percentiles. Productivity dispersion is from the Bureau of Labor Statistics' DiSP dataset.

ability and willingness to participate in financial markets (e.g. portfolio return uncertainty). Moreover, works such as Bekaert et al. (2013), have highlighted that the VIX is a function of agents' risk aversion and variance premium, in addition to uncertainty. Therefore, my empirical exercises using the VIX are limited in their ability to isolate the source of uncertainty correlated with concentration, which motivates the use of additional measure. The Ludvigson et al. (2021) and Jurado et al. (2015) measures are both "econometric" measures of uncertainty, in the language of Cascaldi-Garcia et al. (2023). That is, the Ludvigson et al. (2021) measure of financial uncertainty is defined as the expected variance in the unforcastable component of financial variables in a large panel of financial asset returns, as estimated by a statistical forecasting model. The Jurado et al. (2015) measure of macroeconomic uncertainty is similarly defined for a large panel of real economic activity indicators.

Figure 18 presents a comparison of the Ludvigson et al. (2021) measure of financial uncertainty, Jurado et al. (2015) measure of macroeconomic uncertainty, and the VIX. It is apparent that the Ludvigson et al. (2021) measure and the VIX are highly correlated (the correlation coefficient is 0.809), but the former has a much longer history, making it more advantageous when studying the persistent impact of uncertainty shocks on concentration. In comparison, macroeconomic uncertainty is less correlated with the two financial asset based measures, with correlation coefficients of 0.633 and 0.611 with financial uncertainty

and the VIX respectively. In particular, the difference between the two financial asset based measures and the real activity measure is in 1980, when macroeconomic uncertainty spikes and remains elevated with no corresponding rise in financial uncertainty.

A.2.2 Industry-level uncertainty

I compliment my empirical exercises based on macro-level measures of uncertainty with a study of industry-specific firm-level uncertainty. While I leave a detailed description of the firm-level uncertainty measure to Section 3.2, Figure 19 shows the evolution of the inter-quartile range of uncertainty shocks in the panel of US manufacturing industries. Firm-level uncertainty has been rising since 1990, and appears to reach its maximum in 2008 during the global financial crisis.

B Empirical evidence from a structural BVAR

In this appendix I study how concentration responds to contractionary demand, supply, monetary policy, and uncertainty shocks, within the context of a parsimonious Bayesian VAR with standard sign and zero restriction identification strategies. These exercises highlight two patterns: uncertainty shocks lead to large and persistent increases in market concentration, and that uncertainty shocks have the most potent effect on concentration of any business cycle shock estimated in this model. The following will detail this exercises in more detail, presenting the data, methodology, and results in turn.

Table 4: Sign and zero restrictions

	GDP	Inflation	Policy Rate	Uncertainty	HHI
Demand shock	-	-	-	0	?
Supply shock	-	+	-	0	?
Rate shock	-	-	+	0	?
Uncertainty shock	-	-	-	+	?

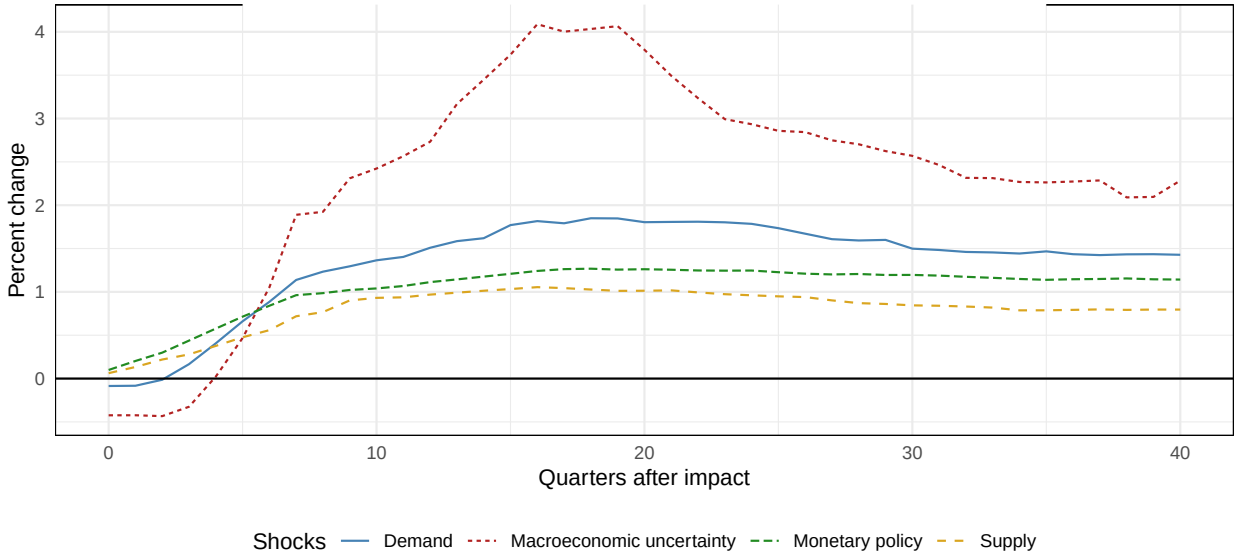
Methodology

My goal is to study the dynamic conditional correlations between concentration and business cycle shocks in the most standard setting possible. To this end, I estimate a six variable Bayesian VAR with standard Minnesota priors, quarterly data, and 4 lags.

I study business cycle shocks: aggregate demand, aggregate supply, monetary policy, and uncertainty. In the absence of direct measures or proxies for each shock, I turn to identification via a combination of sign and zero restrictions. The restrictions are as follows:

- Following quantitative studies, such as Bloom (2009), Bloom et al. (2018), and Alfaro et al. (2023), uncertainty will be treated as exogenous, so no shock but uncertainty shocks will impact financial uncertainty. Motivated by the same work, uncertainty is a contractionary shock and has a negative impact on output, thus I will allow uncertainty shocks to negatively impact GDP, inflation, and induce a monetary policy easing.
- Using GDP as quantities and inflation as prices, demand and supply shocks will be standard, such that a negative supply shock decreases quantities and increases prices while a negative demand shock will decrease both quantity and prices. It then follows from any standard Taylor rule that monetary policy will ease, thus the policy rate will fall. See Uhlig (2017) for a discussion of the supply and demand shock restrictions, as well as a discussion on sign restrictions more broadly.

Figure 20: Concentration's response to contractionary business cycle shocks, a sign restriction approach



Notes: Lines depict the response of sales concentration to a one standard deviation, contractionary, business cycle shock. Responses are the median of the posterior impulse response function, estimated with a Bayesian VAR based on standard Minnesota priors and four lags of each endogenous variable. The full set of endogenous variables include: inflation, GDP growth, policy rate, uncertainty, and HHI. Structural shocks are identified via zero and sign restrictions. Data is quarterly from 1980 through 2022.

- The tightening monetary policy shock will be identified as an increase in the policy rate, decrease in activity (GDP), and decrease in inflation, following work such as Uhlig (2005).
- Lastly, there will be no restrictions regarding shocks' impact on concentration, given this is the response we are interested in studying.

The full collection of zero and sign restrictions are summarized in Table 4.

Data

The data is also standard. Activity is measured by the annualized quarterly change in real GDP, inflation is the core PCE, the policy rate is proxied by the one year Treasury yield, as in Gertler and Karadi (2015). Concentration is measured as the HHI among US public firms and uncertainty is the Jurado et al. (2015) one-quarter ahead macroeconomic uncertainty series. Data is quarterly from 1990 through 2022.

Results

Figure 20 shows that while all of the contractionary business cycle shocks persistently increase the level of concentration, the uncertainty shock induces the largest response. A one

Table 5: Concentration's variance decomposition by business cycle shocks

	Forecast Horizon (Quarters)			
Shock	4	8	16	40
Demand	0.0352	0.0512	0.0639	0.0707
Macro Uncertainty	0.487	0.462	0.442	0.431
Monetary policy	0.0168	0.0324	0.0457	0.0515
Supply	0.0360	0.0518	0.0651	0.0714

Notes: Each column presents variance decomposition of innovations in concentration (in either levels or changes) across standard business cycle shocks for a given forecast horizon. Each row marks the variance attributed across forecast horizons to an individual structural shock. Proportion of variance explained by each shock represent the median of the posterior forecast error variance decomposition, estimated with a Bayesian VAR based on standard Minnesota priors and four lags of each endogenous variable. The full set of endogenous variables include: inflation, GDP growth, policy rate, uncertainty, and HHI. Structural shocks are identified via zero and sign restrictions. Data is quarterly from 1980 through 2022.

standard deviation uncertainty shock raises concentration by approximatley four percent by four years after impact, and the increase remains greater than 2.2 percent even ten years after the shock. Note that qualitative dynamics of the response are similar to those presented in Figure 1, which was estimated with a different model, sample, and identification scheme.

Table 5 then shows that uncertainty shock accounts for the plurality of historical variation in concentration at each horizon considered. Uncertainty shocks appear to explain approximatley 49 percent of the variation in concentration at the one year horizon, while the next most influential structural shock, supply shocks, only account for 3.6 percent.

C Empirical Robustness: alternative shocks and model

This appendix presents a series of robustness checks on the empirical facts presented in Section 3. I find that the facts presented in the main text are robust to using alternative measures of uncertainty (the CBOE VIX and the Ludvigson et al. (2021)), firm entry (business applications), and the intensive margin of the firm sales distribution (relative sales growth of public firms), as well as to controlling for contemporaneous productivity and monetary policy (i.e. financial) shocks.

To begin, I will focus on perhaps the most commonly used proxy for uncertainty, the CBOE VIX, and estimate two measures of the relationship between uncertainty and concentration: 1) the simple dynamic correlation between an increase in uncertainty and an aggregate measure of sales concentration in the United States and 2) the conditional dynamic correlation between the same uncertainty impulse and measure of concentration, controlling for contemporaneous innovations in utilization-adjusted productivity (as a proxy for contemporaneous first moment productivity shocks) and monetary policy shocks (as a proxy for contemporaneous financial shocks). Both correlations will be estimated via Jordà (2005) style local projections:

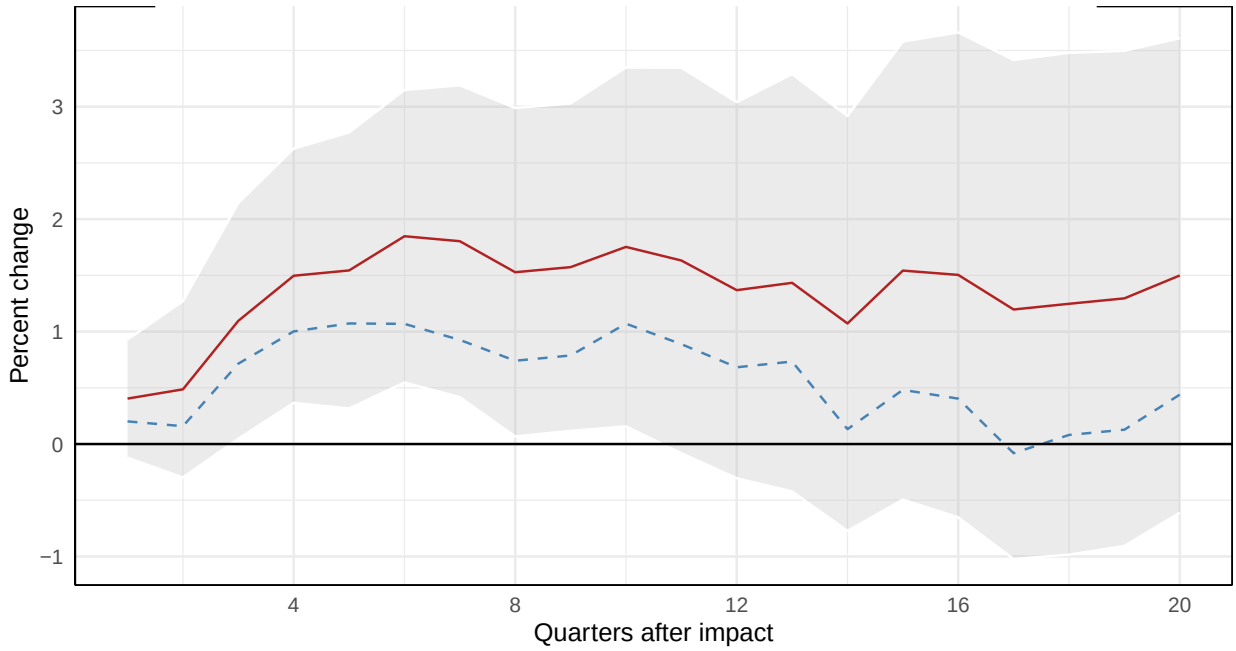
$$log(HHI_{t+h}/HHI_{t-1}) = \alpha^h + \beta^h(uncertainty_t) + \theta^h X_t + \epsilon_t^h \tag{26}$$

where t is the time index and h the forecast horizon; concentration will be measured as the Herfindahl–Hirschman index (HHI) of sales among all US public firms, uncertainty will be proxied by the CBOE VIX, and X is the vector of contemporaneous controls, the Fernald (2014) utilization-adjusted TFP and Jarociński and Karadi (2020) monetary policy shocks. Data is quarterly from 1990 through 2022.[44]

The VIX is positively correlated with a persistent rise in concentration, both conditionally and unconditionally. Figure 21 shows the estimated unconditional correlation (dashed blue line) and conditional correlation (solid red line) between uncertainty and concentration, β^h from Equation 1 above. The conditional dynamic correlation shows that concentration rises by approximately half a percent on impact of an unanticipated increase in uncertainty, rises to approximately 1.9 percent within a year and a half after impact, and then remains elevated by 1.5 percent compared to its pre-shock level for at least five years. The conditional correlation is both economically significant, as well as statistically significant at the 10 percent confidence level through two and a half years after impact.

[44] I will note, and discuss in more detail in Appendix A, that basing my measure of concentration on public firms is currently a limitation of this paper that should be addressed by future research.

Figure 21: Concentration's response to an uncertainty shocks, the VIX



Notes: This plot presents the response of sales concentration in US public firms to uncertainty shocks, as measured by the CBOE VIX. The blue dashed line is the dynamic correlation between a one standard deviation increase in the VIX and the US public firms revenue HHI at a quarterly frequency. The solid red line is the the same correlation, conditioning on innovations in the Fernald (2014) utilization-adjusted TFP measure (as a proxy for shocks to the real economy) and Jarociński and Karadi (2020) monetary policy rate shocks. The gray shaded region denotes the 90 percent confidence intervals around the conditional correlation (the solid red line) and are Newey-West adjusted. Data is quarterly from 1990 through 2022.

I next turn to re-estimating the model with the Jurado et al. (2015) measure of macroeconomic uncertainty (the same series used in the baseline exercises presented in Section 3) and the Ludvigson et al. (2021) measure of financial uncertainty. While the former attempts to capture the estimated variance in the unforcastable portion of a large panel of variables tracking real economic activity, and the latter measures an analogous object for a large panel of financial returns.[45] The goal of these additional exercises is two-fold: first to show that the baseline results presented in Section 3 are robust to only changing the information set and data frequency, and second to showing that the relationship is robust to using an alternative, but similarly measured, proxy for uncertainty.

Both financial and real activity uncertainty are conditionally correlated with concentration. Figure 22 shows that concentration rises by approximately two percent one year after an unanticipated increase in financial uncertainty, and remains elevated through five years after the fluctuation. Concentration's response to the change in financial uncertainty is similar to

[45]Both are presented alongside the VIX in Figure 18 in Appendix A.

Figure 22: Concentration's response to financial and activity uncertainty shocks



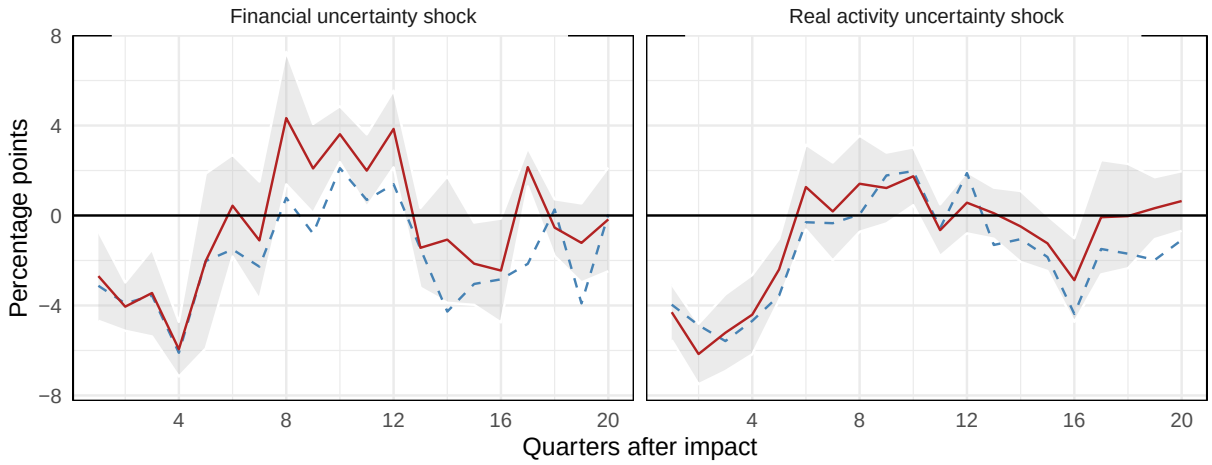
Notes: This plot presents the response of sales concentration in US public firms to one standard deviation financial and real activity uncertainty shocks. Financial uncertainty is proxied by the Ludvigson et al. (2021) 3-month ahead financial uncertainty series, and the real activity uncertainty series is proxied by macroeconomic uncertainty series presented in Jurado et al. (2015). The blue dashed line is the dynamic correlation between a one standard deviation increase in the VIX and the US public firms revenue HHI at a quarterly frequency. The solid red line is the the same correlation, conditioning on innovations in the Fernald (2014) utilization-adjusted TFP measure (as a proxy for shocks to the real economy) and Jarociński and Karadi (2020) monetary policy rate shocks. The gray shaded region denotes the 90 percent confidence intervals around the conditional correlation (the solid red line) and are Newey-West adjusted. Data is quarterly from 1990 through 2022.

its response in the VIX, although less persistent. In comparison, concentration immediately increases after an increase in real activity uncertainty, and continues to increase through five years after impact. A one standard deviation increase in real activity uncertainty is conditionally correlated with an approximately 6.5 percent increase in concentration five years after impact.

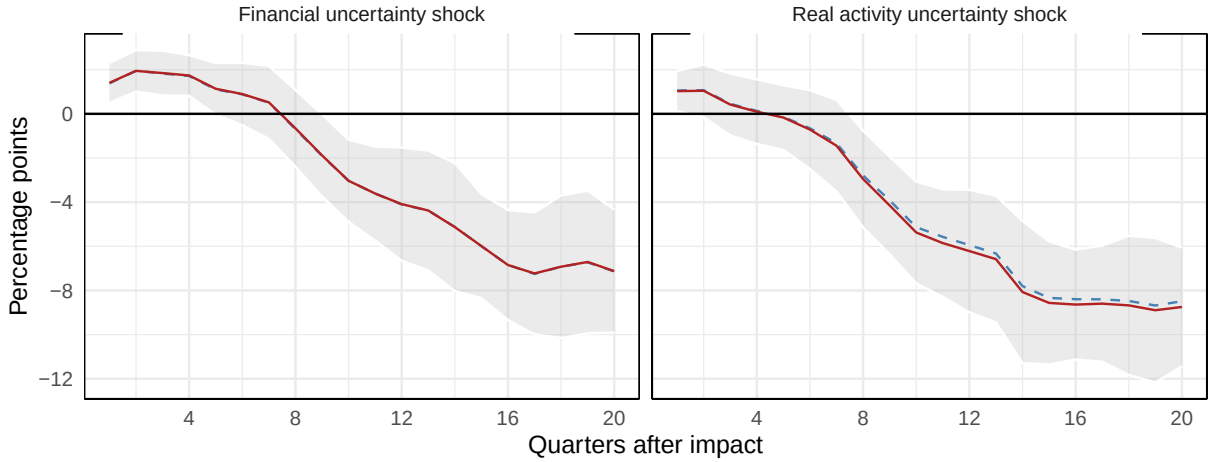
I lastly turn to examining how uncertainty impacts alternative measures of the intensive and extensive margins of concentration with alternative measures of firm entry and exit. That is, I ask the question: does an increase in uncertainty increase the sales growth of large firms relative to small firms (the intensive margin of concentration), and decrease the number of business applications? To answer this question I continue to use the same framework as before to estimate the conditional correlation between uncertainty and business applications. However, when studying the conditional correlation between uncertainty and

Figure 23: The firm sales distribution's response to financial and activity uncertainty shocks

(A) The extensive margin: firm entry rate



(B) The intensive margin: differential sales growth between small and large incumbent firms



Notes: This plot presents the response of the firm sales distribution among US firms to financial and real activity uncertainty shocks. Financial uncertainty is proxied by the Ludvigson et al. (2021) 3-month ahead financial uncertainty series, and the real activity uncertainty series is proxied by macroeconomic uncertainty series presented in Jurado et al. (2015). The top panel shows the response of the extensive margin of the distribution, measured by the annual growth of business applications for all NAICS. The bottom panel shows the change in the intensive margin of the distribution, measured by the differential annual growth rate of firm revenue among the top and bottom quartiles of firms, by revenue. The blue dashed line is the dynamic correlation between a one standard deviation increase in uncertainty and either outcome of interest. The solid red line is the the same correlation, conditioning on innovations in the Fernald (2014) utilization-adjusted TFP measure (as a proxy for shocks to the real economy) and Jarociński and Karadi (2020) monetary policy rate shocks. The gray shaded region denotes the 90 percent confidence intervals around the conditional correlation (the solid red line) and are Newey-West adjusted in panel A and are clustered at the date-level in panel B. Data is quarterly from 2006 through 2019. The panel of firms underlying the local projections presented in panel B includes approximately 166 thousand observations, 7800 firms, and 23 two-digit NAICS industries.

the relative sales growth between large and small firms, I can extend the framework to a panel local project:

$$log\left(\frac{sales_{i,t+h}}{sales_{i,t-1}}\right) = \alpha^h + \beta^h(uncertainty_t) + \sum_{q=1}^{4} \mathbf{1}(i \in F_q)[\gamma_q^h + \delta_q^h uncertainty_t] + \theta^h X_t + \epsilon_t^h \tag{27}$$

where γ_q is the fixed effect of a firm being in the q-th quartile of the firm sales distribution for a two-digit NAICS industry in a given date, $\mathbf{1}(i \in F_q)$ is the indicator function that firm i is in the q-th quartile in period $t-1$, so that $\mathbf{1}(i \in F_q) \cdot uncertainty_t$ is the interaction term between an uncertainty shock in period t and a firm's quartile ranking in period $t-1$. With the top quartile as the reference group, the coefficient δ_1^h is our statistic of interest, and is interpreted as the percentage point difference in the growth rate of firms in the bottom sales quartile compared to firm sales growth in the top sales quartile. A negative value in this difference indicates that large firms are growing faster than small firms, thus an indicator of increasing concentration, holding the number of firms constant.

Firm entry is proxied by the total number of high propensity business applications, reported by the US Census Bureau's Business Formation Statistics, going back weekly starting in 2006.[46] Firm sales are measured as firm revenue reported by US public firm income statements, and are collected via Compustat. The panel of firms underlying includes approximately 166 thousand observations, 7800 firms, and 23 two-digit NAICS industries. Data is quarterly from 2006 through 2019. As before, I will examine both financial and real activity uncertainty.

Both margins of the firm sales distribution are conditionally correlated with both types of uncertainty. Figure 23, panel A, shows that firm entry falls for at least a year after an unanticipated increase in either type of uncertainty. On the one hand, entry decreases approximately three percent on impact and then further to a six percent decline in the by one year after a financial uncertainty shock. On the other hand, entry decreases by four percent on impact, then to six percent two quarters after impact, before fully recovering within six quarters of an unanticipated increase in real activity uncertainty. The negative correlation between firm entry and both types of uncertainty are economically and statistically significant. Importantly, neither entry rate subsequently increases enough to compensate for the initial decline in entrants. This means that, on net, the total number of firms that will ever enter the market declines following an increase in uncertainty.

Panel B shows that the differential sales growth between small and large firms is either

[46] See Bayard et al. (2018) for details on the Business Formation Statistics data and the high propensity application label.

slightly positive in the short run after an increase in uncertainty —although this is statistically indistinguishable from zero after a spike in real activity uncertainty— but then negatively correlated by two years after the impulse. Moreover the differential becomes quite stark by four years after impact, growing to approximately seven percent and eight percent for financial and real activity uncertainty respectively. Overall, this indicates that there is a weakly negative or no correlation between uncertainty and the intensive margin of concentration in the near term, but a strongly positive correlation in the medium term.

D Characterizing the Venture Capitalist's behavior

I first present solutions to the Venture Capitalist's consumption-saving and portfolio allocations problems, then turn to demonstrating how a shock that impacts financial uncertainty —the variance of expected risky asset returns— such as productivity uncertainty shocks, may decrease start up funding and firm entry in turn.

D.1 Characterizing the solution to the VC's problem

Samuelson (1969) analyzes this class of problem (albeit augmented with a dynamic consumption-saving problem) and finds that 1) the portfolio allocation problem is completely separable from the investor's consumption-saving problem and 2) the investor's portfolio allocation problem is time invariant.[47] I will take advantage of these results and focus on the agent's dynamic portfolio allocation problem.

The VC's portfolio allocation problem becomes time-invariant, thus identical to the canonical Markowitz (1959) mean-variance investor problem. The investor's portfolio allocation problem follows:

$$\max_{\delta} \int_{R \in \mathbb{R}^+} U(\delta W_t R_t^e + (1-\delta) W_t R^d) d_{F_{R^e}} \tag{28}$$

where R^e is the risky asset return, F_{R^e} is the known distribution of risky asset returns, R^d is the risk free asset return, W_t is the investor's level of wealth, and δ is the percent of the portfolio invested in the risky asset.

It follows that, using the investor's FOC, I can implicitly define the Capitalist's portfolio allocation to risky assets as:

$$\bar{\delta} \equiv \int_{R \in \mathcal{R}^+} U'(\bar{\delta} W_t R_t^e + (1-\bar{\delta}) W_t R^d) \cdot (R^e - R^d) d_{F_{R^e}} = 0 \tag{29}$$

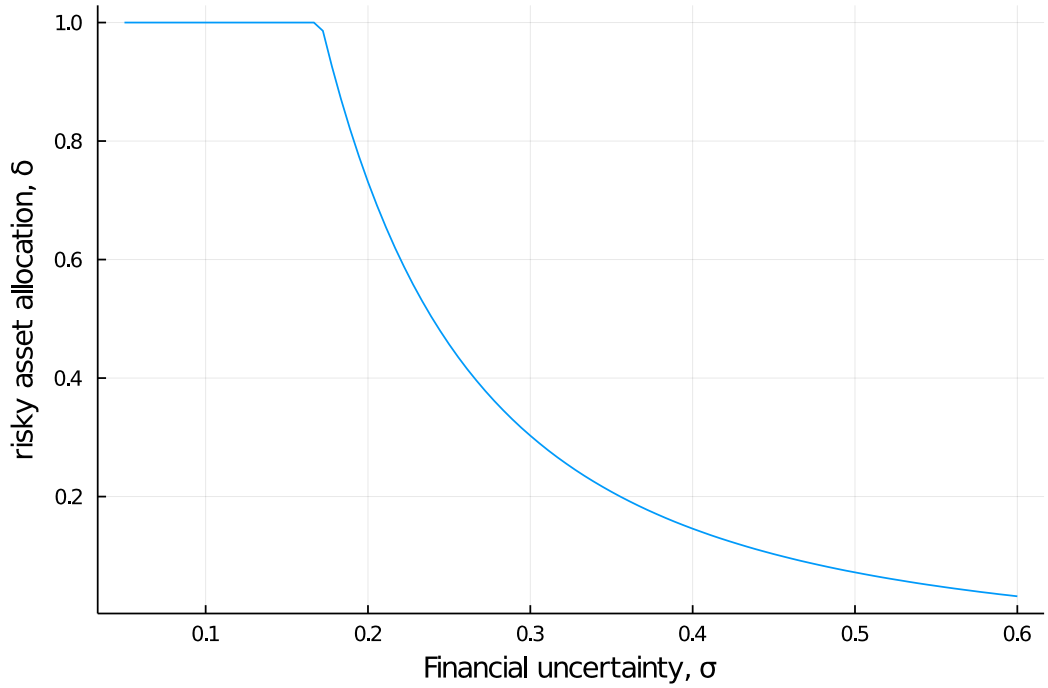
However, the FOC is only an implicit function, depending on the distribution of gross risky returns, F_{R^e}. As a result, I will have to solve the portfolio allocation problem numerically when calibrating the model.

D.2 Uncertainty and the Venture Capitalist's risky portfolio allocation

I next present the a numerical example demonstrating how an uncertainty shock may impact the Venture Capitalist's entrepreneurial funding decision. It will show that as financial

[47] Samuelson (1969) studied the same problem but proved results for a more general isoelastic utility function. Merton (1969) analyzed the same problem and proved these conclusions hold in a continuous time setting as well.

Figure 24: Numerical example of the VC's risky portfolio allocations by uncertainty level



Note: The portfolio allocation decision of an investor with log utility. The risky asset has an average return of 7% and the risk free return is 1%.

uncertainty (the variance of expected financial returns) increases, for example, due to an increase in productivity uncertainty, may lead to a decrease in the VC's portfolio allocation to risky investments, such as funding startups.

As a numerical example of the VC's risky asset allocation policy, take the net risky asset return, r, to follow a Normal distribution with mean 0.05, and the risk free rate to be 0.01. These values approximately correspond to the average return of the US equity market and short-term Treasury bonds, respectively. Then Figure 24 shows the agent's allocation to risky assets for a given level of uncertainty, a given standard deviation of the net risky return.

The VC's portfolio allocation is non-linear in financial uncertainty. Figure 24 shows that for low levels of uncertainty, up to approximately $\sigma = 0.16$, the allocation to risky assets is constant and is 100 percent. However, at $\sigma = 0.16$ the allocation to risky assets begins to decline sharply, but at a decreasing rate. That is, investors maintain a constant rate of investment in risky assets in nonvolatile times but then sharply decrease their flow of capital to risky assets in volatile times. This behavior is similar to the non-linear investment response of financial intermediaries during normal versus crises times in macro-financial models such as

He and Krishnamurthy (2013) or Brunnermeier and Sannikov (2014). I present this numerical example to show that 1) financial uncertainty will determine the relative flow of funds to the Bank versus the VC Fund, and 2) that the starting location of the financial uncertainty and the size of the financial uncertainty shock will be important states in understanding the response of the economy to a financial uncertainty shock.

Note that at this point I cannot analytically prove how a financial uncertainty shock will impact firm entry due to the response's sensitive to the size of the shock and original state of financial uncertainty. However, I have shown that there exists a subset of the parameter space in which productivity uncertainty shocks (via their impact on financial uncertainty) will decrease start up funding. The extent to which the real economy falls into this parameter space, thus the impact financial uncertainty has on firm entry, is left as an open quantitative question to be answered in a careful calibration.

E Steady state outcomes: Uncertain economies are concentrated economies

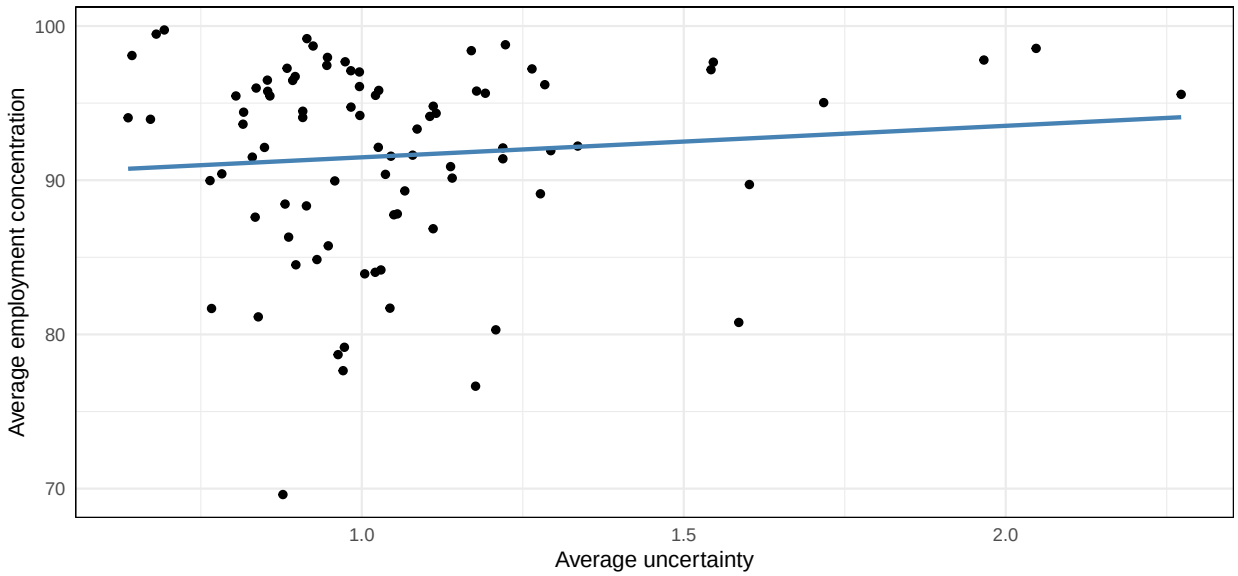
This appendix briefly describes the relationship between industry concentration and permanent differences in uncertainty levels.

Empirical. There is an empirical correlation between the average level of industry-level uncertainty and industry-level concentration. Using the same industry-level concentration and uncertainty data described in Section 3.2, I plot the average uncertainty and concentration levels of 86 US manufacturing sub-industries. Figure 25 shows a clear positive correlation between average uncertainty and concentration. This in itself does not argue for causality, but it does highlight that uncertainty industries are also concentrated industries.

Quantitative. With this empirical evidence in hand, one may ask if the fundamental level of uncertainty has a causal effect on the ergodic distribution of the economy, and more specifically, on the steady state concentration of an economy?

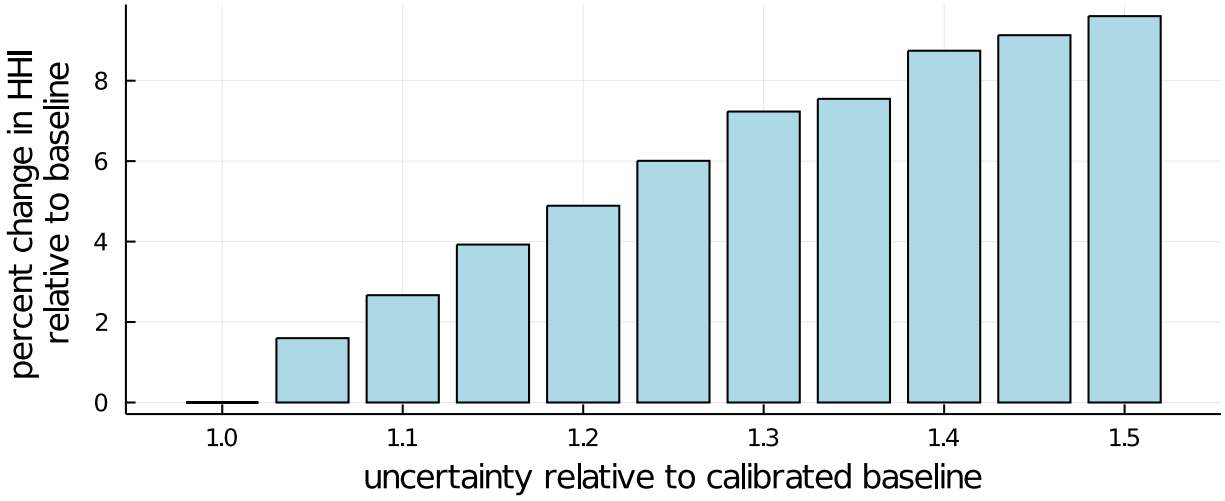
To answer this question, I start with the baseline calibrated economy and then simulate a set of economies, each for a 500 year period, and each with an increasing level of uncertainty

Figure 25: Average industry-level concentration and uncertainty



Notes: This figure shows the average employment concentration (industry-level employment share in firms with 20+ employees) and average uncertainty (measured as the difference in 90th and 10th percentile productivity shocks). The data represents 86 industries at the 4-digit NAICS, and averages are based on annual data from 1990 through 2019.

Figure 26: Steady state concentration by level of fundamental uncertainty



Notes: This figure shows the average concentration and fundamental uncertainty level across a sample of simulated economies. Average concentration is based on 400 years of simulated data after an initial 100 year burn in period. Concentration in a given period is measured as the average of 5000 bootstrapped HHI values from an industry with $2000 \cdot |\mu_t|$ firms and sales drawn from the population weighted vector of possible firm sales. Uncertainty is measured as a relative multiple of the uncertainty levels σ^L and σ^H in the calibrated baseline model. Each economy is simualted 10 times.

in the fundamental firm-level technology process (multiplying the uncertainty both levels σ^L and σ^H by a constant greater than one). I then take each economy's average level of concentration after a 100 year burn in period, and take this average as a reasonable estimate of the economy's steady state level of concentration.

Figure 26 shows that the long run level of concentration in an economy is increasing in the underlying level of uncertainty. However, the increase in concentration appears to be increasing at a decreasing rate. Take, for example, an economy with a technological process 10 percent more uncertain than the baseline economy is approximately three percent more concentrated than the baseline, while an economy that is 50 percent more uncertain is approximately nine percent more concentration. A linear relationship instead would imply an economy with 50 percent more uncertainty would be 15 percent more concentrated.

F Calibrating parameters

I divide parameters into three groups: standard values set in line with the extant literature, the parameters dictating the exogenous productivity processes, and parameters governing firm entry and exit rates. For the first group, I primarily assign parameters according to their corresponding values in the baseline model of Khan and Thomas (2013) (which is also largely followed by Bloom et al. (2018)). Although I refer to Cooper and Haltiwanger (2006) when setting capital adjustment costs. For the second group, I directly estimate the productivity law of motion Equations (10 - 12) with econometric methods and an empirical measure of firm-level TFP shocks. Lastly, I calibrate the remaining parameters, namely the firm entry cost and probability of an exogenous exit shock, with a Simulated Method of Moments (SMM) approach. The model is calibrated to an annual frequency. The following outlines the data and methods used to calibrate the second and third groups of parameters.

F.1 Directly estimating the productivity process

Given I postulate that firm-level productivity is solely driven by exogenous shocks, I estimate the productivity law of motion independent of other model parameters and irrespective of other model moments. The intuition is that as an exogenous process, no endogenous outcomes of the model should be allowed to influence the estimation of the shock process.

Recall that firms face an exogenous productivity path, determined by idiosyncratic, $\epsilon_{i,t}$, and common, A_t, components that follow independent AR(1) processes:

$$\epsilon_{i,t} = (1-\rho_\epsilon)\mu_\epsilon + \rho_\epsilon \epsilon_{i,t-1} + \sigma^\epsilon_t e_{i,t} \quad e_{i,t} \sim \mathcal{N}(0,1)$$

$$A_t = (1-\rho_A)\mu_A + \rho_A A_{t-1} + \sigma^A e_t \quad e_t \sim \mathcal{N}(0,1)$$

where μ_x is the long run average, ρ_x the persistence, and σ^x_t the volatility of the $x \in \{\epsilon, A\}$ productivity component. I assume that evolution of the firm-level conditional volatility, $\{\sigma^A_t, \sigma^\epsilon_t\}$, is time-varying and follows a two-state Markov process, governed by transition probabilities

$$\sigma_t \in \{\sigma^L, \sigma^H\}, \quad \text{where} \quad Pr(\sigma^H_t | \sigma^H_{t-1}) = \iota^H, \quad Pr(\sigma^L_t | \sigma^L_{t-1}) = \iota^L$$

where $\sigma^L < \sigma^H$, and the volatility levels or transition probabilities do not have to be the same for idiosyncratic versus aggregate productivity processes.

I estimate the parameters of the aggregate and firm-level productivity processes described above in 6 steps:

1. first estimate a panel of firm-level TFP as the cost-share implied TFPR, for US public firms with their available revenue and costs data,
2. calculate the aggregate TFP measure as the mean firm-level productivity per period,
3. estimate the AR(1) law of motion for the aggregate TFP process, yielding parameters ρ_A and σ^A,
4. subtract the aggregate TFP from firm-level TFP and treat the resulting difference as the firms' idiosyncratic productivity,
5. estimate the idiosyncratic firm productivity law of motion as an AR(1) process with firm-specific means, yielding the persistence parameter ρ_ϵ, and lastly
6. estimate a two-state Markov Switching model in means for the residuals of firm-level productivity shock model estimated in step (5), yielding the uncertainty parameters $\sigma^L, \sigma^H, \iota^L, \iota^L$.

Data. Public firm data is from Compustat, and is annual from 1980 through 2019. Firm-level TFP is proxied by $TFPR_{f,t} = log(Revenue_{f,t}) - log(Cost_{f,t})$, analogous to the common "cost-share" approach employed in the productivity measurement literature. Note that since I am studying total factor productivity, I do not discriminate among source of revenue or type of input. I rely on this broad approach in accounting for the numerator and denominator because I am studying a broad group of industries that may contain firms that differ widely in how they report revenue from their primary business activities as well as rely on differing types of inputs. Ideally future work would move to using Census data for a more representative measure of firm productivity, and/or incorporate estimating these parameters in the SMM procedure. Since the productivity measures are based on a cost-share implied TFPR measure the resulting estimates should be taken with caution, as the productivity proxies will also include demand shocks and firm-level distortions. See section 5.1 for further discussion of these potential limitations.

F.2 Calibrating entry and exit with Simulated Method of Moments

Now taking the productivity process parameters as given in the model, I calibrate the two parameters most closely related to firm entry and exit, the firm entry cost and the probability that a firm will experience an exogenous exit shock. Neither parameter is easily observed or directly estimated from the data, so I use a SMM approach to calibrating their values.

In this procedure I target five macro-moments based on US data: the average firm entry rate, average firm exit rate, average output growth, standard deviation of output growth,

and the average firm-level leverage. Where the first two are meant to align the model with US entrepreneurial and bankruptcy dynamics, the second two proxy for the US business cycle, and firm-leverage to replicate the average level of financial risk which in turn may inform both the level of bankruptcies and volatility of business cycle fluctuations.

The SMM algorithm to calibrate the two parameters of interest by matching the five selected moments is straightforward. I first narrow the domain of candidate parameter draws to $(0, 1]$ for the entry cost (1 is the VC's total initial wealth) and $[0, 0.0891]$ for the probability of receiving an exogenous exit shock (0.0891 the average firm exit rate in the data sample). I then simulate 1000 economies solved with randomly drawn candidate parameters, calculate each simulation's model-implied moments, and store them in the vector $J(\bar{\theta}_i)$, where $\bar{\theta}_i$ is the vector of the i-th economy's randomly drawn parameter values. Note that model-implied moments are calculated with 100 periods of data that begins after a 100 period burn-in length. Having created a sample of model-implied moments, I find the set of candidate parameters, $\bar{\theta}_i$, that minimizes the standard GMM loss function:

$$\mathcal{L} = \sum_{i=1}^{N} \left((J(\bar{\theta}_i) - J(\theta))' \, W^{-1} \, (J(\bar{\theta}_i) - J(\theta)) \right)$$

where $J(\theta)$ is the vector of target moments and θ the parameters of the data generating process I am trying to characterize, W is the weighting matrix, and N is the number of simulated economies. I use the standard weighting matrix $W^{-1} = J(\theta)^{-1}$, which recasts the GMM problem as minimizing the (equally-weighted) squared percent deviations from the targeted moments.

It is standard to use a two-step procedure to update W based on the variance-covariance matrix of moment errors from a first stage estimation, typically using the $W = J(\theta)^{-1}$ as the first stage weighting matrix. This two-step procedure implicitly reduces the weight of noisy moments and increases the weight on moments tightly linked, thus informative for identification, to parameters of interest. However, I leave this extension to future work.

Data. Firm entry and exit data are collected from the Census Bureau's Business Dynamism Statistics dataset, which (almost) covers the entire universe of US firms. Output growth is measured as the annual percent change in real GDP. Firm leverage is collected from Compustat and measured as the median total liabilities to total (book) asset ratio among US public firms. Data is annual from 1980 through 2019.

G Solving the model

This appendix discusses the computational details behind solving the quantitative model described in Section 4 and generating the impulse response functions presented in Section 5. I generally follow the algorithm developed in Krusell and Smith (1998) and Khan and Thomas (2008, 2013), which was also the foundation used by Bloom et al. (2018). Because the algorithm is outlined in great detail in these previous works, I will keep the discussion of my implementation focused on my model and exercise specific details.

G.1 Solving agent problems

I first start by discussing how I solve each agent's problem for the computational exercises.

Banks. The Bank is assumed to be sufficiently capitalized such that there is no risk of bankruptcy —the Bank being unable to pay the deposit rate to households— or funding shortage to make loans. Additionally, the Bank is indifferent between making loans to incumbent firms versus leaving capital in storage. As a result, the Bank will lend to firms according to the firm-specific rate schedule defined by Equation 22, and can otherwise be ignored in the solving the quantitative model.

Firms. I solve the firm-level problem with a standard value function iteration.[48] However, instead of tracking the entire joint-distribution of firms, each firm instead tracks the productivity-weighed capital stock. This aggregate variable is a sufficient statistic for determining the prevailing wages and aggregate consumption (thus marginal utility of consumption) in the economy, and yields a more tractable state-space than following the entire distribution. This assumption of bounded rationally is standard (see the cited works above), and reduces the firm-level state space to $(\hat{K}, \epsilon, A, k, b, \sigma^A, \sigma^\epsilon)$. I discuss $\hat{K}$ in more detail below.

Venture Capitalist. As outlined in Section D.1, I have to solve the VC's portfolio allocation problem numerically. To start, note that I solve the firm's problem first, thus already have the firm value function in hand. Therefore, I begin solving the VC's problem already knowing the stock market value of entrants for the entire grid of aggregate states, and the probability of transitioning to any aggregate state (via productivity's known law of motion and the aggregate capital forecasting rule). It also follows that I calculate the expected value

[48] I initially solve the value function over a 4-dimensional array with a coarse 10 point grid per non-productivity state. I then use a cubic interpolation to project the value function over a new 100 point per non-productivity state grid. Both firm-level and aggregate productivity grid points are kept constant between initial and interpolated state-spaces. Aggregate productivity is discritized to a two-state process, and firm-level productivity to a three state process by Tauchen's method.

and standard deviation of an entrepreneurial portfolio for a given period. With the first two moments of the risky asset and the return of the risk-free deposit known, I can find the VC's portfolio allocation for any state by searching over a discrete grid of candidate allocation values, selecting the allocation that minimizes the absolute value of the first order condition defined by Equation 29.

Households. After the previous three agents have been solved, then the Household's allocations can be deduced from market clearing conditions. Moreover, the Household's influence on market clearing prices can be made analytical and is discussed in Section 4. As a result, I do not have to computationally solve for the Household's explicit policy functions.

G.2 The distribution of firms

I next discuss the distribution of firms, both how bounded rational agents forecast prices without tracking the entire distribution of firms —a limitation necessary to make the model's state space to a tractable size for computational solution methods— and how the true distribution is updated and tracked during simulations.

G.2.1 Bounded rationality and forecasting prices

Agents are forward looking and need to know two future objects, wages and the marginal utility of aggregate consumption in period $t+1$, as they make allocation decisions in period t. However, to properly forecast these two objects, agents need to track the full joint-distribution of firms. Following in the spirit of Krusell and Smith (1998) and Khan and Thomas (2008, 2013), agents will be assumed to suffer from bounded rationality and only track the productivity weighted level of firm capital, $\bar{K} = \int_{i\in\Delta} \epsilon_i K_i^{(1-\alpha)\chi} \mu_i \, di$. Then, using a forecast of the level of productivity weighted capital, $E_t \bar{K}_{t+1} = \hat{K}_{t+1}$, the agents can then forecast the wage rate one period ahead as:

$$\hat{w}_{t+1} = E_t A_{t+1} \alpha \chi \hat{K}_{t+1}$$

where $E_t A_{t+1}$ is given by the aggregate productivity law of motion. Agents, following Bloom et al. (2018), also forecast the marginal utility of consumption, $\hat{U}'(C_{t+1})$, as an auxiliary forecast of the productivity weighted level of aggregate capital. That is, agents perceive a linear mapping between the forecast $\hat{K}_{t+1}$ and the next period's marginal utility of consumption, such that $\hat{U}'(C_{t+1}) = a + b\hat{K}_{t+1}$, where a and b and coefficients to be estimated with standard OLS. Agents then form the stochastic discount factor

$$\Xi = \beta \frac{\hat{U}'(C_{t+1}))}{U'(C)}$$

where C_t is the period t aggregate consumption.

G.2.2 Updating the distribution of firms

Rather than simulate a discrete, large, sample of firms, similar to Krusell and Smith (1998), I instead track the distribution masses, more akin to the histogram approach proposed by Young (2010) as well as employed by Bloom et al. (2018).

G.3 Solving the model

I solve the model following a Krusell and Smith (1998) style algorithm. I initialize the forecasting rule for the productivity weighted aggregate capital, solve each agent's problem as outlined above, simulate a large number of periods, keeping track of the distribution of firms and aggregate states as outlined above, and ex post check for internal consistency in the forecasting rule throughout the simulation (i.e. check the forecasts are indeed an accurate forecast of the realized outcome). If the forecasting rule is accurate then the model is solved, if not, then I estimate the forecast rule based on the simulated data and rerun the simulation, repeating this process until the forecasting rule is accurate.

I currently use an AR(1) as the functional form of the forecasting rule and simulate 500 periods per forecast estimation loop. The literature, in particular Bloom et al. (2018), use a more sophisticated forecasting rule that allows for a state-dependent AR(1) forecasting process, and I plan on adopting this functional form in iterations of this research. I initialize the forecasting rule as a "random walk" forecast, that is with an auto-regressive coefficient of one. Future work should test the robustness of results to the forecast initialization point.

G.4 Current limitations

There are two primary limitations to current implementation of quantitative model that will need to be updated moving forward. First, agents forecast future aggregate capital with an AR(1) rather than the more customary state-dependant AR(1) in the literature. This constitutes a stricter form of bounded rationality than is standard and further work should be done to evaluate the robustness of the results to more sophisticated forecasting methods and stricter forecast accuracy tolerances. Second, I do not interpolate values off of any grid, which limits the state-space that can be explored during simulations and can potentially lead to jumpy impulse response functions, although this second consequence does not appear to be a problem in the current implementation.